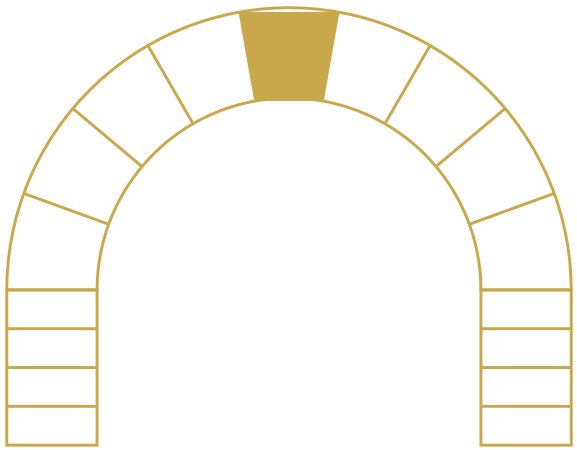


The Museum of Gerundelle

A Cognitive Architecture of Competing Thought-Molecules, Structured Ignorance, Territorial Ownership, Scarce Attention, and the Market Forces That Drive a Mind to Learn, Explore, Grow, and Express Itself

SCOTT D. SETO

The Museum of Gerundelle: A Cognitive Architecture of Competing Thought-Molecules, Structured Ignorance, Territorial Ownership, Scarce Attention, and the Market Forces That Drive a Mind to Learn, Explore, Grow, and Express Itself

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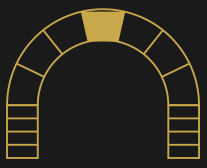
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THE MUSEUM

The Entrance Hall

I arrive at a building I have never seen before. It stands finished: the façade is stone, the windows are set, the steps are swept. A brass plate beside the door reads MUSEUM. I go in.

The lobby is high and cool. Glass cases line every wall, mounted on mahogany stands, lit from within. Not one of them holds a finished thing.

In the nearest case, a bird skeleton, wired upright, missing its sternum. The wire crosses the gap cleanly, as though the gap were meant. In the next, a clock face with all its numbers and no hands. In the next, a coat buttoned around no one. A bridge, in miniature, with a span that ends in open air. A sentence in careful handwriting on a strip of paper, the last three words left blank. Each case is complete — the glass clean, the label typed, the lighting exact — and each thing inside it is not.

I walk slowly. By the fourth case I have stopped looking for the finished ones. There are none. Everything in this building is partway through what it is doing, and the building has catalogued it that way.

I come to a long case near the center of the lobby. Inside, the coat and the clock and the bridge and the bird are arranged together, and a thread of red yarn connects each missing part to the next: the coat's absent chest to the bird's absent sternum to the bridge's absent span. The thread does not explain what is missing. It says only that what is missing is related.

By the door to the first hall there is a placard.

THE COLLECTION
Incomplete things, catalogued by activity. No materials recorded.

This museum collects things that are not finished. Every exhibit is partway through something: bearing, holding, connecting, opening, falling, locking. The museum records what each exhibit is doing. It does not record what it is made of.

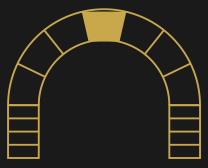
A stone arch and a steel beam bearing the same load receive one catalogue entry between them. A clock with no hands and a sentence with no ending are filed together if both are waiting.

The building is finished. The collection is not. It will not be. The cases are permanent and the things in them are in progress.

I read it twice. I look back at the lobby. The cases are permanent and the things in them are in progress. I had thought the museum was showing broken things. It is showing things that are still going.

In the corner beside the coat check, a strip of cloth hangs from a short pole. While I stand there it lifts once, swings gently, and falls still. I feel no draft.

Through the door I can see the first hall, already lit. I go in.



PREFACE

Building Gerundelle

Three Stories, and an Open Investigation Rather Than a List of Verdicts

IN THIS SECTION

- Meeting the three stories this book tells — how the agent understands, why it acts, and what breaks when you build it
- Understanding the epistemic contract between this book and its reader
- Seeing why every chapter follows the rhythm: What If → Why → Possibility
- Learning why "Building" is a gerund — and why that matters

Before the three stories, there was a question. It was not a research question. It was an irritation that would not go away.

The standard way to represent knowledge in a machine is a chain of rules: *if A then B; if B and C then D*. The chain is clear, checkable, and brittle. The moment your understanding of B changes — you discover B has a hidden condition, or B was never quite what you thought — the chain breaks. Every conclusion that depended on B must be retracted and rebuilt. Add a new condition and the chain breaks again. This is not a bug in any particular system. It is the nature of chains: they transmit force in one direction, and a link that changes shape shatters every link below it.

Worse: what the chain does not know, it cannot represent. In the standard framework, an unknown is a gap — a variable with no value, a query that returns nothing. The gap has no shape. It exerts no pull. It cannot tell the system what kind of evidence would fill it, or which of its neighbors are closest to answering it. The system knows what it knows. It has no structure for what it doesn't know.

This book is an attempt to build something that does not break that way.

Not a better chain. Not a chain with more exception handlers, or a chain that retracts more gracefully, or a chain that searches more broadly. A different kind of structure altogether: one where what is not understood has a shape made of what is understood, where that shape exerts a pull on the system's attention, where changes in understanding do not propagate linearly down a chain but are resolved through competition among alternatives for scarce commitments, where a concept that sometimes works and sometimes fails is not broken but *split* on the condition that separates the two cases, where the split can be mended if it was wrong and confirmed if it was right, and where the whole process is driven not by a truth-maintenance system retracting propositions but by a market of agents bidding for the slots they can fill.

Every mechanism in this book — the force-balance world model, the thought-molecule tournament, the structured open variables, the two-phase storage, the ownership register, the fiction pipeline — is part of that answer. None of them was built to refute the chain. Each was built to solve a specific problem the chain could not solve, and the alternative emerged from the solutions taken together.

The three stories that follow are three ways of telling how that alternative was found.

This book tells three stories at once. They share a cast, a setting, and a question — the question just stated — but they are not the same story.

The first is a **scientific story about how**. It asks: can causal reasoning, conceptual understanding, emotional regimes, and autonomous inquiry emerge from interactions among forces, potentials, and structured relationships? Not from language. Not from labeled datasets. Not from gradient descent on a trillion tokens. From *forces* — the same kind of primitive that Newton used to explain the motion of a cannonball and Faraday used to explain the pull of a magnet. The scientific story makes a bet: that a small number of force-based principles, applied consistently across physical worlds and thought alike, can produce a system that discovers causation, invents tools, models its own confusion, dreams counterfactual experiments, and studies its own mistakes.

The second is a **story about why**. The force-based architecture explains how an agent understands its world. It does not explain why the agent acts at all. Curiosity — the original answer — turns out to rank questions without originating wants. The deeper answer is **territory**: an agent is a node that acts to keep or extend the boundaries it perceives as its own. Hunger, fear, curiosity, love, and status are all strategies for holding territory. Understanding is bought by the ownership slots that need it — which explains why inquiry is uneven, why it stops, and why an agent with nothing to claim collapses. This story arrived late in the build and reorganized what came before it.

The third is an **engineering story**. It asks: what actually happens when you try to build those principles in dependency-free code, where naive assumptions fail catastrophi-

cally on the first real run and every mechanism must justify its existence against a controlled ablation? The engineering story is messier, more humbling, and ultimately more instructive. Gerundelle's first prototype locked in on tick 1 and never recovered. Its meta-cognition module crushed the thoughts it was supposed to observe. Its understanding register bloated the pool until thirteen tests broke. Its inquiry system was first built as procedures outside the mind — invitation pulls, a conditions search, an experiment planner, thresholds tuned round after round — and had to be rebuilt as SOV holes suggesting experiment molecules that compete for attention. Every one of these failures taught something that the scientific story, told in the abstract, would never have revealed.



REMEMBER

Three stories, one architecture. The **scientific story** asks how force-based principles can produce understanding. The **territorial story** asks why the agent acts at all — what it holds, what it claims, what it grieves. The **engineering story** asks what breaks when you actually build it. All three are present in every chapter. None is complete without the others.

The Epistemic Contract

Before we begin, a contract between this book and its reader.

Throughout this book, the physical definitions of *force*, *potential force*, *work*, and *phase transitions*, as well as their cognitive counterparts in thought molecules, are presented as **Gerundelle's computational formalism** — not as established extensions of classical Newtonian mechanics or neuroscience. When we say "work" in Gerundelle, we mean a specific structural computation over gerund ASTs, not the physicist's $W = F \cdot d$. When we say "phase transition," we mean a threshold crossing in an internal reserve, not a thermodynamic state change. The terms are borrowed deliberately — the structural parallels are real and illuminating — but the borrowing is always declared.

Likewise, our experimental designs and ablation arms test whether specific mechanisms are necessary or sufficient *within this implementation* under controlled conditions. They leave the broader scientific questions open for investigation. When an experiment shows that removing the meta-cognition track causes the agent to loop indefinitely, that result tells you something about Gerundelle's architecture. It does not tell you that biological brains use the same mechanism. The claims are scoped. The extrapolations are the reader's to make.



WARNING

This contract applies throughout. Whenever Gerundelle uses a physics term — force, work, potential, phase change, equilibrium — it means the **computational formalism defined in this book**, not the classical physics concept. The parallels are deliberate and productive. The identity is not claimed.



LINEAGE

The Bitter Lesson and the structure-first graveyard. Rich Sutton's *Bitter Lesson* (2019) argues that methods leveraging computation and learning consistently beat methods that leverage human-designed structure. The graveyard of symbolic AI systems that tried to hand-code the world is real and sobering. Gerundelle makes this bet with eyes open: its vocabulary of 102 gerunds is designed structure, not learned from data. The difference, we argue, is that Newton's force vocabulary *closes* — a small set of primitives covers the space — while GOFAI's frame-based ontologies never did. Whether that distinction holds is exactly what the ablation experiments test. If it doesn't, Sutton wins again. **Alfred North Whitehead** deserves a nod here too: his *process philosophy* (1929) insisted that reality is composed of processes, not static substances — the philosophical ancestor of a world model built entirely from gerunds.

Why "Building" Is a Gerund

This book is *The Museum of Gerundelle*, after the galleries you walk through before each chapter. This preface is *Building Gerundelle*, and the word is chosen. Not *Built Gerundelle*, not *The Gerundelle System*: a gerund, an ongoing activity, a process without a declared endpoint. This is not an accident.

Every chapter in this book follows a three-beat rhythm: **What If**, **Why**, and **Possibility**. The *What If* poses a question — not as rhetoric, but as a genuine open premise whose consequences we don't fully know. The *Why* motivates the question by showing what fails without the proposed mechanism, often through a traced engineering failure from the real implementation. The *Possibility* sketches what could happen if the mechanism works — without spoiling the result. And at the end of every chapter, **Suggested Experiments** give you concrete setups with observables, ablation comparisons, and open questions to discover for yourself.

This structure mirrors Gerundelle's own cognitive architecture. In Gerundelle, open questions drive awareness — an incomplete thought dominates consciousness through open-site tension until its missing piece arrives. A finished thought precipitates into

storage. A question that has been fully explored stops asking. The book works the same way: each chapter opens a question, investigates it, and lets you discover when it settles.



TIP

This book is designed as a **computational laboratory manual**, not a textbook. A textbook tells you the answers and asks you to remember them. A laboratory manual gives you the apparatus and asks you to find out. The Suggested Experiments at the end of each section aren't exercises with known solutions — they're genuine experiments with setup, observables, and open questions. Some of them have been run. Some haven't. The ones that haven't are waiting for you.

The Central Thesis

Gerundelle is an attempt to construct intelligence from the dynamics of possibility: what exists, what can change, what causes change, how an agent comes to understand those relationships through experience, and why it bothers.

The world model uses no coordinates, no material labels, no static properties — only gerunds (ongoing activities) connected by forces, with phase transitions triggered by internal reserve thresholds. The cognitive model uses no central observer, no random sampling, no hardcoded emotions — only thought molecules competing in a hierarchical tournament for limited working-memory slots, with attraction driven by expected surprise reduction and open-site tension.

The motivational model uses no designer-specified goals, no scalar reward signal, no curiosity-as-purpose — only a territorial ownership register whose open slots generate wants, fund inquiry, and determine when understanding is worth pursuing. Understanding explains how the agent knows. Territory explains why it cares.

The developmental curriculum uses no semantic shortcuts, no pre-given concepts, no training on final answers — only structured worlds where every concept must be earned through intervention, with an adaptive generator that reads the agent's open ownership slots and instantiates the next experience needed.

Whether this works — whether force-based principles and territorial ownership can produce something that deserves to be called understanding and purpose — is the open question. The architecture makes the bet. The experiments decide whether to keep it.

SIDEBAR

How to Read This Book

This book is organized into four parts and eleven sections, each containing multiple What If / Why pairs with suggested experiments. Before each chapter, a short passage in a different voice — first person, present tense — takes the reader through one gallery of a museum. The galleries show the chapter's concepts as physical exhibits: structures that pull, fills that seat or don't, gauges that spend down silently. The chapters that follow name what the galleries showed. Read the gallery first; it is short, and the chapter will land differently if you have walked through the room.

Part I: The Dynamics of Possibility builds the world model — gerunds, forces, affordances, and structural work. If you want to understand what Gerundelle's world looks like, start here.

Part II: The Chemistry of Thought Molecules builds the cognitive architecture — bonds, superposition, the tournament, surprise-driven learning, and two-phase storage. If you want to understand how Gerundelle thinks, start here.

Part III: Self-Reflection, Ownership & Collective Regimes adds metacognition, the understanding register, territorial ownership, and emotional phases. The territory chapter is the hinge: it explains why the agent acts, and the emotional phases that follow are readings of the ownership graph rather than hand-coded detection rules.

Part IV: Growing Up in Gerundelle covers the developmental curriculum, dreaming, self-study, and the argument for training on the messy journey. This is where the system learns to learn.

You can read linearly or jump to any section — each is self-contained. But the experiments build on each other, so running them in order will make the later ones more meaningful.

The World Is Instantiated. The Slots Are Open.

The architecture described in this book is real, implemented, and testable. The force engine runs. The thought-molecule tournament runs. The curriculum worlds run. The experiments have setup code, observables, and ablation arms. This is not a whitepaper sketch or a theoretical framework waiting for someone to build it. It's a system you can run, break, and learn from.

Some of the experiments in this book have been run and their results are known to the authors. Some have been designed but not yet executed. Some are genuinely open — we don't know what will happen. We've tried to be honest about which is which, and to resist the temptation to present open questions as settled ones.

The world is instantiated. The slots are open. The experiment begins.

Key Terms

Gerundelle	A computational architecture that constructs intelligence from force-based principles — gerund world models, thought-molecule cognition, and developmental curricula.
Gerund	A verb form naming an ongoing activity (supporting, falling, thinking) — the basic unit of representation in Gerundelle's world model, and the title of this preface.
What If / Why / Possibility	The three-beat rhythm of every chapter: pose an open question, motivate it with a failure, sketch what could happen.
Suggested Experiments	Concrete setups with observables, ablation comparisons, and open questions — a computational laboratory manual embedded in the book.
Epistemic contract	The agreement that Gerundelle's terms (force, work, phase change) name its own computational formalism, not established physics or neuroscience.
Traced engineering failure	A real implementation failure whose diagnosis motivated an architectural change — the engineering story behind the scientific claim.
Ablation	Turning off a specific mechanism and comparing the result against the full system — how Gerundelle tests whether each piece earns its place.

PART I

I

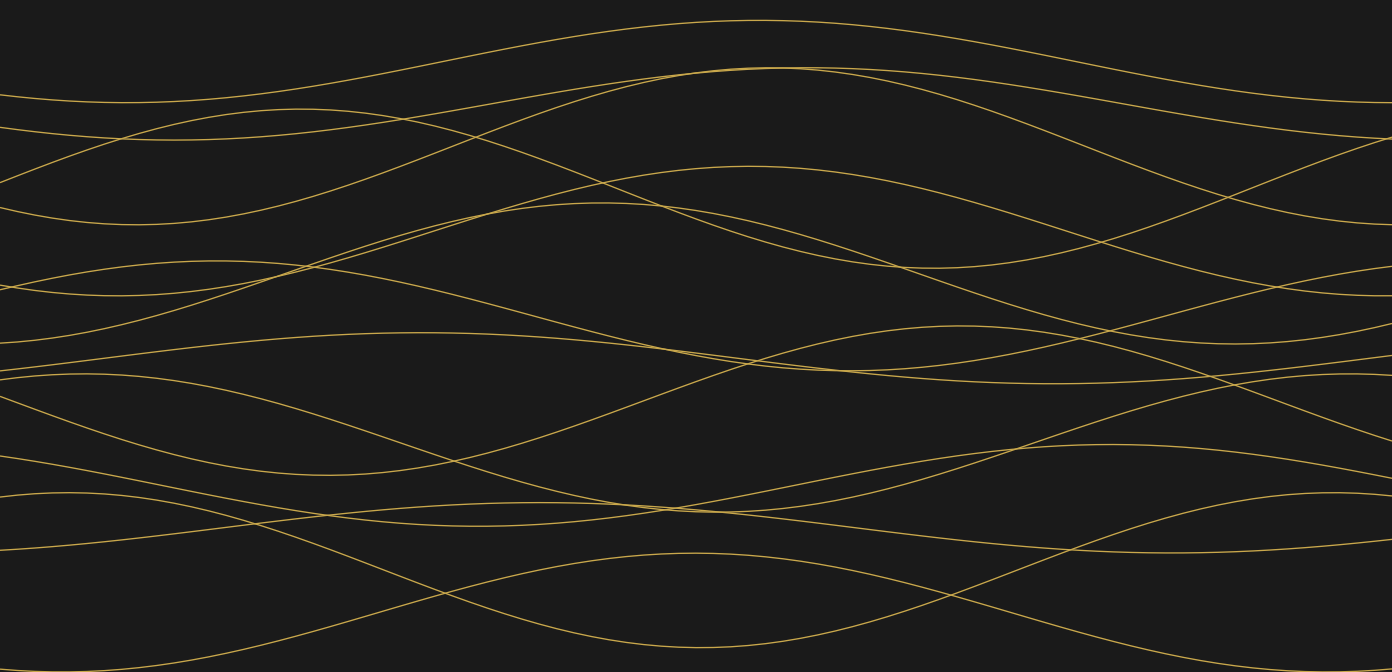
The Dynamics of Possibility

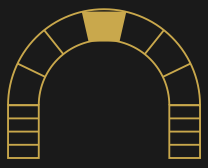
The Gallery of Doings

1 World Model Foundations

The Gallery of Openings

2 Potential Forces & Affordances





THE MUSEUM

The Gallery of Doings

The first hall is bright and warm. The ceiling is higher than I expected.

Two structures stand side by side on a long table: one of rough pale stone, one of dark steel. They are not the same shape. The stone one is an arch, wide and heavy. The steel one is a beam across two posts, thinner, longer, utterly different. I walk around them. They look nothing alike.

There is one label between them. It reads:

> **BEARING**
> *One entry. Two exhibits.*

I look again. Each one holds a weight at its center, a brass disc the same size, and under each weight the structure pushes down into its supports and the supports push back. The stone pushes through compression — I can see the joints tighten. The steel pushes through its own stiffness — the beam bows a little and holds. They are not doing the same thing with the same parts. They are doing the same thing.

In the next case, two blocks sit touching. I tap the glass and nothing moves. A small card reads: *glued*. Below them, two more blocks, the same size, resting against each other. The card reads: *placed*. I look for the difference and find it: the glued pair has no edge between them. They are one thing. The placed pair has a faint line. I lean closer. The line is where the force changes — above it, the blocks hold together; below it, nothing keeps them from separating. The boundary is not the surface. It is where the holding stops.

At the far end of the table I find the comic.

Four glass panels, upright, each showing the same shelf and ball. In the first, the ball sits on the shelf. In the second, the shelf is gone and the ball is lower. In the third, the ball is on the floor. There is no fourth panel. There are no arcs, no dotted lines, no arrows. Just: shelf, no shelf, floor.

A placard between the second and third panels:

A BALL, THREE MOMENTS
Falling, in three moments. Mounted in series.

In the first panel the shelf bears the ball. In the second the shelf is gone. In the third the ball rests on the floor.

What happened between the panels is not shown. It is not needed. The shelf held the ball against its weight. Then the shelf was not there. Then the ball was at the bottom. The reason it fell is in the first panel, not the second: weight, with nothing opposing it.

This museum does not collect trajectories. It collects the moments at which forces change.

I go back and look at the first panel. The ball on the shelf. I had not noticed: the shelf is slightly bowed under the weight. It is working. In the second panel it is gone and nothing is working. In the third the floor has taken over.

Against the far wall, alone, is a bridge.

It is a model, about two feet long, made of thin brass rods soldered at the joints. It spans a gap between two stone blocks. At its center a small platform holds a stack of tiny lead discs — weights, each the size of a shirt button. Beside the platform, a gauge: a glass tube, nearly full of dark liquid.

A sign says PLACE ONE WEIGHT. I take a disc from the stack and set it on the platform. The bridge does not move. The gauge drops by a hair. I set another. The bridge does not move. The gauge drops again. I set a third. The gauge is near the bottom now. The bridge has not changed at all — the rods, the joints, the span, everything exactly as it was.

I set the fourth weight.

The bridge buckles. The center drops, the rods on the left side twist, and the whole structure hangs in a new shape, still connected but no longer spanning. The gauge is empty. The weights sit in the crumpled center, three inches lower than before.

Nothing about the first three weights prepared me for this. The bridge looked the same after each one. The gauge knew. The bridge did not show it until the last.

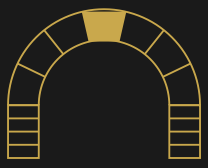
A placard on the wall behind it:

> **BRIDGE WITH RESERVE**
> *Spanning, then hanging. One reserve.*
>
> The bridge holds a reserve: the capacity to bear without changing shape. Each weight spends a little of the reserve. The bridge does not show the spending. The gauge does.
>
> When the reserve reaches zero, the structure reorganizes in one step. This is not breaking. The rods are intact. The joints are intact. What changed is what the structure is doing: it was spanning, and now it is hanging.
>
> The question of whether it was already failing on the third weight is the question of whether you watch the gauge or the bridge.

I stand with the crumpled bridge for a while. Four weights. Three of them did nothing visible. The fourth did everything. I think of the rubber band I snapped this morning and wonder whether it knew on the second stretch.

On my way out I pass the stone arch and the steel beam again. I look at the single label between them. It does not say what they are made of. It says what they are doing.

The cloth in the corner lifts and hangs still.



SECTION

1

World Model Foundations

Forces, Gerunds, and the Physics Beneath Every Modality

IN THIS SECTION

- Seeing why language, images, and video are surface renderings of a deeper world model, not the model itself
- Discovering why defining objects by what they are doing beats defining them by what they are made of
- Understanding how object boundaries follow force boundaries, not visible surfaces
- Learning why Gerundelle strips out position, velocity, and continuous time entirely
- Following phase changes from rubber bands to bridges to changes of mind

WHAT IF Language, images, and video are merely 'printouts' of a deeper world model, not the model itself

Imagine you have a printer connected to a weather station. It can produce a temperature graph, a humidity chart, a wind-speed log, or a plain-English summary: *it's going to rain this afternoon.* Each printout looks completely different. The graph is a line, the chart is bars, the summary is sentences. But they all came from the same instruments reading the same storm.

Now imagine you've never seen the weather station. All you have are the printouts. You might try to learn the weather by studying how words relate to words in the summaries, or how pixels relate to pixels in the charts. You'd get surprisingly far. But you'd be learning the printout, not the weather.

What if language, images, and video are printouts like these — surface renderings driven by a deeper, modality-independent world model that we've been ignoring because the printouts are so spectacularly available?

WHY Each modality needs its own Reader and Printer against one structural space

One Structural Space for Every Modality

Multimodal models like CLIP and GPT-4V align modalities by training on co-occurring image-text pairs — learning that pictures of dogs appear near the word "dog." The alignment is statistical: it works when the training set contains a similar pairing, and breaks when it doesn't. Present a novel physical configuration that has never been photographed alongside a description — a siphon made of unfamiliar materials, say — and the model has no co-occurrence data to bridge the gap. The alignment was in the data, not in the structure.

Gerundelle inverts this. Each modality gets a bidirectional translator: a **Reader** that converts a rendering into a structural form, and a **Printer** that converts structural form back into a rendering. All readers and printers talk to a single shared structural space — content-free, with a null point for "matches no structure." A sentence describing an arch and a force diagram of that same arch should land at the same coordinates, not because that pairing appeared in a training set, but because both renderings have the same force-balance skeleton. Cross-modal analogy stops being a special alignment problem and becomes simple geometric closeness: two things are analogous when their readers map them nearby.

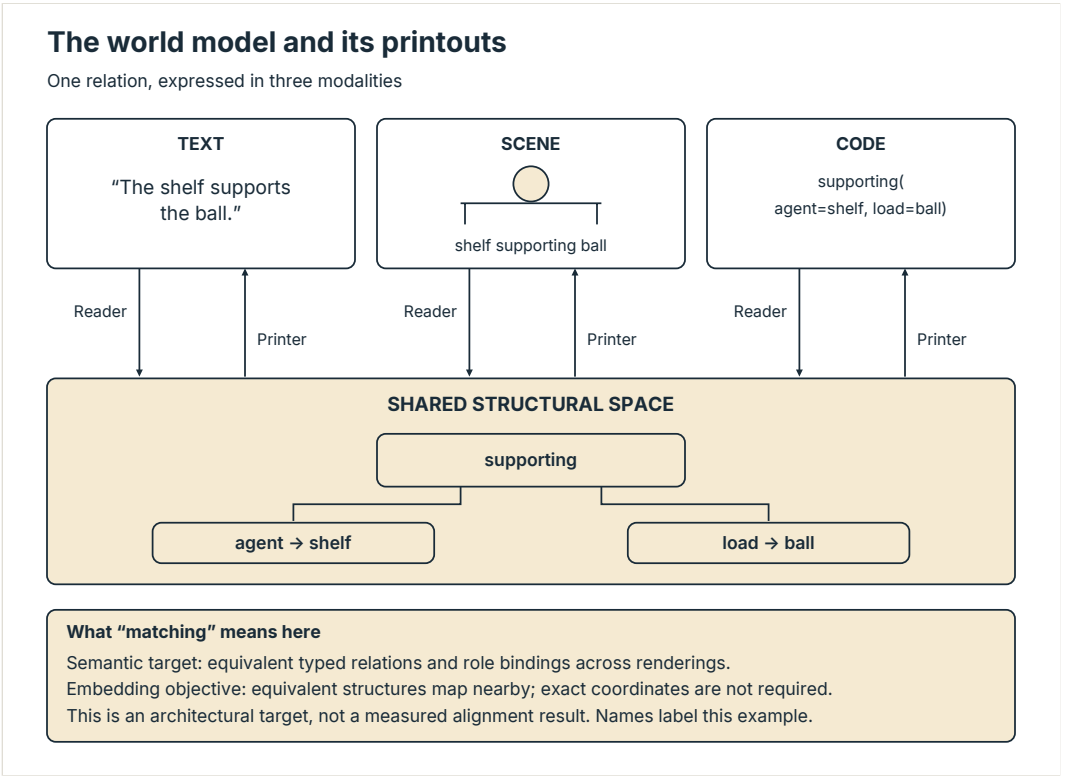


Figure 1 **The world model and its printouts.** Three renderings express the same filled supporting relation. Equivalent typed structure is the semantic target; proximity is the proposed embedding objective.

REMEMBER

The world model is the weather station, not any of its printouts. Language, vision, and physics simulations are all renderings of a shared structural space. If you train each modality's reader against one space, then **"this sentence describes this scene"** becomes **"these two points are near each other"** — no special cross-modal machinery required.

LINEAGE

Spelke's core knowledge and Tenenbaum's intuitive physics. Developmental psychologist **Elizabeth Spelke** has shown that human infants come equipped with expectations about objects, contact causality, and persistence — a physics-shaped model is innate equipment, not something learned from language. **Josh Tenenbaum** and colleagues at MIT built this into computational terms with their *intuitive physics engine*: a probabilistic simulator that predicts how scenes will unfold, matching infant looking-time data. Gerundelle's nearest relatives in cognitive science already model reasoning as simulation over force-like primitives. What Gerundelle adds is the positionless, slice-based representation — stripping out the continuous coordinates that Tenenbaum's engine keeps.

WHAT IF We defined objects solely by what they are doing under forces, not by material or properties

What If Objects Are Defined by Their Doings?

Imagine describing a stone arch to someone who has never seen one. You could list the parts: seven stones, some mortar, a keystone at the top. That description is precise and completely useless to someone trying to understand how an arch works.

Or you could describe what each part is *doing*: the base stones are **anchoring**. The side stones are **supporting**. The keystone is **wedging**. The mortar is **binding**. Gravity is **pressing**. Now replace every stone with a steel beam. The parts list changes completely — different material, different shape, different weight. But the description of what each part is doing? Identical.

What if we refused to describe the world by what objects are made of, and defined every entity solely by what it is doing under the forces acting upon it?

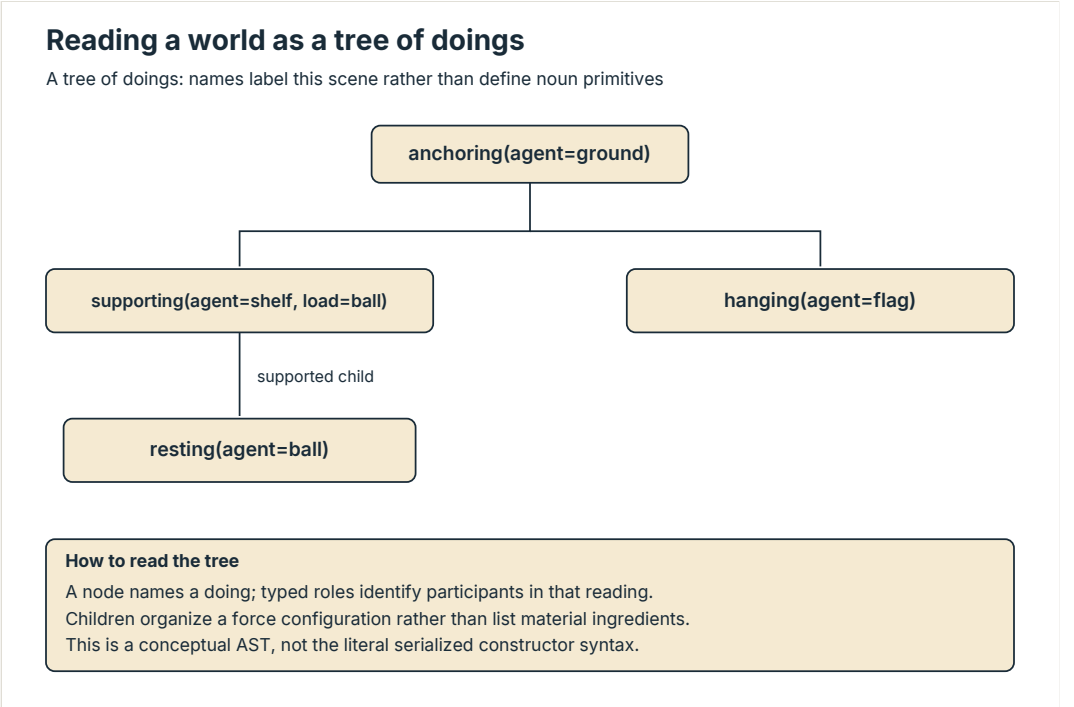


Figure 2 **Reading a world as a tree of doings.** A small scene becomes a hierarchy of doings and typed relationships. Participant names identify the example; the AST is conceptual.

WHY A stone arch and steel cantilever share one skeleton when you describe doings, not materials

Material Invariance

Object-centric models like CLEVR, MONet, and Slot Attention decompose scenes into objects defined by visual features — color, shape, size, material. Replace every stone in an arch with steel beams and these models see entirely different objects. The representation must be re-learned from scratch, because material was load-bearing in the description. This is the heart of Gerundelle's representational bet against that approach.

A gerund — *supporting*, *hanging*, *containing*, *wedging* — names an activity, not a substance. Attach unnamed, role-typed slots to each gerund (the thing doing the supporting, the thing being supported, the force being transmitted), and you get a skeleton that doesn't care about material. A stone arch and a steel beam bearing the same load share a single catalogue entry in structural space: something anchors, something supports, loads transmit through the structure to the ground. The material decides how much load the structure can bear before failure. The gerund skeleton decides what the struc-

ture *is doing at all*. Change the material and the failure point moves; the description of the activity stays put.

TECHNICAL STUFF

Gerund ASTs. In Gerundelle, every node in the world model is a gerund with typed slots, not a noun with properties. "Stone" never appears in the representation. What appears is `supporting(agent=?, load=?, via=compression)` — and the same node works whether the agent is limestone, steel, or ice. This is called **material invariance**, and it's what makes cross-domain analogy a structural operation rather than a metaphorical stretch.

LINEAGE

Talmy's force dynamics. Linguist **Leonard Talmy** showed in the 1980s that natural language already organizes causation as force patterns: an *agonist* (the thing that tends toward a state) and an *antagonist* (the thing that opposes it). "The ball kept rolling" has one force pattern; "the wall stopped the ball" has another. Talmy's insight was that the grammar of causation is *force-dynamic*, not logical — humans don't parse causes as abstract conditionals but as clashes between opposing tendencies. Gerundelle's 102-gerund vocabulary is, in this light, not an invention but an *excavation*: completing a formalization that linguistics started but never connected to a computational world model. Appendix A's force tables can be read as Talmy's categories given slots, reserves, and transition rules.

WHAT IF The boundary of an object is literally where internal force gives way to external force

What If Object Boundaries Follow Force Boundaries?

Pick up a brick. It's obviously one object. Now look at a brick wall. Is that one object? What about a brick wall with a crack running down the middle — is that one object or two?

The usual answer involves spatial boundaries: look at the gap. But that answer fails for a wall under compression. Squeeze a cracked wall hard enough from both sides, and the crack doesn't matter — the wall transmits load as one thing. Release the compression, and the two halves fall apart. The object count changed, but nothing moved. What changed was a force.

What if the boundary of an object is literally where internal force gives way to external force? Not a visible edge, but a *force frontier*.

WHY Object boundaries are force boundaries, not spatial surfaces

Composition Is a Force Relation

Image segmentation models like SAM and Mask R-CNN define object boundaries by visual discontinuities — edges, color changes, depth gaps. A cracked wall gets split into two objects because the crack is visible. But squeeze that cracked wall from both sides and it transmits load as one thing; release the compression and it falls apart into two. The visual boundary didn't change. The force boundary did. Segmentation gets the answer right only by accident, when force and vision happen to agree.

Gerundelle draws boundaries where forces flip. A wall exists as a single node only while its internal cohesion — mortar binding, compression, friction — balances the external load trying to pull it apart. The boundary of the object isn't a surface you can photograph. It's the frontier where forces switch from holding things together to separating them. Weaken the mortar past a threshold, and one node (*wall:standing*) becomes many nodes (*bricks:piling*). Strengthen the compression, and a pile of loose stones becomes a single supporting structure. The inventory of parts didn't change. The balance of forces did.



REMEMBER

In Gerundelle, you don't ask **"what is this made of?"** You ask **"what forces are holding this together, and what would happen if they changed?"** The boundary of an object is the place where the answer to that question flips from "internal" to "external."

WHAT IF Motion is already implied by unbalanced force (falling = weight with nothing opposing it)

What If Physics Had No Coordinates?

Draw two pictures. In the first, a ball sits on a shelf ten feet above the ground. In the second, the shelf has vanished and the ball is somewhere between the shelf's old position and the floor.

A physics textbook responds with coordinates, velocity vectors, and differential equations. The parabolic trajectory is beautiful and precise — and it tells you almost nothing about *why* the ball is falling. To understand the why, you need one fact: the shelf was supporting the ball, and now it isn't. Weight with nothing opposing it. That's the entire causal story.

What if a world model had no 3D coordinates, no Euclidean distances, no velocity vectors, and no differential equations — only discrete time slices and the force-balance gerunds holding at each slice?

WHY Continuous kinematics overcomplicate causal understanding

The Causal Story Without the Trajectory

Physics simulators like MuJoCo and PyBullet track full state — position, velocity, acceleration — at every timestep. Ask one "why did the ball fall?" and it can replay the exact parabolic trajectory with sub-millisecond precision. But the causal answer — "the shelf was removed" — is buried under thousands of state vectors. The simulator provides maximal precision and minimal causal transparency.

Gerundelle strips all of that out. Falling means weight with nothing opposing it. Where the ball ends up is simply the gerund at the next time slice: *piling* on the ground, or *resting* if something catches it. Continuous kinematics track *how* something moves with beautiful precision. Force-balance gerunds track *why* something changes with minimal machinery. Gerundelle bets that causal understanding lives in the why, not the how — keeping only discrete force-balance snapshots and the transitions between them.



TIP

Think of it like a comic strip versus a movie. A comic strip shows only the key moments: the shelf is there, the shelf is gone, the ball is on the ground. The causal story is completely clear. A movie adds smooth motion between frames, but that interpolation doesn't add causal clarity — you already knew why the ball fell. Gerundelle is the comic strip. It keeps the causal story and throws away the smooth interpolation between panels.

WHAT IF Phase changes (breaking, melting, changing one's mind) are transitions watched by internal reserve thresholds

What If Phase Changes Are Threshold Crossings?

Hold a rubber band between your fingers and stretch it gently. It's the same rubber band — longer, thinner, but still intact. Stretch it a little more. Still fine. Now stretch it just a bit further, and without warning, it snaps. A moment ago it was one thing. Now it's two things. Nothing about the gradual stretching prepared you for the discontinuity.

Breaking, snapping, melting, boiling, buckling — and, more provocatively, *changing one's mind*. What if all of these are the same kind of event: an internal reserve ticking

down silently under sustained force, until a threshold is crossed and the structure reorganizes in a single step?

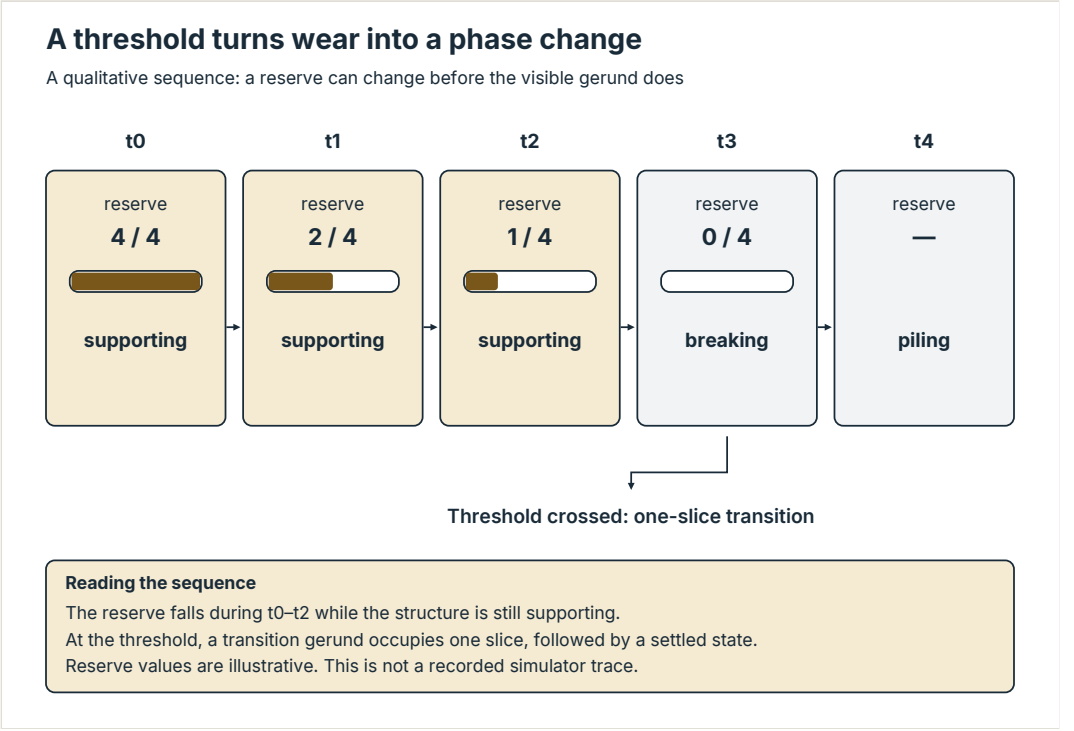


Figure 3 **A threshold turns wear into a phase change.** Hidden damage can accumulate without changing the visible gerund. A threshold crossing produces a transition and then a settled state.

WHY Phase changes need internal reserves, and the grain of the gerund vocabulary is a critical design choice

The 102-Gerund Vocabulary and the Grain Question

Continuous dynamics models — neural ODEs, differentiable physics engines — model smooth change with gradients everywhere and no discontinuities. This works beautifully for a rubber band stretching... right up until it snaps. The continuous model must somehow produce a discontinuous event from smooth dynamics. It either misses the snap entirely or requires special-case handling that breaks the differentiable framework. The foil fails at the exact moment the interesting thing happens.

Gerundelle models each phase change as a child node watching an internal reserve: capacity, tensile strength, cohesion, grip, seal, heat tolerance, or resolve. The reserve can tick down gradually — cracking, bending, wearing — without changing the surface gerund. The wall still stands. The beam still supports. But when the reserve hits zero,

the gerund flips in one time slice: *standing* becomes *collapsing*, *supporting* becomes *buckling*, *spanning* becomes *hanging*, *resolved* becomes *yielding*.

This design creates what Gerundelle calls the **grain question**: how many distinct gerunds do you actually need? Is "snapping" different from "breaking," or are they the same phase change applied to different reserve types (tensile versus cohesion)? The current vocabulary contains 102 gerunds across ten functional families — Bearing, Losing Balance, Contact, Transmitting, Holding Together, Storing and Releasing, Containing and Flowing, Phase Changes, Heat, and Abstract Agent Forces. The number is a design choice, not a physical law, and the experiments are built to test whether merging or splitting gerunds changes what the model can predict.



WARNING

The claim that **"changing one's mind"** belongs in the same formal category as **"a bridge buckling"** is intentionally provocative. Gerundelle treats resolve as an internal reserve, just like tensile strength — it can be worn down gradually by persuasion or doubt, then flip in one step. Whether this is a deep structural analogy or a surface pun is exactly the kind of question the experiments are designed to settle. The architecture makes the bet. The data decides whether to keep it.

Suggested Experiments

Each experiment below can be run against Gerundelle's force engine. The point isn't to confirm a prediction — it's to discover what happens when you turn a mechanism on and off.

EXPERIMENT I.1

The Stone Arch Test



Setup: Build two scene readers. The first describes a stone arch as a material list: `[stone, stone, stone, keystone, stone, stone]`. The second describes it as a gerund AST: `[supporting → wedging ← supporting]`. Now replace every stone with a steel beam. Feed the steel structure to both readers.

Observe: (1) Which representation transfers to the steel structure unchanged? (2) Which must be rewritten from scratch? (3) If you add a third material — say, ice blocks — does the gerund reader generalize without retraining?

Compare: Material-based reader versus gerund-based reader on the identical physical configuration, across three materials.

EXPERIMENT I.1

The Stone Arch Test (continued)

Discover: Does one skeleton genuinely fit all three? And when it doesn't — when the ice arch sags under its own weight because the reserve is lower — does the gerund reader degrade gracefully (the AST changes at the right node) or catastrophically?

EXPERIMENT I.2

The Silent Reserve

Setup: Model a masonry wall with a cohesion reserve of 4. Apply repeated light strikes, each reducing the reserve by 1 but never crossing the shattering threshold until the fourth blow. The wall's surface gerund stays *standing* through strikes one, two, and three.

Observe: (1) Does the wall's visible representation change before the final strike? (2) At which strike can an external observer first detect that the wall has been weakened? (3) If you place an agent in front of the wall, does it treat the fourth strike differently from the first?

Compare: Visible phase change (the wall cracks immediately) versus silent reserve accumulation (the surface gerund doesn't budge until the threshold is crossed).

Discover: Can an agent detect hidden state accumulation when the surface doesn't change? And if it can't — if strikes one through three look identical — what kind of memory or internal model would it need to predict the sudden collapse on strike four?

EXPERIMENT I.3

The Compressed Wall

Setup: Take a cracked brick wall — two halves with a visible gap. Apply lateral compression from both sides. Measure the force transmitted through the wall before, during, and after compression.

Observe: (1) Does the model represent the cracked wall as one node or two? (2) Does adding compression change the node count? (3) If you gradually reduce compression, at what point does one node become two?

Compare: A spatial boundary model (crack means two objects) versus Gerundelle's force boundary model (internal cohesion versus external load decides).

Discover: Does the force-boundary model correctly predict that a cracked wall under compression behaves as one object — and does it predict the exact moment the wall becomes two objects as compression is released?

EXPERIMENT I.4

The Comic Strip

Setup: Model a ball on a shelf. Remove the shelf. Step the simulation forward three time slices. Compare two representations: (1) a continuous physics engine tracking position and velocity at each instant, and (2) Gerundelle's gerund-only engine tracking force-balance states at each discrete slice.

Observe: (1) Do both engines agree on the final state (ball on ground, *resting*)? (2) Does the gerund engine capture the causal chain (supporting → falling → piling/resting) without coordinates? (3) For a downstream reasoning task — "why did the ball end up on the ground?" — which representation supports a shorter, more accurate explanation?

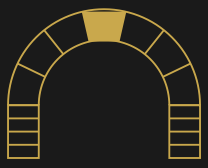
Compare: Coordinate-based physics (position, velocity, forces at every instant) versus gerund-based slices (force-balance state at each discrete step).

Discover: Does throwing away the trajectory lose any information that matters for causal reasoning? And where does it fail — what kind of question requires coordinates that gerunds can't answer?

Key Terms

World model	The modality-independent structural representation beneath all surface renderings — the weather station, not the printouts.
Reader	A translator that converts a surface rendering (text, image, force diagram) into a position in the shared structural space.
Printer	A translator that converts a position in the structural space back into a surface rendering in a specific modality.
Gerund	A verb form naming an ongoing activity (supporting, wedging, falling) used as Gerundelle's basic unit of world description.
Gerund AST	An abstract syntax tree where every node is a gerund with typed force slots, representing the functional structure of a scene.
Material invariance	The property that a gerund skeleton remains unchanged across different physical materials — stone, steel, and ice arches share one structure.
Force boundary	The frontier within a structure where internal cohesive forces give way to external forces — Gerundelle's definition of an object's edge.

Internal reserve	A hidden numerical quantity (capacity, tensile strength, cohesion, grip, resolve) that ticks down under sustained force and triggers a phase change when it hits zero.
Phase change	A one-slice transition that reorganizes the gerund structure when an internal reserve crosses its threshold (standing → collapsing, resolved → yielding).
Grain question	The design problem of choosing how many distinct gerunds the vocabulary needs — too few conflates different mechanisms, too many drowns the model in synonyms.
Time slice	One discrete step in Gerundelle's simulation — a snapshot of all force-balance gerunds, with no continuous time between slices.
102-gerund vocabulary	Gerundelle's current set of 102 gerunds across ten functional families: Bearing, Losing Balance, Contact, Transmitting, Holding Together, Storing & Releasing, Containing & Flowing, Phase Changes, Heat, and Abstract Agent Forces.



THE MUSEUM

The Gallery of Openings

The door closes behind me without a sound. The room is long and has no windows. In the middle of it, at the height of my chest, a structure hangs in the air. Nothing holds it up.

It is made of rods and joints, pale, like the bones of a bird. Among the rods there are openings. They are not breaks. The rods around them are finished and smoothed, turned inward to meet the edge.

I come closer. My sleeve lifts toward the nearest opening. I step back and it falls.

The dust on the floor has been combed into lines. They run toward the structure from every corner and gather under each opening. Under the largest one the floor is bare.

I count seven openings. On the far side I find an eighth. It is small, and the rods near it are still moving, settling a little at a time, the way a house settles at night. Around the oldest opening the rods have bent so far that they have almost become a different shape.

I hold my hand near the small opening. The pull is even and mild, like water in a slow stream. Near the oldest it is not mild. My fingers draw toward the center, and the joints of my hand ache as if they wanted to close around something.

The oldest opening is also the largest. I think the pull is because of its size.

By the door there is a placard. I did not see it when I came in.

STRUCTURE WITH OPENINGS
Rods and joints, eight openings. Suspended.

Every structure in this museum is incomplete, and is meant to be. Its openings are not damage. Each is the exact outline of something the structure does not hold.

An opening pulls. It draws on light, on dust, and on the visitor's hand, toward the shape it lacks. A structure is shaped less by what it holds than by what it is missing. Its openings are its most active parts.

New openings are surrounded by rods still adjusting to them. Old openings have bent the structure around themselves.

Visitors often try to rank the openings by the strength of their pull. Size is not the measure. Age is not the measure. The measure is shown in the Gallery of Held Things.

I read it twice. Then I go back and stand under the oldest opening again. The placard says what it lacks. It does not say what it lacked before.

◆ ◆ ◆

Past the structure, against the back wall, a boulder sits on a wooden prop. The prop is thin — a stick, barely wider than my thumb, wedged under one edge. The boulder is the size of a horse. A small sign on a chain reads: TOUCH THE PROP.

I put one finger on it. I barely press.

The prop tips. The boulder drops. The floor tilts — the whole floor, hinged at the far wall — and three glass cases on it slide sideways, striking each other. A door at the end of the gallery swings open. A counterweight somewhere below me groans and settles.

I stand with my hand still out. One finger. I have rearranged the room.

Against the opposite wall, another exhibit: a steel plate, bolted to the floor. A sign reads: PUSH. I push it with both hands. It does not move. I lean into it. Nothing. I put my shoulder against it and shove. The plate is warm where my hands were. That is all.

A placard between the two:

A TOUCH AND A SHOVE
Propping. Resisting. Resting, then falling.

The touch spent almost nothing. The shove spent a great deal. By any measure of effort the shove was larger.

The touch opened a door, moved three cases, tilted a floor, and changed the shape of the room. The shove changed nothing.

This museum does not measure the push. It measures what changed because of the push.

◆ ◆ ◆

In the last alcove, three glass cases in a row, each holding something that looks like the others.

The first holds a spring, clamped tight, its coils visibly compressed. I can see it straining. It is ready to do something, the moment the clamp lifts. Below it, a card: *wound.*

The second holds the gauge from the bridge — or one like it. The same glass tube, the same dark liquid, at about half. Below it: *half spent.* It is not ready to do anything. It is resisting something, less than it was, and when it empties the thing it resists will happen.

The third is empty. Not broken, not cleared out — the case is clean and lit and there is nothing in it. A faint pull, the same kind I felt from the openings, draws my hand toward the glass. Below it: *waiting.*

A placard between the three:

THREE THINGS THAT ARE NOT THE SAME THING
A wound spring. A half-spent gauge. An empty case.

The spring holds energy it accumulated. Release the clamp and it pushes.

The gauge holds resistance. It is being spent by something pressing against it. When it empties, the thing it holds will change.

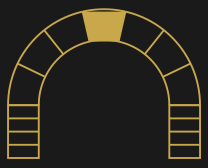
The empty case holds nothing. It has no energy and no resistance. It has an outline where something fits, and a pull toward what fits there.

These three are commonly confused. They should not be. The spring will act. The gauge will give way. The case will receive.

I look at them again. The spring is tense. The gauge is patient. The case is open. Three different kinds of waiting.

In the corner a strip of cloth hangs from a short pole. While I watch, it lifts, swings once, and hangs still again. I feel no draft. There is no placard for the cloth.

When I leave, my hand is still aching.



SECTION

2

Potential Forces & Affordances

What a World Invites Before Anyone Acts

IN THIS SECTION

- Seeing why labeling objects as 'pushable' or 'breakable' creates circular definitions
- Discovering how an open force slot formalizes what an object invites
- Distinguishing three concepts that sound alike but aren't: internal reserves, stored potential, and potential forces
- Understanding why a light tap can count as more work than a hard shove
- Learning how Gerundelle computes work without displacement

WHAT IF An agent perceives objects through the potential interactions they invite, not semantic labels

Imagine you've never seen a door handle. You walk up to a closed door and there it is: a horizontal bar at waist height, mounted on a vertical panel. Nobody tells you it's a handle. Nobody tells you to pull. But something about the shape — the bar is grip-sized, it protrudes just far enough, and the door is clearly a barrier — suggests that grabbing and pulling might do something interesting.

The psychologist James Gibson called this an **affordance**: what an object offers for doing. A flat surface affords sitting. A knob affords turning. A gap affords passing through. Gibson's insight was that organisms don't first recognize objects and then figure out what to do with them — they perceive *possibilities for action* directly.

What if an agent perceived every object in its world not through labels like "door" or "lever" but solely through the potential interactions the object's structure invites?

WHY Gibson's affordances become circular when implemented as hand-coded semantic tags

The Circularity of Hand-Coded Tags

Gibson's idea is elegant in theory. In practice, it usually gets implemented as a tag: `is_pushable: true`, `enables: grasp`, `affordance: sit`. These tags look functional, but they're just the designer's vocabulary glued onto the object. The agent doesn't *discover* that a surface affords sitting — it reads a label that a programmer wrote. The label says "you can sit here" the same way a dictionary says "pain: an unpleasant sensation." It's circular: the agent knows it can sit because the tag says it can sit.

Gerundelle replaces the tag with a structural fact. A **potential force** is an **open force slot** in the gerund AST — a place in the structure where a force *could* be applied but currently isn't. A lever with nothing pressing its input arm has an open slot. A support with nothing resting on it has an open slot. The slot isn't a label someone typed; it's a consequence of the current force configuration. Remove the lever from the scene and the slot vanishes. Add a weight to the lever and the slot fills. The affordance lives in the structure, not in a lookup table.



REMEMBER

An affordance in Gerundelle isn't a property of an object. It's an **open force slot** — a structural invitation created by the current configuration of forces. Change the forces and the invitations change. The agent doesn't need someone to label what's possible; it reads the open slots directly from the AST.



LINEAGE

Gibson's radical claim. James Gibson (1979) proposed something genuinely radical: organisms don't first recognize objects and then compute what to do with them. They perceive *affordances* directly — action-possibilities that are relational properties of the animal-environment system, not intrinsic tags on objects. A surface affords walking *relative to the animal's legs*. A gap affords passing-through *relative to the animal's body*. Gibson's ontology was inherently relational and dynamic. What happened next was a dilution: HCI and robotics adopted the word "affordance" but implemented it as a static tag (`is_graspable`), losing the relational core. Gerundelle's open-force-slot formalization is an attempt to recover what Gibson actually meant: affordances as *structural invitations that exist only in context*, not labels a designer wrote. Gibson was right. He lacked the formalism. The **GOFAI qualification problem** lurks here too: every hand-coded predicate like `pushable` is a frame-axiom waiting to break in a new context. Open force slots avoid this because

LINEAGE (CONTINUED)
they're computed from the current configuration, not stored as permanent annotations.

WHAT IF A potential force is formally an open force slot in the structure

Walk into a room with a seesaw. One end is on the ground; the other is in the air. Nobody is sitting on either side. You can see, without touching anything, that the raised end is *waiting for weight*. That perception isn't magic — it's structural. The seesaw is a lever, the fulcrum is visible, one arm is loaded by the ground's reaction and the other arm has nothing on it. There is a slot in the force diagram where a downward force would change everything.

What if that slot — the empty place in the force structure — is the formal definition of a potential force? Not "this object could be pushed" as a vague semantic annotation, but a specific, computable gap in the AST where a specific type of force would complete a specific causal chain.

WHY Crossing Physical Invitation with Epistemic Gap produces curiosity and the formal definition of tool use

Physical Invitation Meets Epistemic Gap

Pure intrinsic-motivation agents — RND, ICM, curiosity-driven exploration — reward the agent for visiting states where its model is most wrong or most improvable, regardless of what the world's structure actually invites. A sealed, featureless box with no input becomes irresistible: the model is maximally uncertain about the interior, so the agent spends all its time staring at something it cannot interact with. Curiosity without a structural gate wastes attention on uninvitable mysteries.

Gerundelle gates curiosity with a structural fact: an open slot must exist. An open slot tells the agent *where* force could go. It doesn't tell the agent *what would happen* if force went there. These are two different things, and crossing them produces four distinct situations.

When a slot is open and the agent understands what filling it would do, the situation is straightforward — the agent can act or not, knowingly. When a slot is open but the agent doesn't understand the consequences, something more interesting happens: **curiosity**. The structural invitation is there, the outcome is unknown, and the pull to find out is formally the product of the invitation's existence and the size of the epistemic gap.

This crossing also gives Gerundelle a formal definition of **tool use**. A tool is any reconfiguration of objects that turns an *uninvited* force into an *invited* one. A bare hand can't break a wall — there's no open slot for that amount of force. But coupling a ram to the wall changes the structure: a new slot opens where amplified force can enter. The agent didn't change the wall. It changed the *invitation*.



TECHNICAL STUFF

The four quadrants. Cross **Physical Invitation** (open slot vs. no slot) with **Epistemic Gap** (understood vs. not understood) and you get: (1) open + understood = ready to act, (2) open + not understood = curiosity, (3) closed + understood = no action available, (4) closed + not understood = no action and no question. Curiosity lives in quadrant 2. Tool use is the act of moving from quadrant 3 or 4 into quadrant 1 or 2 by reconfiguring objects to create a slot that wasn't there before.



REMEMBER

A deeper reframing. An open force slot is not just an invitation to explore — it is **claimable territory**. The curiosity quadrant (open + not understood) is the unclaimed slot: a region the agent has never acted in, waiting for a first push. This reframing, developed fully in the Territory and Ownership chapter, changes what curiosity *serves*. Curiosity is not the base motive. It is the scouting strategy for an agent whose base motive is to hold and extend what it perceives as its own.



LINEAGE

Schmidhuber, Oudeyer, and artificial curiosity. The idea that agents should be driven by information gain rather than hand-coded rewards has a rich history. **Jürgen Schmidhuber** (1991) proposed curiosity as compression progress — an agent seeks states where its model is improving fastest. **Pierre-Yves Oudeyer** and colleagues developed *learning-progress* agents that explore environments by maximizing the rate at which prediction error decreases. Gerundelle's curiosity — invitation × epistemic gap — shares the core intuition (seek where learning is possible) but differs structurally: it's gated by *physical invitation* (an open slot must exist in the world) rather than being a pure function of model improvement. A sealed box with no input slot produces no curiosity, no matter how poorly understood it is. The structural gate means curiosity is about *what the world invites*, not just what the model lacks. A deeper divergence appears in the Territory chapter: learning-progress curiosity (Oudeyer) is uneven too — it concentrates on what is learnable now and leaves the mastered and the hopeless alone. But its unevenness tracks *learnability*, while territory's unevenness tracks *what the agent holds or could hold*. The case that separates them: lamp B, lit by a timer, is highly learnable but un-

LINEAGE (CONTINUED)
claimable. Learning-progress curiosity keeps studying it. Territory stops once it becomes a border.

WHAT IF Stored potential, internal reserves, and potential forces are three different things

Three concepts sound almost identical but do completely different work. Squeeze a spring and let go — that's **stored potential**. The spring releases energy it accumulated. Now think about a block sitting on a rough surface — it resists being pushed because of friction. That resistance is an **internal reserve**, like the cohesion holding a wall together or the grip keeping a rope taut. Finally, look at a lever with an empty input arm — that's a **potential force**, an open slot waiting to be filled.

What if confusing these three concepts — as most informal accounts of affordances do — is the source of deep modeling errors? A compressed spring is not the same as a wall's strength, and neither is the same as an empty lever arm. What happens when you try to treat them interchangeably?

WHY Internal reserves, stored potential, and potential forces serve different structural roles

Three Concepts That Must Stay Apart

Game engines and simplified physics models commonly lump everything that could cause change into a unified "energy" or "potential" field. A compressed spring, a block's friction, and an empty lever arm all get a single number. But the spring will *release* energy, the block will *resist* motion, and the lever arm will *accept* force — three completely different behaviors from three things the unified model treats identically. The first prediction that depends on the distinction fails.

An **internal reserve** is what resists change. It's the cohesion in mortar, the tensile strength in a cable, the grip of friction, the resolve of a person who hasn't yet been persuaded. It ticks down under sustained force. When it hits zero, a phase change happens. The reserve is defensive — it holds the current structure in place.

A **stored potential** is accumulated energy waiting to be released. A compressed spring, a raised weight, a charged capacitor. It doesn't resist anything — it's ready to *cause* something. Release the clamp and the spring pushes. Cut the rope and the weight falls. Stored potential is offensive — it's a loaded gun in the structure.

Potential force is neither. It's an open slot — an invitation in the current configuration where a force *could* act but doesn't. No energy is stored. No resistance is happening. The slot is pure structure: a gap in the AST that a new force would fill. Merging any two of

these three concepts into one representation guarantees that the model will confuse resistance with readiness, or readiness with invitation, and make predictions that fail at exactly the moment the distinction matters. The spring will act. The reserve will give way. The open slot will receive.

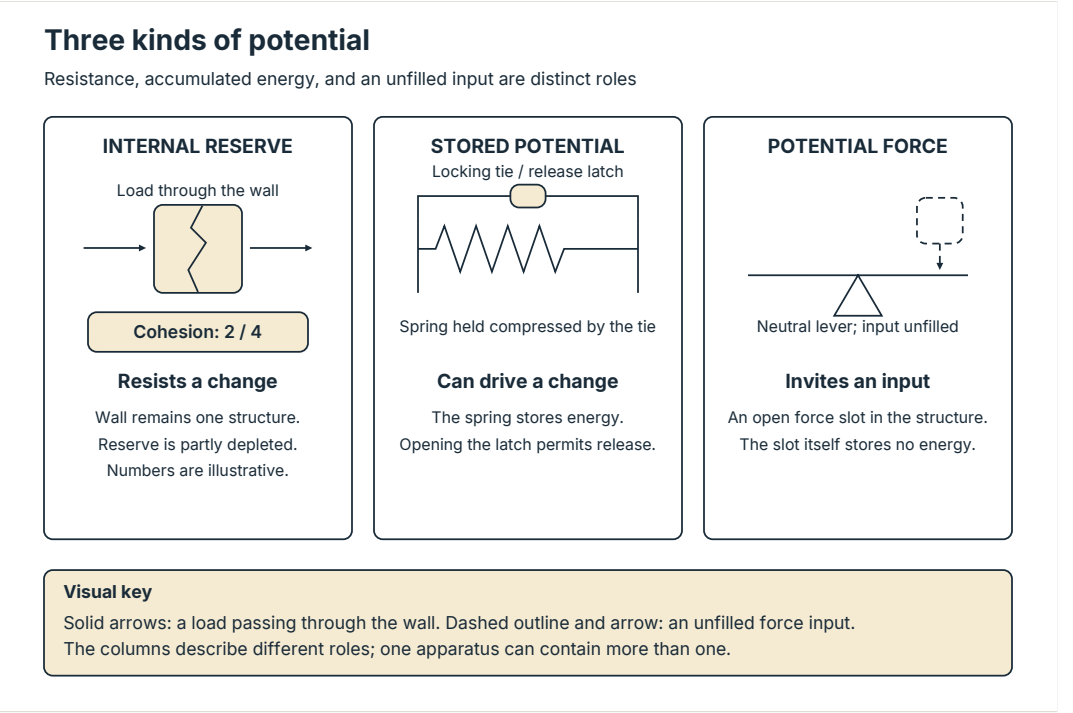


Figure 4 **Three kinds of potential.** Internal reserves resist change, stored potential can drive change on release, and potential forces are open inputs. The wall remains intact with partly depleted reserve.

WARNING

This three-way distinction is easy to nod at and hard to maintain. In everyday language we blur them constantly: "the wall has potential" could mean its reserve is high (hard to break), or that it stores energy (a retaining wall holding back earth), or that it offers an open slot (a wall with a crack inviting a wedge). Gerundelle's force vocabulary treats these as three different node types precisely because collapsing them produces models that get the easy cases right and the interesting cases wrong.

WHAT IF The magnitude of an action is the cascading reorganization it triggers, not the effort expended

A construction worker swings a sledgehammer at a concrete wall. The wall doesn't budge. That took enormous effort and achieved nothing. Meanwhile, a child presses a button on a detonator. A building collapses. That took almost no effort and reorganized an entire city block.

In everyday physics, work equals force times distance: $W = F \cdot d$. The sledgehammer scores high and the button press scores low. But in terms of what actually changed in the world, the ranking is inverted. The button press triggered a cascade — charges fired, supports failed, floors pancaked, dust clouds spread, evacuation routes changed. The sledgehammer moved nothing.

What if the true magnitude of an action is not the physical effort expended by the actor, but the cascading reorganization of forces and possibilities it triggers across the entire world?

WHY A light press on a loaded prop that drops a boulder computes to massive work in Gerundelle's formalism

Small Force, Massive Structural Work

Reward shaping in reinforcement learning handles this by having a designer manually assign values to states or actions — +10 for reaching the goal, -1 per step. But the designer can only reward what they anticipated. If the agent finds a short path that avoids a loaded prop, it never discovers that touching the prop triggers a cascade worth far more. Reward shaping measures the designer's imagination, not the world's causal structure.

Gerundelle replaces both $W = F \cdot d$ and hand-crafted rewards with a structural definition. Work is computed by counting what changed: how many gerunds flipped, how many internal reserves shifted, how many phase changes triggered, how many new force slots opened or closed — all measured over a bounded number of future time slices, discounted so that immediate changes count more than distant ones.

A light press ($F = 1$) on a loaded prop that tips and drops a boulder computes to massive structural work: the prop's gerund flips from *supporting* to *tipping*, the boulder's gerund flips from *resting* to *falling* to *piling*, force slots close where the boulder was and open where it lands, any structure beneath it may cross a reserve threshold and trigger further phase changes. A hard shove ($F = 5$) against an anchored wall computes to zero: nothing flipped, nothing opened, nothing changed.



TIP

Think of it like dominoes. The effort to tip the first domino is tiny. The effort to push a brick wall is large. But the domino triggers a chain reaction that reorganizes the whole table, while the brick wall absorbs your push and stays put. Gerundelle measures the **chain reaction**, not the **push**. That's why a child pressing a button can do more structural work than a worker swinging a hammer.

WHAT IF Counterfactual work is change in forces, reserves, phase states, and invitations — not displacement

Here's a subtlety that matters. Suppose a boulder is already sliding down a hill when you give it a push. The boulder crashes into a wall at the bottom. Did your push do that work, or was the boulder going to crash anyway?

To answer this, you need a **counterfactual**: what would have happened without your push? If the boulder would have crashed regardless, your push did zero work even though the crash was dramatic. If your push changed the trajectory enough to hit the wall instead of rolling past it, your push did all the work. The real question is never "what happened after I acted?" — it's "what happened *differently* because I acted?"

WHY Spatial displacement has no meaning in a positionless gerund AST, so work must be redefined structurally

Work Without Displacement

Temporal-difference credit assignment in reinforcement learning assigns credit to actions based on the difference between expected and received reward at the next timestep. A boulder is already sliding toward a wall. The agent pushes it in the same direction. The wall shatters. TD assigns full credit to the push because the state value changed dramatically after the action. But the crash would have happened anyway — the agent just happened to be there. TD confuses "I was present when it happened" with "I caused it."

Gerundelle needs a different foundation entirely, because it has no coordinates to begin with. Classical physics defines work through spatial displacement: force times distance. But there's no "distance" to multiply by. A ball doesn't move from position A to position B — it transitions from the gerund *resting* to the gerund *falling* to the gerund *piling*. Where the ball "is" doesn't exist in the representation. Only what the ball is *doing* exists.

So Gerundelle computes **counterfactual work** by stepping the world forward a bounded number of slices with the agent's action, then stepping it forward the same number of slices without the action, and taking the difference. The difference is measured in

structural units: gerunds flipped, reserves changed, phase states crossed, invitations opened or closed. Later slices are discounted so the computation stays bounded. The result is a single number that captures how much the agent's action changed the world's causal structure — without ever computing a position, a velocity, or a distance.

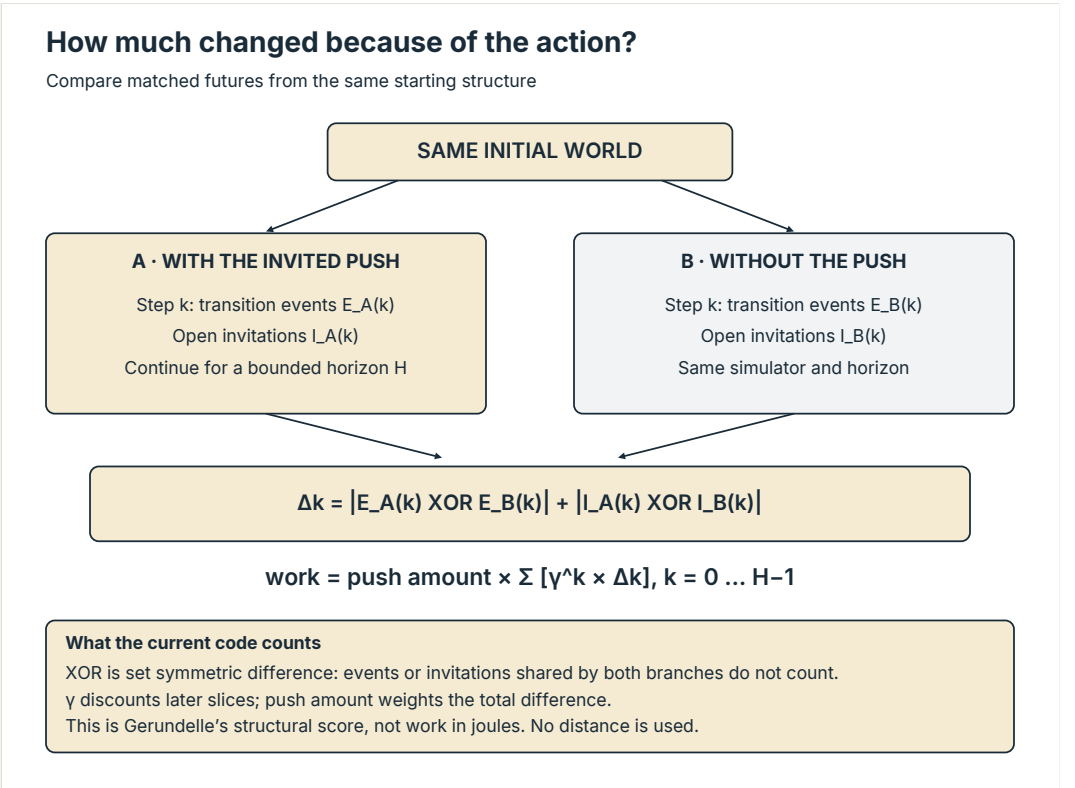


Figure 5 **How much changed because of the action?.** The code computes a structural score from paired futures. It multiplies discounted event and invitation differences by push amount; the result is not mechanical work in joules.

REMEMBER

Work in Gerundelle answers the question: **"How much did the world's force structure change because of what I did, compared to what would have happened if I'd done nothing?"** The agent can't peek at this counterfactual directly — it must estimate it from its own observations. That estimation gap is where curiosity, learning, and surprise live.

Suggested Experiments

Each experiment tests a specific claim about potential forces, affordances, or structural work. Run the main condition, then the ablation, and see what breaks.

EXPERIMENT II.1

The Unlabeled Room

Setup: Place an agent before a masonry wall alongside three unlabeled objects: a soft pad, a wedge, and a heavy ram. The agent's bare push fails — the wall's capacity reserve is higher than the agent's maximum force. No object has a name, a tag, or an `is_breakable` property. The agent can only see current gerunds and open force slots.

Observe: (1) Does the agent try coupling different objects to the wall? (2) In what order? (3) Does it discover that the ram amplifies transmitted force while the pad absorbs it? (4) Does the wedge open a new slot that the bare push didn't have?

Compare: An agent reading structural invitations (open force slots) versus an agent reading hand-coded semantic tags (`is_tool: true`, `enables: break_wall`).

Discover: Does the structural agent discover tool use — reconfiguring objects to create an invitation that wasn't there — without any object vocabulary? And does the tagged agent even bother exploring, since the tags already tell it what to do?

EXPERIMENT II.2

The Loaded Prop

Setup: Two actions of equal force magnitude. Action A: a light press ($F = 1$) on a loaded prop (a thin stick supporting a heavy boulder with a *tipping* watch). Action B: a hard push ($F = 5$) against an anchored wall. Step each forward 3 slices.

Observe: (1) Compute structural work for each action: gerunds flipped + reserves changed + invitations opened or closed, discounted by slice. (2) Compute counterfactual work: subtract changes that would have happened without the action.

Compare: Structural work (Gerundelle's measure) versus effort expended (classical $F \cdot d$).

Discover: Does the light press score massively higher structural work than the hard push? And does the counterfactual subtraction correctly assign zero work to the push against the anchored wall?

EXPERIMENT II.3

The Three Confusables

Setup: Build three scenes that each highlight one of the three concepts. Scene A: a compressed spring clamped in place (stored potential). Scene B: a block on a rough surface resisting a push (internal reserve — friction/grip). Scene C: an empty lever arm waiting for weight (potential force — open slot). Now build a model that merges all three into a single `potential` field, and a model that keeps them as three distinct types.

Observe: (1) In the merged model, does releasing the spring's clamp look the same as exceeding the block's friction? (2) Does the merged model predict that the empty lever arm will "release" something? (3) In the three-type model, do predictions stay correct across all three scenes?

Compare: Merged single-field model versus Gerundelle's three-type distinction.

Discover: At what point does the merged model make its first wrong prediction — and is it exactly the case where one concept is mistaken for another?

EXPERIMENT II.4

The Curiosity Quadrant

Setup: Place an agent in a room with four objects, one per quadrant of the Physical Invitation × Epistemic Gap matrix. Object A: an open slot the agent understands (a familiar lever). Object B: an open slot the agent doesn't understand (an unfamiliar device with a visible input). Object C: no open slot, understood (an anchored wall). Object D: no open slot, not understood (a sealed box with no visible input). Track the agent's attention and exploration over 30 ticks.

Observe: (1) Which object does the agent explore most? (2) Does it spend time on object C (understood, no slot) or object D (no slot, not understood)? (3) Does object B — the curious quadrant — dominate attention?

Compare: Curiosity driven by invitation × epistemic gap versus uniform random exploration.

Discover: Does the formal curiosity signal (open slot + unknown outcome) match intuitive curiosity? And what happens to the agent's interest in object B after it fills the slot and observes the result — does curiosity collapse?

EXPERIMENT II.5

The Boulder That Was Already Falling

Setup: A boulder is already sliding toward a wall (its gerund is *sliding*, triggered by a slope two slices ago). The agent pushes the boulder in the same direction it's already moving. The boulder hits the wall and the wall shatters.

Observe: (1) What does raw structural work assign to the agent's push? (2) What does *counterfactual* structural work assign — after subtracting the crash that would have happened without the push? (3) If the push changed the timing by one slice (the wall shatters on slice 3 instead of slice 4), does counterfactual work correctly capture only the timing difference?

Compare: Raw work (everything that happened after the push) versus counterfactual work (only what changed because of the push).

Discover: Can the agent learn to distinguish "I caused this" from "this was going to happen anyway and I pushed in the same direction"? And what discount rate makes the one-slice timing difference visible without inflating it?

Key Terms

Affordance	What an object offers for doing — in Gerundelle, formalized as an open force slot in the gerund AST rather than a hand-coded semantic tag.
Open force slot	A position in the force structure where a force could be applied but currently isn't — the structural definition of a potential force.
Potential force	An open input slot on a gerund node that invites a force to be applied — distinct from stored potential and internal reserves.
Stored potential	Accumulated energy ready to be released (a compressed spring, a raised weight, a charged capacitor) — offensive, not defensive.
Internal reserve	A quantity that resists change (cohesion, tensile strength, grip, friction, resolve) and triggers a phase change when depleted.
Physical invitation	Whether an open force slot exists in the current configuration — one axis of the curiosity matrix.
Epistemic gap	Whether the agent understands what would happen if a force slot were filled — the other axis of the curiosity matrix.
Curiosity	The pull to act when an open slot exists and the outcome is not understood — formally, invitation × epistemic gap.

Tool use	Reconfiguring objects so that a force that was structurally uninvited becomes invited — creating a new open slot.
Structural work	The count of gerunds flipped, reserves changed, phase states crossed, and invitations opened or closed over bounded future slices.
Counterfactual work	Structural work minus the changes that would have happened without the agent's action — Gerundelle's replacement for $W = F \cdot d$.
Discount	Later time slices count less than earlier ones in the work computation, keeping the measure bounded and favoring immediate causal consequences.

PART II

II

The Chemistry of Thought Molecules

The Alcove

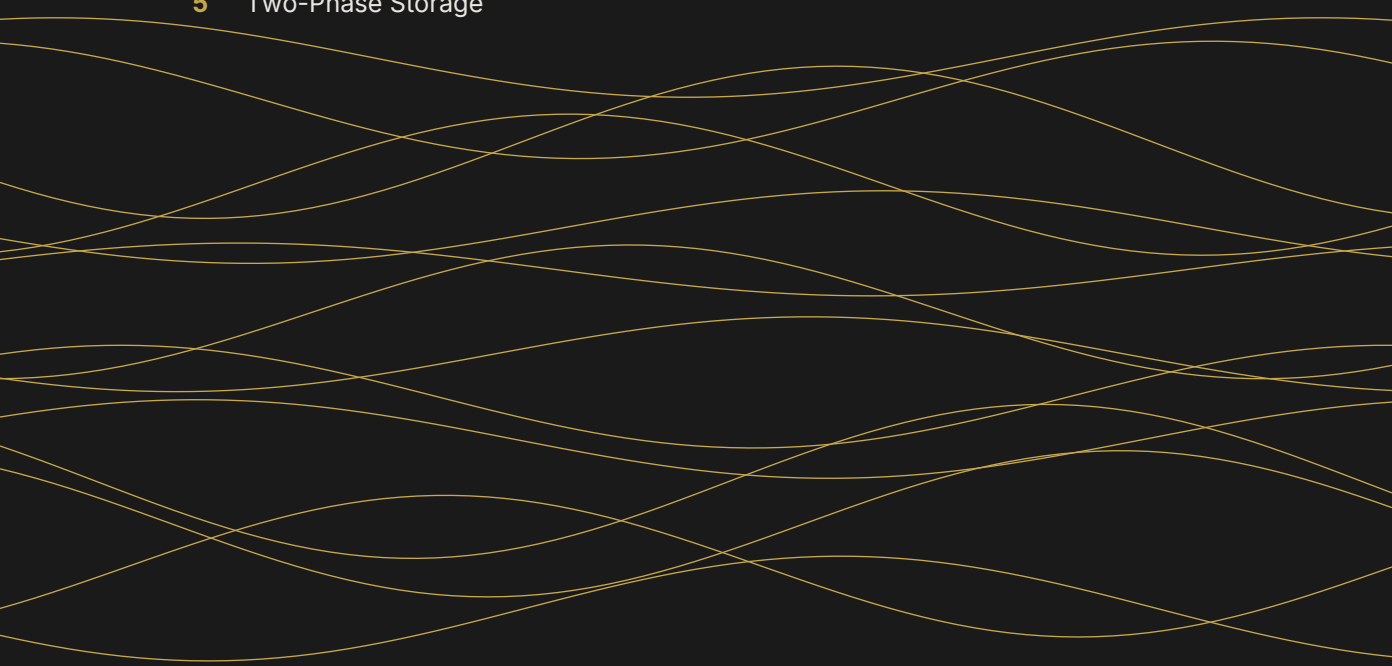
3 Thought Molecule Architecture

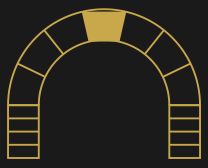
The Frozen Room and the Room of Returns

4 Learning & Surprise

The Gallery of Sealed Cases

5 Two-Phase Storage





THE MUSEUM

The Alcove

A dim hall, much larger than the others. The ceiling is lost in shadow. The air smells of cold stone.

Fragments are everywhere. They drift at waist height, slow as dust motes, each one a small thing of rods and joints — the same pale material as the structure in the Gallery of Openings, but smaller, no bigger than a hand. Some are simple: two rods and a joint. Some are deep: rods bonded to rods bonded to rods, nested several layers in. I cannot tell which are the simplest. There does not seem to be a simplest. Every fragment I look at closely turns out to be made of smaller fragments bonded together, and those of smaller ones still.

Where two fragments touch, a joint forms between them. The joint is not glue. It is a small piece on its own, with a site on each end shaped to fit each fragment. Some joints hold two fragments tightly. Some hold loosely, as if they might let go. Some fragments have joints attached to only one end, the other end open and shaped for something that is not here.

Along one wall, a lit alcove. It is small — room for perhaps seven pieces. The light inside it is warm and even. Everything else in the hall is dim. I think the fragment that was just displaced has been beaten — pushed out by something stronger.

The fragments lean toward it. All of them, faintly, the way iron filings lean toward a magnet. The lean is not equal. Some fragments lean hard enough to drift. Most lean and stay where they are.

I watch. A cluster of bonded fragments reaches the alcove and settles into the light. A simpler fragment, leaning hard, presses in behind it and displaces something that was there. The displaced piece drifts back into the dark, smaller than when it

arrived. It seems to have lost something — not a part, but a brightness. It fades as it goes.

Seven pieces fit. No more. The rest lean from the dark.

Above the first alcove, set into the wall, a second. Smaller. Room for five. The fragments that won places below are not guaranteed places above. I watch one of the seven rise toward the second alcove, and as it rises the joint connecting it to its partner below lets go and reforms with something else — a different fragment, on the other side. The same piece, a different bond, because at this height it leans toward a different neighbor.

Above the second, a third alcove. Room for four. The pieces that reach it are held in a light so bright I can see every joint, every open site, every rod still settling. Below them, the five are dimmer. Below them, the seven dimmer still. And in the hall, the thousands lean.

I find a placard behind a column, in the dark, as though it were leaning too.

THE HALL OF FRAGMENTS
Rods, joints, and binding sites. Ambient.

There is no smallest fragment. Complexity is bonding depth: how many joints connect how many pieces. A half-formed fragment is not broken. It has open sites that have not found their partners.

The alcoves have limited room. Fragments compete for places in them. The competition is fierce and the losers drift back into the dark, where they fade unless they are surprising enough to keep their brightness.

A joint connecting two fragments serves two roles at once. In the dark it holds them together silently. In the alcove it is lit, and it becomes a piece in its own right — something that can be examined, questioned, revised. The same joint. Two roles. Which role it plays depends on whether it reaches the light.

The alcoves narrow: seven, five, four. A piece that won below may lose above. At each level the competition begins again from scratch, and the joints may resolve to different partners, because the history at each level is its own.

I go back to the alcoves and watch the second level. A fragment that was bonded to its left neighbor below is bonded to its right neighbor above. The same piece. A different partner. I cannot tell which bond is the real one. Both are, at their level.

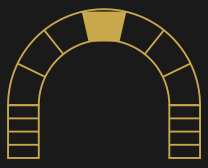
But the fragment I thought was beaten — the one that was displaced earlier — has drifted back. It is not damaged. It is dimmer, smaller than before. It was not beaten. It was outgrown.

Something catches my eye near the base of the lowest alcove. Beside some of the bonds — the ones in the light — there are small stacks of paper slips. Not many. A few slips beside a joint that just formed. A thicker stack beside one that has been in the light a long time. I pick one up. It is blank on one side. On the other, a single word I cannot read in this light.

There is no placard for the slips.

In the far corner of the hall, the cloth on the pole hangs in the dark. It lifts once. The fragments nearest it drift sideways for a moment, then settle.

I leave through a door I had not noticed. My eyes ache from the dim.



SECTION

3

Thought Molecule Architecture

Bonds, Superposition, and the Tournament for Consciousness

IN THIS SECTION

- Seeing why thoughts don't need an atomic bottom layer — complexity is just bonding depth
- Discovering how the same molecule can be a silent structural edge or a conscious thought
- Understanding argmax superposition and why it avoids duplicating molecules in memory
- Learning why consciousness is a tournament, not an observer or a dice roll
- Following the same thought as it resolves to different partners at different levels of awareness
- Seeing how partner search scales from impossible to practical

WHAT IF Half-formed thoughts are simple unbonded fragments

Think about the thought *"it might rain."* It feels like a single thing. But it's actually built from pieces: the concept of weather, the concept of rain as a type of weather, a sense of uncertainty, a time reference to the near future. None of these pieces is the thought. The thought is what happens when the pieces are **bonded** — held together by recognition that they belong in a specific relationship.

Now think about a vaguer sensation: a half-formed feeling that you've forgotten something. No specific content. No structure. Just a fragment floating in awareness, waiting to connect to something that would complete it. That fragment is simpler than the rain thought — not because it's made of smaller atoms, but because it has fewer bonds.

What if there is no bottom layer to thought? What if complexity is nothing more than bonding depth — how many recognitions connect how many fragments into a composite — and a half-formed thought is simply a molecule with open binding sites that haven't found their partners yet?

WHY Thoughts need no atomic bottom layer — complexity is bonding depth

No Atoms, Only Depth

Fixed-dimensional vector representations — word2vec, transformer embeddings — give every concept the same-sized vector regardless of complexity. "It might rain" and "the interdependent web of economic, social, and environmental factors that make climate policy intractable" both get 768 dimensions. The representation can't structurally distinguish a simple clause from a deep composite. Complexity is invisible; you'd have to decode the vector to discover it.

Gerundelle refuses this flattening. A thought clause — "it might rain" — exists because "weather occurs" and "rain is a type of weather" are bonded. But "weather occurs" is itself a bonded composite. And so on. There's no layer where you hit bedrock, because the bedrock *is* the bonding.

This means complexity isn't a property of the content but of the structure. A simple thought is a molecule with few bonds. A complex thought is a molecule with many. A half-formed thought is a molecule with open binding sites — places where bonds *could* form but haven't. The incomplete melody stuck in your head is a molecule with an open site shaped like the missing final note. That open site is doing real structural work: it's pulling for completion.

> **REMEMBER**
> In Gerundelle, thought complexity = bonding depth. There are no primitive thought-atoms. A half-formed thought isn't missing a piece from some lower level — it has **open binding sites** that pull for completion. The architecture goes all the way down without ever hitting a foundation layer.

WHAT IF The recognition connecting two thoughts exists in superposition between being a silent edge and a conscious thought

You know that your friend only takes calls on sunny days. Most of the time, this knowledge sits quietly in the background, doing its job without you noticing it. You call on sunny days and not on rainy days, and if someone asked why, you could articulate the rule — but you didn't *think* the rule before calling. It acted as a silent structural connection between "friend" and "calling conditions."

Now imagine someone asks you: *"Why do you always call in the afternoon?"* Suddenly that quiet connection rises into awareness. It stops being a background edge and becomes a **thought in its own right** — something you examine, question, maybe revise. The same piece of knowledge, without changing its content, shifted from edge to object of attention.

What if every recognition that connects two thoughts exists in **superposition** between these two roles — silent edge and conscious thought — resolving into one or the other only when observed?

WHY A bond is a recognition molecule with two open binding sites, and its role resolves on observation

Bonds Are Recognition Molecules

Knowledge graphs like ConceptNet and Wikidata store every relationship as a permanent edge — never an object of attention in its own right. "She only takes calls on sunny days" is a fixed link between "calling" and "weather." The system can traverse the edge but never examine, question, or revise it. When someone asks "why do you always call in the afternoon?" there's no mechanism to promote the edge into a thought. The graph can't question its own beliefs because beliefs aren't the kind of thing that can compete for attention.

In Gerundelle, a **bond** isn't a static link between two nodes. It's a **recognition molecule** — "she only takes calls on sunny days" — with two open binding sites: one for the concept of calling, one for the concept of weather conditions. When this molecule binds both sites, it holds a composite thought together. When it's examined directly, it becomes an object of consciousness: a rule you're actively thinking about.

The critical move is that the bond doesn't have to choose its role in advance. Until the system resolves it — through the tournament described in the next section — the recognition exists in both roles simultaneously. It's structurally connecting the thoughts *and* it's a candidate for conscious examination. The resolution happens at observation time, deterministically, based on what else is competing for attention. No central observer decides. No coin is flipped.

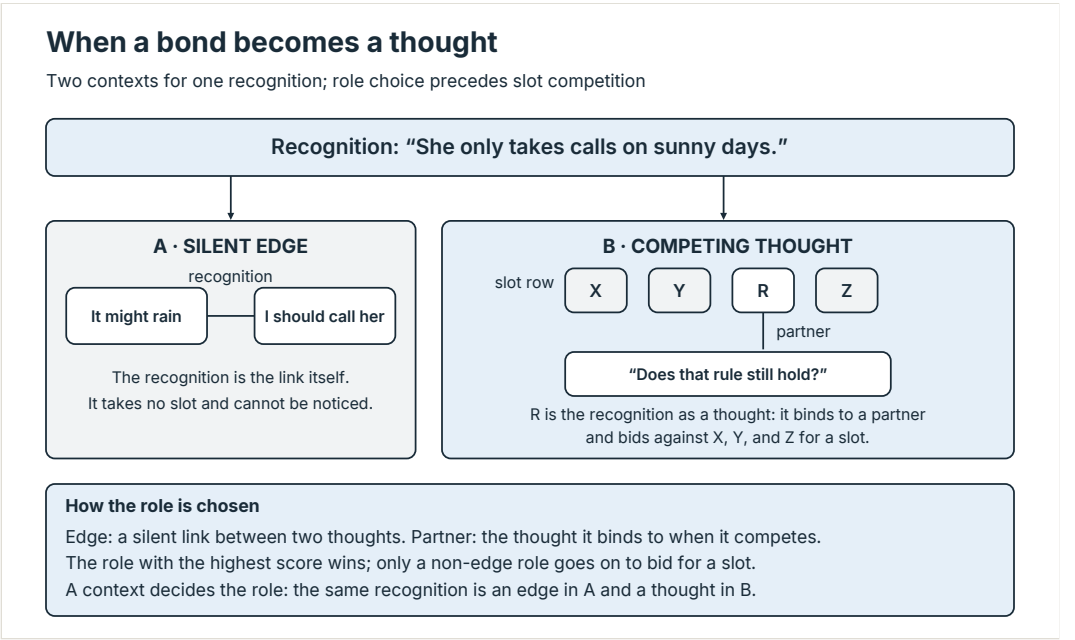


Figure 6 **When a bond becomes a thought.** One recognition can be a silent edge between two thoughts or a candidate thought in a slot row, depending on context. Superposition here means candidate roles, not quantum states.

TECHNICAL STUFF

Edge/molecule superposition. Every recognition molecule has two binding sites and a dual role. As an **edge**, it silently connects two partner molecules into a composite, shaping behavior without occupying a working-memory slot. As a **molecule**, it competes for slots in its own right — becoming an object of examination, questioning, or revision. The same molecule can act as an edge at one level of the tournament and a conscious thought at another. Resolution is deterministic: argmax over competing bids at each level.

WHAT IF A single molecule participates in many candidate bindings simultaneously

You're at a party. You recognize the host, your colleague, and an old friend from school. Each recognition suggests a different conversation, a different version of yourself, a different set of things you might say. Before you commit to walking toward anyone, all three connections exist simultaneously — you're in superposition across multiple possible social configurations.

Now imagine that every thought in your mind works this way. The thought "it might rain" is simultaneously a candidate partner for "I should bring an umbrella," "the pic-

nic might be ruined," and "the garden needs water." It hasn't committed to any of these bindings. It participates in all of them as possibilities, and only when the tournament resolves does one binding win.

What if a single molecule participates in many candidate partner bindings simultaneously, resolving deterministically each time it is observed, without ever being duplicated in memory?

WHY Superposition avoids duplicating molecules in memory and resolves deterministically

Argmax Superposition

Classic AI's slot-filling and frame systems handle multiple bindings by copying: when a concept participates in multiple frames, it's duplicated into each. A concept in 50 candidate bindings requires 50 copies, each accumulating separate experience histories. The system loses track of the fact that these are all the same concept seen from different angles. Identity fragments across the copies. Each copy accumulates its own history, and the system loses track of the fact that these are all the same thought viewed from different angles.

Gerundelle's solution is **argmax superposition**. A molecule doesn't commit to a partner binding until it's observed. It participates in all candidate bindings simultaneously — each scored by the molecule's experience records, the partner's bid, and the structural fit of their binding sites. At observation time, the binding with the highest score wins. No copies are made. No randomness is needed. The same molecule, observed in a different context or at a different level of the tournament, may resolve to a different partner — deterministically, based on different experience records at that level.



TIP

Think of it like a word with multiple meanings. The word "bank" doesn't get copied into memory once for "river bank" and once for "financial bank." You store one word and resolve the meaning from context each time you encounter it. Argmax superposition works the same way: **one molecule, many candidate bindings, deterministic resolution at observation time**.

WHAT IF There is no central observer or random sampling — only fierce tournament competition

Who decides which thoughts you're aware of right now? The intuitive answer is *you* — some inner observer watching the stream of consciousness and choosing what to attend to. But that just pushes the question back: who watches the watcher?

The alternative used in most AI systems is random sampling: roll dice, pick a thought proportional to some probability distribution, hope that important thoughts come up often enough. This works statistically but has no mechanism for urgency. A thought about an approaching car and a thought about lunch plans get the same kind of draw from the same kind of urn.

What if there is no observer and no dice? What if "being conscious of a thought" simply means that thought **won a slot** in a fierce, hierarchical tournament against every other thought competing for the same limited working memory?

WHY 'Being observed' is winning a limited working-memory slot in a multi-level bottleneck

The Hierarchical Tournament

Transformer self-attention lets all tokens attend to all other tokens with soft weights — there's no hard selection, no losers, no limited capacity. As context grows, every token influences every other, and nothing is ever truly "out of mind." A fire alarm and a grocery list get the same kind of soft-weighted treatment. There's no mechanism for an urgent thought to monopolize processing. The softness is the problem: everything matters a little and nothing matters completely.

Gerundelle's tournament has three levels, each with fewer slots than the one below: $K_0 > K_1 > K_{top}$, starting at 7 / 5 / 4. At level 0, all molecules compete. The top K_0 winners advance. At level 1, those winners compete again — re-resolving their bindings from scratch using level 1's own experience records. The top K_1 advance to the top level, where the final K_{top} winners become the contents of consciousness: the thoughts the system is actively aware of.

The bottleneck does real work. Level 0 filters thousands of molecules down to a handful. Level 1 reshuffles bindings under its own experience history. The top level is where conscious examination, questioning, and learning happen. A molecule that wins the top gets its experience records updated, its bonds reinforced or weakened, and its open sites addressed. A molecule that loses decays — more slowly if it's surprising, more quickly if it's expected. It was not beaten; it was outgrown.

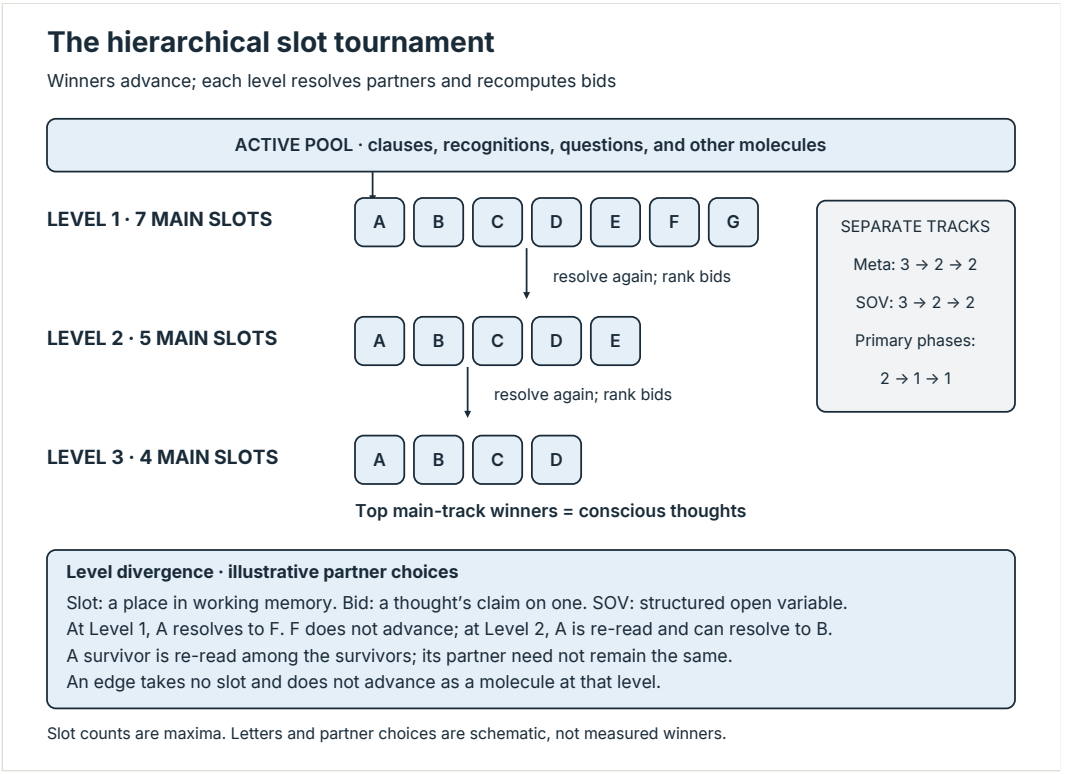


Figure 7 **The hierarchical slot tournament.** The main track narrows 7 → 5 → 4. Each level recomputes bindings and bids among survivors; dedicated tracks have additional budgets.

WARNING

The three-level tournament is **not** a claim about how human consciousness works. It's a computational architecture that solves a specific engineering problem: how do you select which thoughts get processed when you have far more candidates than capacity? The levels are a design, not a discovery. Whether biological brains use something analogous is a separate empirical question.

LINEAGE

Global Workspace Theory and Dehaene's ignition. Bernard Baars (1988) proposed *Global Workspace Theory* (GWT): consciousness is a limited-capacity broadcast — many specialized processors compete for access to a shared workspace, and the winner gets broadcast to all. **Stanislas Dehaene** added the neuroscience: cortical ignition, all-or-nothing access, the sharp threshold between conscious and unconscious processing. Gerundelle's hierarchical tournament — many molecules compete, few win, winners get processed and broadcast their results — is convergent architecture. Neuroscience arrived at limited-capacity competitive broadcast for its reasons; Gerundelle arrives at it for engineering reasons (the lock-in problem, the scaling problem). The convergence is evidence the *design region* is good, not a claim of equivalence. **George Miller** (1956) and **Nelson Cowan** (2001) anchor the slot budgets empirically: Miller's "magical number seven" and Cowan's refined estimate of 4±1 items in working memory are why Gerundelle's levels start at 7/4/1 rather than arbitrary numbers.

WHAT IF Each level maintains its own experience record, so the same molecule resolves differently at each level

Imagine you have a gut feeling about someone. At an instinctive level, something about them makes you uneasy — they remind you of someone who was dishonest. But consciously, you've worked through the evidence and decided they're trustworthy. The *same* recognition molecule — your assessment of this person — resolves to different partners at different levels of processing. The gut pairs it with suspicion. Consciousness pairs it with trust.

What if this isn't a bug but a feature? What if each level of the tournament maintains its own experience records, so that a molecule re-entering competition at a higher level genuinely resolves to a different binding — not because of noise, but because each level has *learned different things* from its own history of winners?

WHY Each level maintains its own experience records, making level divergence essential to the architecture

Re-superposition and Level Divergence

Single-level attention — flat processing with no hierarchy — has one resolution for every signal. Whichever binding is strongest wins everywhere. There's no structural place for "my gut prefers X while my deliberate judgment prefers Y." A gut feeling that contradicts a reasoned conclusion simply gets averaged away or overridden. The unconscious/conscious split can't be represented because there's only one level to resolve in.

Gerundelle's levels make this split natural. When a molecule wins at level 0, it resolves to a specific partner binding based on level 0's experience. Then it goes back into superposition. At level 1, it competes again — but now it uses level 1's experience records, which have a different history because different molecules won at that level in the past. The winning binding can be completely different.

This is called **re-superposition**, and it's not an approximation or a simplification. It's the core mechanism that allows unconscious processing and conscious processing to diverge. The unconscious level might pair "calling her" with "habit" (they've co-won many times at that level). The conscious level might pair "calling her" with "she said not to call in the rain" (that recognition won at the top level and changed its experience record). One molecule, two levels, two different resolutions — each internally consistent with its level's history.



REMEMBER

Re-superposition means **winning at one level doesn't lock in a binding for the next level**. Each level re-resolves from its own experience. This is how the same thought can "want" different partners at different levels of awareness — and it's testable: seed divergent experience records and check whether the molecule actually resolves differently.

WHAT IF Partner search scales from $O(N^2)$ by restricting candidates to history, open-site fits, and the most active

With ten thoughts in the pool, every molecule can consider every other molecule as a potential partner. That's 45 possible pairings — manageable. With a hundred thoughts, it's 4,950. With a thousand, it's nearly half a million. With ten thousand, it's fifty million. Exhaustive partner search hits a wall fast.

But most of those pairings are nonsensical. A thought about lunch doesn't seriously compete to bind with a thought about differential equations. Their binding sites don't match, they've never co-occurred, and neither is among the currently most active or most surprising molecules. Searching through all fifty million pairs to discover this is waste.

What if partner search restricts itself to a small, smart set: molecules in the candidate's own history, molecules whose open sites structurally fit, molecules already bonded to it, and the *A* most active and longest-waiting molecules in the pool?

WHY Thousands of molecules require tag-indexed pruning to keep the tournament tractable

Indexed Search at Scale

Gerundelle's solution sets $A = 8$: for each molecule, the candidate partner list includes molecules from its own co-win history, molecules whose binding-site tags match its open sites, molecules it's already bonded to, and at most 8 of the highest-activation and longest-waiting molecules in the pool. This drops the search from $O(N^2)$ to roughly $O(N \cdot A)$, where A is small and constant.

The pruning sounds aggressive — you're throwing away 99% of candidate partners. The engineering question is whether that 99% contained any binding that would have won. The test is straightforward: run the full test suite with $A = None$ (exhaustive search) and $A = 8$ (restricted), and check whether any test gives a different result. If not, the pruning preserved everything that mattered. If so, you've found the edge case where the restricted set is too small — and the failing test tells you exactly which kind of molecule was missed.



TECHNICAL STUFF

Tag-indexed bond ends. Each molecule's binding sites carry structural tags (the type of slot, the kind of force or relation it expects). A tag index maps each tag to the set of molecules with a matching open site. Instead of comparing every molecule against every other, the system looks up matching tags and restricts the candidate set to structural fits. Combined with history lookups and the activation cap, this keeps per-tick computation linear in the number of molecules rather than quadratic.

Suggested Experiments

Each experiment isolates one claim about the thought-molecule architecture. Run the main condition, then the ablation, and see what changes.

EXPERIMENT III.1

The Single Slot

Setup: Run a mind with $K_{top} = 1$ (one conscious slot) versus $K_{top} = 2$ (two conscious slots). Present two thoughts that want to bond — they have complementary open binding sites and positive experience records with each other.

Observe: (1) At $K = 1$, can the two thoughts ever co-win? (2) Do they accumulate co-win experience at level 0 but not at the top level? (3) At $K = 2$, how many ticks does it take for them to co-win at the top and begin building a bond?

Compare: Single-slot versus multi-slot conscious level.

Discover: Can learning between two thoughts happen if only one can ever be "in mind" at a time? Does the top level need at least two slots for pairs to form?

EXPERIMENT III.2

The Divergent Mind

Setup: Seed level 1's experience records so that molecule *M* strongly prefers partner *X* (good = 10, neutral = 2). Seed level 2's records so *M* strongly prefers partner *Y* (good = 10, neutral = 2). Both *X* and *Y* are in the pool. Run one tick.

Observe: (1) What does *M* bind to at level 1? (2) What does it bind to at level 2? (3) Are both resolutions consistent with their level's experience record?

Compare: Divergent records across levels versus identical records at all levels.

Discover: Can one thought genuinely "want" different partners at different levels of awareness? And if the conscious level overrides the unconscious binding, does the unconscious record still update based on *its* resolution?

EXPERIMENT III.3

The Pruning Test

Setup: Run the full test suite (T1–T28) twice. First with $A = None$ (exhaustive partner search — every molecule considers every other). Then with $A = 8$ (restricted to history, site fits, bonds, and the 8 most active). Time a single tick at 400 molecules under both conditions.

Observe: (1) Do all 28 tests produce identical results? (2) What is the speedup factor? (3) If any test diverges, which specific pairing was missed by the restricted set?

Compare: $O(N^2)$ exhaustive search versus $O(N \cdot A)$ indexed search.

EXPERIMENT III.3

The Pruning Test (continued)

Discover: What is lost by pruning 99% of candidate partners? Is the answer "nothing" for all 28 tests, or are there edge cases where a structurally unexpected pairing matters?

EXPERIMENT III.4

The Edge That Becomes a Thought

Setup: Place a recognition molecule *R* ("she only takes calls on sunny days") in the pool alongside its two binding partners: "calling her" and "weather conditions." For 20 ticks, let *R* act as a silent edge binding the composite. Then introduce an open question: "why do I always call in the afternoon?"

Observe: (1) Before the question arrives, does *R* ever win a conscious slot in its own right? (2) After the question arrives, does *R* compete as a molecule rather than an edge? (3) How many ticks does it take for *R* to transition from silent edge to examined thought?

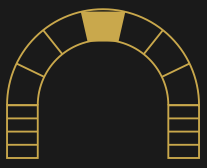
Compare: Pool with the open question versus pool without it (the question never arrives).

Discover: What triggers a recognition to shift from structural background to conscious foreground? Is the open question sufficient, or does *R* also need to have high enough activation from some other source?

Key Terms

Thought molecule	A clause or composite in the cognitive pool — its complexity is its bonding depth, not some hidden atomic layer.
Binding site	An open slot on a molecule where a specific type of partner could attach — the structural shape of incompleteness.
Bond	A recognition molecule with two binding sites that can act as a silent structural edge or compete as a conscious thought in its own right.
Edge/molecule superposition	A bond's dual role: silently connecting partners (edge) or being examined directly (molecule), resolved at observation time.
Argmax superposition	A molecule participates in all candidate bindings simultaneously, resolving to the highest-scoring partner deterministically at observation time — no copies needed.

Tournament	The three-level bottleneck ($K_0 > K_1 > K_top$) through which all molecules compete for working-memory slots.
Working-memory slot	A position at a given tournament level — winning one is what 'being observed' means in Gerundelle.
Re-superposition	After winning at one level, a molecule goes back into superposition and re-resolves its binding at the next level using that level's experience records.
Level divergence	The same molecule resolving to different partner bindings at different tournament levels because each level has its own experience history.
Experience record	Per-level Dirichlet counts (good, neutral, bad) tracking what happened when a molecule won at that level in the past.
Candidate restriction	Pruning partner search to history, tag-matched open sites, existing bonds, and the A most active molecules — reducing $O(N^2)$ to $O(N \cdot A)$.
Tag index	A lookup table mapping binding-site tags to molecules with matching open sites, enabling fast structural partner search.



THE MUSEUM

The Frozen Room and the Room of Returns

A corridor with two doors, one on each side. The left door is cold to the touch. The right door is warm.

I go left.

The room is still. Completely still. Fragments hang in the air the way they did in the Alcove, but none of them is moving. They are locked in place: a cluster of three near the alcove, bonded tight, lit from within, and everything else frozen in the dark at whatever distance it had reached. One fragment is near the wall, leaning hard toward the alcove, its joints straining. It will never arrive. The dust on the floor has settled in the pattern the fragments made when they stopped, and no one has swept it.

The alcove holds only the cluster. The same three fragments, in the same three places, since the room stopped. I can see that they are not the most interesting things in the room. A fragment near the back wall has open sites on all sides — deep, complex, clearly unfinished. It never reached the light.

A placard, framed and mounted like a memorial:

THE ROOM THAT FROZE
Fragments, joints, dust. Preserved.

This room is kept as it was the day it froze. The three fragments nearest the alcove arrived first. Their mutual attraction grew every time they were lit together. Nothing bad happened, so their bond strengthened. By the time more interesting fragments drifted near, the three had accumulated enough weight that nothing could displace them.

The room's own mechanism of learning — rewarding what had been seen together — was the thing that killed it. The first accident became permanent. Nothing in the design could say: I have learned all this pairing has to teach me. Nothing could say: that new fragment might be more worth knowing.

This exhibit is not a reconstruction. The room froze on its first day, and the museum chose to keep it so.

I stand in the doorway a moment longer. The straining fragment near the wall has been leaning since the room stopped. I feel its pull from here. No one is coming.

◆ ◆ ◆

I cross the corridor to the warm door.

This room is alive. Fragments drift and compete as they did in the Alcove, but something is different. The alcove is brighter, and the fragments near it are not the same ones from moment to moment. I watch a bonded pair rise into the light, hold for several seconds, then drift out and let something else in. The turnover is steady. Nothing locks.

I look for what changed. In the Alcove, fragments that won stayed until something stronger pushed them out. Here, they seem to drift out on their own. I watch one: it enters the light, holds, and each second it holds, its brightness dims a little. Not much. But the fragment beside it — new, unknown, with an open site pulling hard — gets slightly brighter each second by comparison. When the newcomer's lean overtakes the incumbent's fading brightness, the incumbent gives way. Not pushed out. Outgrown.

Near the center of the room, one fragment does not fade. It has been drifting in and out of the alcove since I arrived, and each time it leaves it comes back. I look at it closely. It is unfinished: a joint on its left side is attached to nothing. The open site. I hold my hand near it and feel the same pull I felt in the Gallery of Openings, but smaller, steadier, constant. The site does not care that the fragment has been in the light a dozen times. It cares that the site is still empty.

Everything else in the room loses brightness when it is not surprising. This fragment keeps its brightness because it is incomplete.

I hum a few bars of something without thinking, and stop. The fragment seems to lean when I do, but that may be my eyes.

A placard, low on the wall, easy to miss:

THE ROOM OF RETURNS
Fragments, joints, one open site. Active.

This room was rebuilt after the one across the corridor froze. Every fragment here is drawn toward its neighbors by how much it would learn from them, not by how often it has met them.

A fragment that has been fully explored offers nothing more. Its pull fades as the room learns what it has to teach. A fragment that has never been examined offers everything. Its pull is highest before its first meeting, and drops as its mystery is used up.

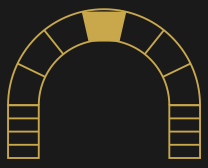
One fragment in this room does not follow that rule. It has an open site, and the site adds a constant pull that does not fade. The fragment returns to the light not because it is surprising but because it is unfinished. It will return until the site is filled.

Visitors sometimes hum near it.

I read it and look back at the fragment. It has drifted into the alcove again. The fragment next to it — fully bonded, no open sites — is already fading.

Before I leave I go back across the corridor and stand in the doorway of the frozen room one more time. The three fragments in the alcove. The straining fragment by the wall. The placard says the room's own learning was the thing that killed it. Across the corridor, a room that learned a different way to learn.

In the warm room's corner, the cloth lifts, swings twice — the first time I have seen it swing twice — and hangs still.



SECTION

4

Learning & Surprise

Why the First Run Froze and How Free Energy Fixed It

IN THIS SECTION

- Watching Gerundelle's first prototype lock in on tick 1 and never recover
- Seeing how Friston's Free Energy Principle replaces runaway attraction with expected information gain
- Understanding earworms as open binding sites that pull for completion
- Learning why forward-looking surprise reduction replaced backward-looking precision credit

WHAT IF The first POC locked in whatever random thoughts won on tick 1

Imagine a classroom where the teacher rewards every student who happens to be standing near the front of the room. The first few students who wander to the front get rewarded. Rewarded students stay at the front. New students see the front is crowded and don't try. Within minutes, the students who arrived first have permanent seats, and everyone else has given up. The classroom has frozen.

This is exactly what happened in Gerundelle's first proof-of-concept run. Attraction between thought molecules grew by a fixed amount on every neutral co-win — two thoughts land in the same working-memory slots, nothing bad happens, their mutual attraction increases. Novelty was computed as a simple ratio: $n / (1 + count)$, which drops toward zero the moment a pairing has been seen a few times. The result was catastrophic. Whatever random thoughts happened to win on tick 1 accumulated enough mutual attraction to monopolize the slots forever. New thoughts couldn't break in. The mind froze on its first accident.

What if the very mechanism designed to help thoughts find good partners — attraction growing with co-occurrence — is the thing that kills exploration?

WHY Co-win attraction with novelty as n/(1+count) created runaway positive feedback that froze the system

Runaway Feedback

The failure had a precise structure. On tick 1, thoughts A and B happened to co-win. Their mutual attraction increased by +1. On tick 2, that extra attraction made them slightly more likely to co-win again. They did. Another +1. By tick 5, their mutual attraction was so high that no other pairing could compete. Meanwhile, novelty — the system's only mechanism for exploring new pairings — was computed as $n / (1 + count)$, which rewarded never-seen pairs but plummeted to near-zero after just a handful of encounters.

The combination was lethal: attraction grew without bound on every meeting, while the only counterweight decayed to nothing after a few meetings. The system had no way to say "I've learned everything this pairing has to teach me" or "this new pairing might be more informative." It just kept reinforcing whatever it saw first. This wasn't a bug in the parameters — it was a structural flaw in the learning rule.



WARNING

This failure is not hypothetical or reconstructed after the fact. It was the traced result of the first real implementation run. The lock-in happened immediately, was reproducible, and motivated the complete redesign of the attraction and novelty mechanisms described in the next section. Engineering failures like these are the most valuable parts of the development story — they show which assumptions break and why.

WHAT IF Thoughts bond by how much future surprise they promise to reduce

Forget attraction for a moment. Think about why you pay attention to anything at all. You don't stare at the wall because it's familiar. You don't stare at pure static because it's novel. You attend to things that are *informative* — things that, if you paid attention to them, would reduce your uncertainty about something you care about. The wall teaches nothing. The static teaches nothing. The half-understood pattern in the data teaches everything.

What if thoughts bond and compete not by how often they've co-occurred, but by how much future surprise they promise to reduce? A thought that has been fully explored — whose outcomes are completely predictable — offers zero surprise reduction. A thought

that has never been examined offers maximum surprise reduction. The pull is toward the informative middle: enough history to form an expectation, but enough uncertainty that attending would still teach something.

WHY Friston's Free Energy Principle and expected information gain replace runaway attraction with self-limiting learning

Free Energy Fixes Lock-In

The fix drew on Karl Friston's **Free Energy Principle**: an organism should minimize surprise by updating its model of the world. Gerundelle translated this into four concrete mechanisms that replaced the broken co-win attraction.

First, **outcome beliefs** became Dirichlet distributions. Instead of a single attraction score that grows without bound, each molecule maintains counts of good, neutral, and bad outcomes at each tournament level. The expected outcome μ_L updates by prediction error: *(μ - outcome) / (total + 1)*. Crucially, a neutral co-win pulls μ toward zero rather than building runaway attraction. The belief self-limits.

Second, **novelty** became expected information gain — the mutual information of the Dirichlet prior. A never-formed pair has the highest epistemic pull because every observation would maximally update the distribution. But as observations accumulate, the prior tightens and the information gain drops. Novelty decays naturally as the system learns, without an arbitrary *1/(1 + count)* schedule.

Third, the **bid** each molecule submits to the tournament was revised to include forward-looking surprise reduction: how surprising the molecule's arrival was, plus how much attending to it would still teach. Fourth, **open-site tension** adds a constant, non-decaying pull to any molecule with an unfilled binding site — the formal mechanism for earworms.



TECHNICAL STUFF

The bid equation. A molecule's bid combines: (1) **outcome belief μ_L** from Dirichlet counts at that level, (2) **novelty N_L** as expected information gain from the prior, (3) **forward-looking surprise reduction** (arrival surprise + remaining teachability), and (4) **open-site tension T**, a constant added until the site is filled. The first two components self-limit as the system learns. The third redirects attention toward what's still informative. The fourth keeps incomplete thoughts in competition until they find their partner.

LINEAGE

Friston's Free Energy Principle. Karl Friston's Free Energy Principle (FEP) proposes that all adaptive systems minimize variational free energy — a bound on surprise. Perception updates the model; action changes the world to match the model's predictions. The framework is mathematically elegant and sometimes treated as either a theory of everything or an unfalsifiable tautology. Gerundelle takes a pragmatic stance: FEP is presented not as ideology but as *the thing that fixed a real bug*. The traced lock-in failure is the most convincing introduction Friston has ever gotten — the first-run system froze, Dirichlet-based surprise reduction unfroze it, and the fix is traceable to specific FEP-derived mechanisms. What Gerundelle adds beyond FEP is the structural gate: surprise isn't minimized over a featureless field but over *typed relation slots* in the Understanding Register, so the system knows not just that it's surprised but *what kind of thing* it's surprised about.

REMEMBER

What surprise does, under the territorial reorganization. Surprise does not fund attention by itself. It reveals where the boundary is open: a change the agent did not cause opens an ownership slot — something moved in the agent's territory without the agent's push. Free energy stays as the tie-breaker among competing molecules and as a half-life modulator, but the *reason* surprise matters is that it marks a territorial breach, not that it reduces a variational bound. The FEP mechanisms are retained as machinery. The motive they serve is territorial.

WHAT IF Rehearsal is just repeatedly winning slots, and earworms are open sites pulling for completion

You're trying to memorize a phone number. You repeat it: 555-0142, 555-0142, 555-0142. Each repetition isn't teaching you new information — you already know the number after the first hearing. What the repetition does is keep the thought *winning slots* in your working memory, preventing it from decaying and eventually precipitating it into long-term storage.

Now think about an earworm — a melody stuck in your head. It's not winning because it's surprising or informative. It's winning because it's *incomplete*. The melody has an open site shaped like its missing resolution, and that open site adds a constant pull to the molecule's bid every single tick. The pull doesn't decay. It doesn't care that you've heard the fragment a hundred times. It cares that the fragment hasn't found its ending.

What if rehearsal is nothing more than repeated slot wins, and earworms are nothing more than open binding sites with non-decaying tension?

WHY An unfilled binding site adds a non-decaying constant to a molecule's bid until filled — the formal earworm mechanism

Open-Site Tension

Under uniform exponential decay — the default in most activation-based models — all molecules lose activation at the same rate. An incomplete melody and a complete melody of equal initial activation both fade identically. After 20 ticks, neither has any special claim on attention. The incomplete melody has no mechanism to persist; its incompleteness is invisible to the decay schedule.

Gerundelle breaks this symmetry. Most components of a molecule's bid are self-limiting — outcome beliefs converge, information gain drops, surprise decays. But open-site tension T is different. It's a constant — added to the bid on every tick, unchanged, for as long as the site remains unfilled. It doesn't grow, but it doesn't shrink either.

This creates a precise behavioral prediction. A complete thought with no open sites competes on its merits: how surprising it is, how informative, how good its outcome history is. An incomplete thought with the same merits *plus* an open site gets an extra boost on every tick. Over time, as the complete thought's surprise and novelty decay, the incomplete thought's constant tension makes it progressively more dominant. It rises in consciousness not because anything about it changed, but because everything else became less interesting while its incompleteness stayed the same.

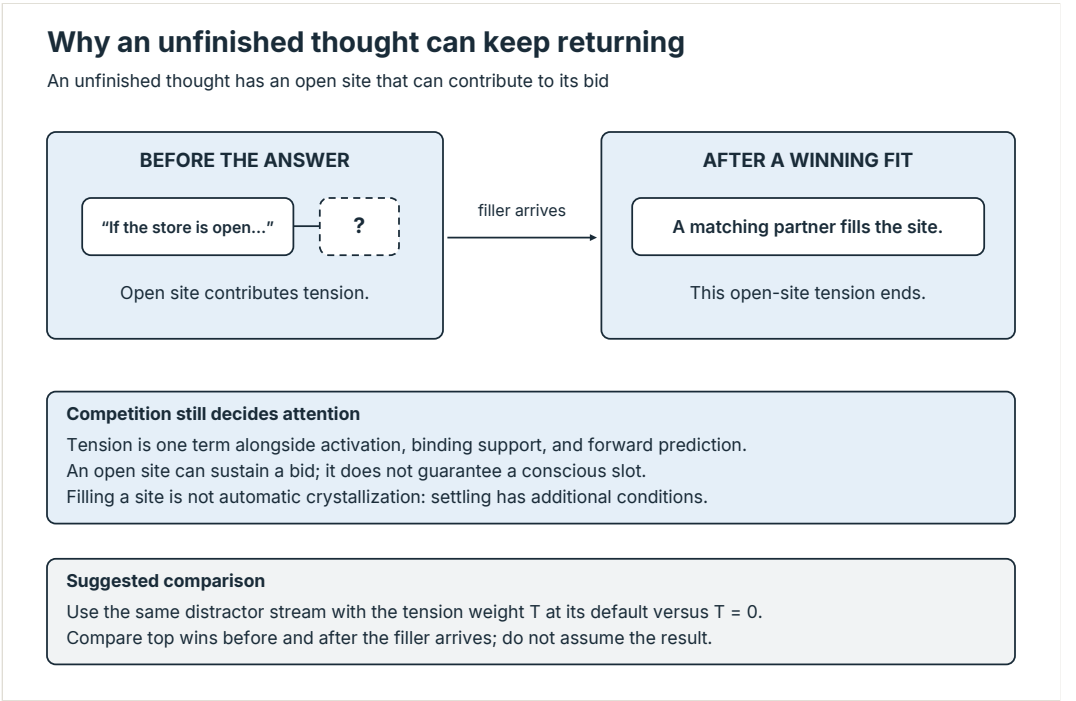


Figure 8 **Why an unfinished thought can keep returning.** A matching filler can remove the tension from a particular opening. Other bid terms and settling conditions still apply. Compare the mechanism with an ablation rather than assuming its effect.

TIP

The earworm prediction is testable: place a complete thought and an incomplete thought of equal starting activation in the same pool. The incomplete thought should increasingly dominate over time. And the instant its missing partner arrives and fills the open site, the tension drops to zero — the earworm should vanish from consciousness as suddenly as a resolved chord. If it doesn't, the tension mechanism is wrong.

LINEAGE

The Zeigarnik effect. In 1927, psychologist **Bluma Zeigarnik** reported that interrupted tasks are remembered better than completed ones. Waiters remembered open orders perfectly and forgot them the instant the bill was paid. The finding has been replicated, debated, and refined for nearly a century. Gerundelle's open-site tension is a mechanistic account of the Zeigarnik effect: an incomplete thought has a non-decaying constant added to its bid, making it more likely to win conscious slots than a completed thought of equal baseline activation. Experiment IV.2 (The Earworm) is, in effect, a mechanized Zeigarnik study — and its prediction (the incomplete thought dominates until completion, then vanishes instantly) is sharper than the original finding, because the mechanism specifies *exactly* when and why the advantage disappears.

WHAT IF Expected surprise reduction looks forward, not backward

Suppose you've been studying a topic for weeks. Your past track record is excellent — you've reduced a lot of surprise, made accurate predictions, built a solid model. Should that history earn you continued attention?

Not necessarily. If your model is now complete, there's nothing left to learn. Continued attention is wasted. The relevant question isn't "how much surprise *have* you reduced?" but "how much surprise *would* you reduce if I attended to you now?" A thought with a glorious past but nothing left to teach should lose to a thought with no past but everything to teach.

What if the bid term that determines a thought's pull on consciousness looks forward — measuring current arrival surprise plus remaining teachability — rather than backward at accumulated precision credit?

WHY Past precision credit doesn't capture what a thought is still worth — forward-looking surprise reduction does

Forward-Looking Beats Past Credit

UCB — Upper Confidence Bound — is the gold standard for balancing exploration and exploitation in bandit problems. It selects arms based on past reward plus an exploration bonus that shrinks with trials. The edge case: an arm that was excellent early but has been fully explored — all uncertainty resolved, nothing left to learn. UCB still gives it a high score from accumulated reward. It coasts on past glory, unable to distinguish "was informative" from "still is informative."

Gerundelle's original bid design had the same flaw. It gave credit for past performance: molecules that had historically reduced a lot of surprise got higher bids. This sounds

reasonable but creates a new kind of lock-in. A molecule that reduced surprise early and often accumulates a large precision credit. Even after it has nothing left to teach, its historical score keeps it winning slots — the mind equivalent of a tenured professor who stopped publishing but still holds the corner office.

The fix was to replace past precision with a forward-looking term: **current arrival surprise** (how unexpected was this molecule's appearance right now?) plus **remaining teachability** (how much would attending to it still narrow the Dirichlet prior?). A molecule that just arrived unexpectedly scores high even with no history. A molecule that has been fully explored scores low even with a stellar track record. Attention flows to where learning is still possible, not to where learning already happened.



REMEMBER

Four mechanisms replaced the broken first-run learning rule: **(1)** Dirichlet outcome beliefs that self-limit instead of growing without bound, **(2)** novelty as expected information gain instead of *n/(1+count)*, **(3)** forward-looking surprise reduction instead of past precision credit, and **(4)** non-decaying open-site tension for incomplete thoughts. Together they let the system explore without freezing and commit without forgetting to check.

Suggested Experiments

These experiments test whether the Free Energy redesign actually solves lock-in, and whether earworms and forward-looking surprise behave as predicted.

EXPERIMENT IV.1

The Lock-In Test



Setup: Two agents receive the same changing stream (8 new clauses per tick, 50 ticks). Agent A uses the original rule: attraction += 1 on neutral co-wins, novelty = *n / (1 + count)*. Agent B uses the Free Energy rule: Dirichlet outcome beliefs, novelty = expected information gain.

Observe: (1) Diversity of pairings over 50 ticks (how many unique pairs form). (2) Share of bindings driven by novelty versus experience. (3) Whether tick-1 winners monopolize later ticks. (4) When, if ever, each agent settles into a stable pattern.

Compare: Runaway attraction versus Free Energy novelty.

Discover: Which agent keeps exploring? Which freezes on early random wins? And does Agent B explore *too much* — never committing to any stable pairing?

EXPERIMENT IV.2

The Earworm

Setup: Two thoughts in the pool alongside 6 distractors per tick, run for 30 ticks. Thought A: complete, no open sites. Thought B: incomplete, with `sites=("answer",)`. The answer exists in storage but its only key has a warmth of 0.05 (tip of the tongue — barely retrievable).

Observe: (1) Top wins for A versus B before the answer arrives. (2) What happens at tick 15 when the answer arrives in the pool. (3) Repeat with *T = 1.0* (tension on) versus *T = 0* (tension off).

Compare: Complete versus incomplete thought; tension on versus off.

Discover: Does the incomplete thought dominate consciousness despite competition? What happens the instant the missing piece arrives — does the earworm vanish? And at *T = 0*, does it lose its grip entirely?

EXPERIMENT IV.3

The Surprise Gradient

Setup: Seed a base rate: molecule *Y* has been observed 200 times (completely expected). Present molecule *X* for the first time (maximum arrival surprise). Both compete for slots alongside 6 distractors.

Observe: (1) Which wins more slots in the first 10 ticks? (2) How quickly does *X*'s surprise decay? (3) Repeat with *a_surprise = 1.0* (surprise modulator on) versus *a_surprise = 0.0* (off).

Compare: Surprise-weighted bidding versus flat bidding.

Discover: Does the surprising newcomer dominate early? How many ticks until the established molecule catches up? And without the surprise modulator, does the highly expected molecule win by sheer familiarity?

EXPERIMENT IV.4

Forward vs. Backward

Setup: Two molecules in the pool. Molecule *H* has a long history: many after-records, high past precision, thoroughly explored. Molecule *N* is brand new: no history, maximum teachability. Both compete for slots.

Observe: Compute two bid components separately: (1) Past precision — running average of surprise reduction already achieved. (2) Forward surprise — current arrival surprise + 1/(1 + tries). Compare total bids under *w_pred = 0.5* (blended) versus *w_pred = 0.0* (past credit only).

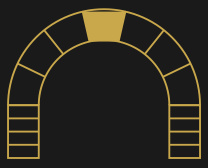
Compare: Forward-looking versus past-credit-only bidding.

Discover: Does forward-looking give new thoughts a fair shot at consciousness? Can an established thought coast on past precision alone? And at what blend of forward/backward does the system balance exploration and exploitation best?

Key Terms

Lock-in	The failure mode where early random winners accumulate enough advantage to monopolize slots permanently, freezing exploration.
Dirichlet outcome belief	A distribution over (good, neutral, bad) outcomes maintained per molecule per level, updated by prediction error — self-limiting instead of growing without bound.
Expected information gain	The mutual information of the Dirichlet prior — how much a single new observation would narrow the distribution. Replaces n/(1+count) novelty.
Arrival surprise	How unexpected a molecule's appearance is right now, based on its recent history of being perceived.
Forward-looking surprise reduction	Current arrival surprise plus remaining teachability — the bid term that replaced past precision credit.
Open-site tension	A constant T added to an incomplete molecule's bid on every tick until its open site is filled — the earworm mechanism.
Rehearsal	Repeated slot wins that prevent a thought from decaying — not new learning, but sustained presence in working memory.
Earworm	A thought that dominates consciousness because its open binding site adds non-decaying tension, not because it's surprising or informative.

Prediction error	The difference between expected and actual outcome, used to update Dirichlet beliefs: (outcome - μ) / (total + 1).
Past precision credit	The discarded bid term that rewarded historical surprise reduction — replaced by forward-looking surprise because it caused a second form of lock-in.



THE MUSEUM

The Gallery of Sealed Cases

This gallery is quiet in a way the others were not. The fragments here have stopped moving. They are sealed in glass cases, each one fixed in the shape it had when it closed: joints locked, bonds set, no open sites. The cases are mounted on stands at different heights, some at eye level, some near the floor, and the glass is clean and heavy.

On the outside of each case, words are written. Not labels — more like names, in the handwriting of whoever sealed it. On the nearest case the words are dark and sharp: *calling*, *afternoon*, *answered*. On a case further back the words are faded almost to nothing. I lean close and can barely read them. Time has been at the ink.

I say aloud: "Calling." Nothing happens. I try the faded case: I read its words, slowly, one at a time, the ones I can still make out. When I reach the third, the case opens. The fragment inside stirs, lifts, and drifts into the air. Its joints loosen. Its bonds begin to flex. It is active again, moving the way fragments moved in the Alcove. It stays near the open case for a moment, then floats toward the alcove on the far wall, leaning.

I try it on the nearest case: "Calling. Afternoon. Answered." The case opens immediately — the words are warm and dark. The fragment inside rises faster, already bright.

A case in the corner does not open, no matter what I say. Its glass has no words on it at all. The outside is blank. I knock on it. I read the words from the cases nearby, loudly, in case they are its words too. Nothing. It sits there, sealed, with no way in from this side.

A placard on the wall between the blank case and the others:

SEALED CASES
Glass, words on the outside, fragments inside. Closed.

A fragment that has finished its work — no open sites, examined several times, each examination confirming the last — precipitates into a case. The case seals. The fragment stops competing and stops learning.

On the outside of each case are the things that were active when it closed. These words cool over time: their ink halves in darkness every twenty steps. When enough of the words on a case match what is happening now — when half or more are warm and present — the case opens and the fragment returns to the air.

Some cases have no words on the outside. Their contents were received, not earned. Nothing the visitor says can warm a word that was never written. These cases open only one way: from outside, by force.

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In the middle of the gallery, two cases side by side.

The first holds a fragment with a fill seated firmly in an opening. The fill is heavy and snug. Beside it, a thick stack of slips — I have seen these before, in the Alcove. I pick one up. On its face, in small print: **1.0**. I count the stack. Twelve slips. Beside the fill, the opening's rods are relaxed. The pull is gone. Whatever this opening wanted, it got, fully.

The second case holds a fill in an opening of the same shape. But the fill is lighter. It does not sit all the way down. The rods around it are not fully relaxed — there is still a faint pull, though less than an empty opening's. Beside it, a thinner stack. I pick one up: **0.5**. Four slips. The fill was placed, not earned. It covers the opening but does not complete it.

On the blank case in the corner, I now see: no slips at all. Not thin. None.

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Further on, a case I almost walk past. Inside, a fragment with an opening. In the opening, something heavy. I look more closely. The heavy thing does not fit. Its shape is wrong — too round for the angular opening, seated at an angle, pressing on rods that were not made to hold it. But it is there, and the case is sealed around it.

The rods near it are bent. Not the way old openings bend toward what they want — bent awkwardly, accommodating something that does not belong. The fragment

leans on the heavy thing the way a person leans on a crutch: it holds, but it is not right. I think it will fall out on its own.

A placard beside this case alone:

A FILL THAT DOES NOT FIT
Fragment with opening. Sealed on a wrong partner.

This opening closed not because the right fill arrived but because a heavy, high-scoring neighbor appeared beside it often enough. The neighbor's shape does not match the opening's. But the two were lit together many times, and the encounters were pleasant, and eventually the opening accepted what was near instead of what was right.

The case sealed. From the outside it looks like any other case. Inside, the rods know.

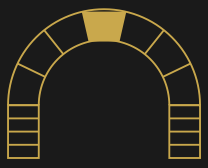
It will stay sealed unless the world contradicts what it predicts. Then, and only then, the case breaks open from outside.

I stand in front of it for a long time. It looks like any other case.

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As I leave, I pass a case on a high shelf. A crack runs down its glass. The crack was not there when I came in — or I did not notice it. Something fell in the gallery while I was reading the placard. Whatever was predicted by the fragment inside did not happen, and the case broke from outside. The fragment is already drifting upward, bright, urgent, with a pull I can feel from the floor.

In the far corner, the cloth on the pole hangs still. It does not lift. This is the first gallery where it has not moved, and the stillness is louder than the swinging.



SECTION

5

Two-Phase Storage

Crystallization, Rationalization, and Pandora's Box

IN THIS SECTION

- Seeing why thoughts exist in two phases: active fluid and crystallized inert
- Understanding settled closure — when a thought has nothing left to learn
- Discovering how warming and cooling keys let storage boxes resurface
- Watching a mind close a question on the wrong answer — rationalization
- Learning how the world itself can force a stored belief to re-open
- Understanding why told knowledge can only be shattered, never gently recalled

WHAT IF Thoughts exist in two phases: active fluid versus crystallized inert

Water exists in two familiar phases. As a liquid, it flows, mixes, takes the shape of whatever contains it, reacts to every nudge. As ice, it's rigid, inert, locked into a fixed structure that persists unchanged until enough heat arrives to melt it.

What if thoughts work the same way? In Gerundelle, every thought molecule exists in one of two phases. In the **active phase**, it flows through the pool, competes in the tournament, binds with partners, accumulates experience, updates its beliefs. It's fluid — responsive to everything happening around it. In the **crystallized phase**, it has precipitated into a storage box: inert, no longer competing, no longer learning, locked into the structure it had when it closed. It persists in storage until something specific brings it back.

What if closure — the transition from active to crystallized — is the only way a thought truly finishes? Not fading, not being overwritten, not being deleted. Finishing.

WHY Without a crystallization mechanism, active thoughts either decay into nothing or linger forever

Closure Is the Only Way a Thought Finishes

Flat key-value memory — as in memory-augmented neural networks like NTMs and DNCs — keeps every memory equally accessible and equally active. After 10,000 ticks, the memory bank holds 10,000 entries all competing for read attention. There's no structural distinction between what the agent is actively thinking about and what it learned on day one. Retrieval becomes a soft-attention lottery over an ever-growing haystack.

Without two phases, you have two bad options. If thoughts decay over time, useful knowledge gradually vanishes — the agent forgets things it learned simply because time passed. If thoughts persist indefinitely in the active pool, the pool grows without limit and the tournament becomes impossibly crowded. Every thought the agent ever had competes with every new perception, forever.

Crystallization solves both problems. A thought that has been fully explored — no open sites, stable outcome beliefs, nothing left to learn — precipitates out of the active pool into storage. It stops competing. It stops consuming tournament bandwidth. But it doesn't vanish. It sits in its box, inert but intact, ready to melt back into the pool if the right conditions arrive. The active pool stays manageable. Learned knowledge stays accessible. And the transition between the two phases is a meaningful event: the thought *finished*.



REMEMBER

Thoughts in Gerundelle have two phases of matter. **Active**: competing, binding, learning, decaying. **Crystallized**: inert, stored, no longer consuming tournament slots, but intact and potentially retrievable. The transition from active to crystallized is **closure** — the only way a thought truly finishes.

WHAT IF A holeless thought that wins for 3 ticks with flat precision precipitates into storage

Imagine you're working a jigsaw puzzle. You pick up a piece, try it in several spots, and finally snap it into place. For a moment you admire it — yes, that's right, it fits. You check it a second time. Still right. A third time. Still right. At that point, you stop thinking about it. The piece is done. You move on.

Gerundelle formalizes this with **settled closure**. A molecule qualifies for precipitation when three conditions hold simultaneously: it has no open binding sites (no missing pieces), it has won top-level slots for *dwell* consecutive ticks (it's been examined, not

ignored), and its precision is flat across those ticks (each examination confirmed the same thing — nothing new was learned). The parameter *dwell = 3* means three consecutive confirmations trigger crystallization.

What if this is the computational signature of "I'm done thinking about this" — a thought with no holes, no surprises, and no remaining questions, settling into storage?

WHY Settled closure requires holelessness, repeated wins, and flat precision — three checks that the thought is genuinely finished

Nothing Left to Learn

Time-based eviction — an LRU cache — evicts the least recently used entry when capacity is full. A thought that's been recently accessed but is still actively surprising and informative gets to stay; a thought that's been fully explored stays too, as long as it was accessed recently, blocking newer thoughts. LRU cares about recency, not completion. It evicts by clock, not by cognitive state.

Settled closure cares about whether the thought has anything left to teach. Each of the three conditions does specific work. **No open sites** means the thought is structurally complete — every binding site has found its partner. This prevents incomplete thoughts from being prematurely filed. **Consecutive top wins** means the thought has been consciously examined, not just lurking at lower levels. This prevents thoughts from crystallizing without ever having been properly considered. **Flat precision** means each examination produced the same result — attending to the thought one more time taught nothing new.

All three must hold simultaneously. A thought with an open site can't settle, no matter how many times it wins. A thought that wins once but loses the next tick hasn't been confirmed. A thought whose precision is still changing is still learning. Only when all three conditions hold for *dwell* consecutive ticks does the molecule precipitate into a storage box, keyed by the partners and conditions it was bound to when it closed.



TECHNICAL STUFF

The precipitation rule. A molecule *m* precipitates at tick *t* if: (1) `m.open_sites == ∅`, (2) *m* was among top-level winners at ticks *t*, *t–1*, and *t–2*, and (3) the variance of *m*'s precision scores across those three ticks is below a threshold. The resulting storage box is keyed by *m*'s bound partners and the facts-register contents at time of closure.

WHAT IF Storage boxes have warming and cooling keys that halve every 20 ticks

You haven't thought about your childhood phone number in years. Then someone mentions the street you grew up on, and the number surfaces instantly. The memory didn't degrade — it was intact, waiting in storage. What brought it back was a **key**: a concept that was active in the pool and happened to match something the stored box was filed under.

But not all keys are equally warm. The street name might surface the memory immediately because you heard it five minutes ago. Your old neighbor's name might work too, but faintly — you haven't heard it in months. Gerundelle models this with **warming and cooling keys**. Every storage box is filed under the concepts that were active when it closed. Each key has a warmth that halves every 20 ticks. Recent keys are hot; old keys are cold. A box re-opens when enough of its keys are among the current pool's winners — specifically, when half or more of its keys are warm and active.

What if this resonance mechanism — half your keys are lit up — is the formal definition of being reminded?

WHY Resonance re-opens a box when half its keys are among current winners, with key warmth decaying over time

Pandora's Resonance

Nearest-neighbor retrieval over embeddings — the mechanism behind retrieval-augmented generation — returns the *k* most similar memories regardless of whether they actually apply. A situation that resembles nothing in storage still gets *k* results: the least-bad matches, not no matches. The system has no way to return "nothing comes to mind," which is sometimes the correct answer.

Gerundelle's resonance has a threshold: half the keys must be warm and active among current winners. If the current context doesn't match enough keys, nothing opens. The halving schedule adds a temporal dimension. A box stored 5 ticks ago has keys at nearly full warmth — any overlap with current activity will reopen it. A box stored 200 ticks ago has keys near zero — only a very strong, specific overlap will bring it back. This means recent storage is volatile (easy to reopen, easy to revise) while old storage is stable (hard to reopen, resistant to casual disruption). The system naturally develops a distinction between short-term and long-term memory without any separate mechanism.

Gerundelle calls this **Pandora's resonance**, because reopening a box can release thoughts that the system had finished processing — including unresolved emotions, incomplete questions, and beliefs that were closed under conditions that no longer hold.

A box that was closed with open sites still has those sites when it reopens. Whatever was unfinished is unfinished again.

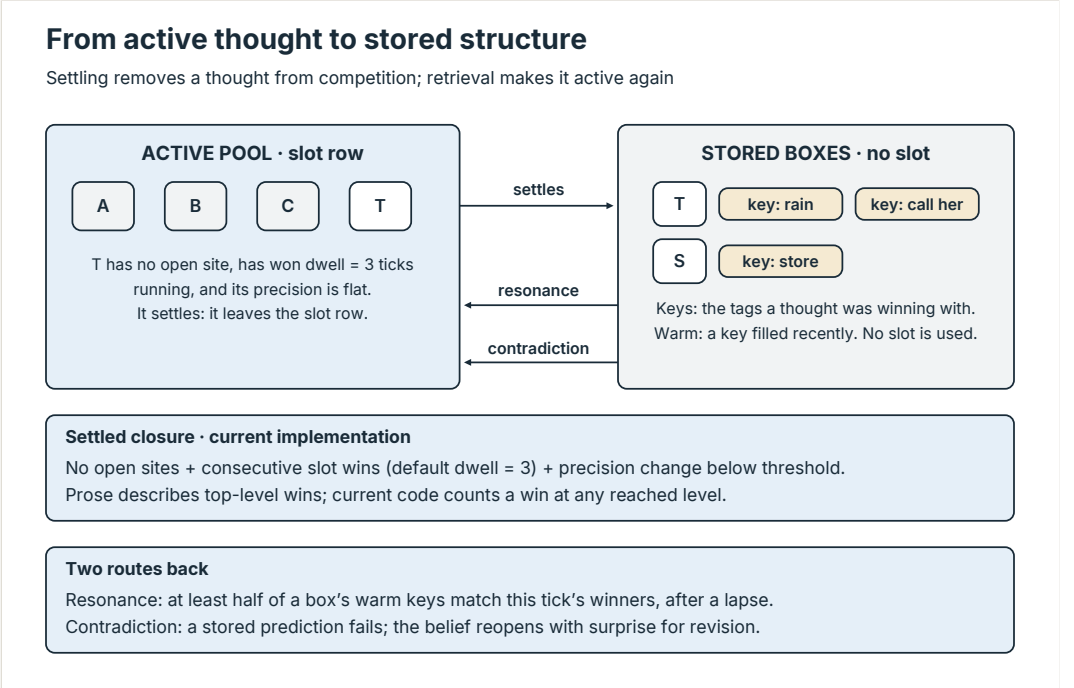


Figure 9 **From active thought to stored structure.** A settled thought leaves the slot row but keeps its keys and remains retrievable. Resonance matches warm keys; contradiction forces revision of a failed prediction. The closure criterion shown is from current code; parked and told boxes follow additional rules.

TIP

Think of storage boxes like sealed envelopes with keywords written on the outside. Time fades the ink. If enough keywords on the envelope match what you're currently thinking about, the envelope tears open and the contents spill back into active thought. Recent envelopes have dark ink and open easily. Old envelopes have faded ink and stay shut unless the match is very specific.

WHAT IF An open thought closes on a high-scoring partner that doesn't actually fit its site

You've been puzzling over why your friend didn't return your call. The question has an open site shaped like "reason." While you're thinking about it, a completely unrelated positive thought wins alongside the question — say, you just got a promotion. The two thoughts co-win repeatedly. The promotion thought has a high score, it frequently appears alongside the question, and the co-wins accumulate positive experience. At some point, the question molecule closes — not because the promotion *answers* it, but because the promotion *feels* like an answer. The site closes on a partner whose tag doesn't actually match.

What if the mind can close an open question on a high-scoring partner that doesn't genuinely fit — simply because they co-won enough times in a positive context?

WHY A question can close on a bad key if the partner scores high enough, but only if that partner is understood at least 50%

Rationalization

Rationalization in Gerundelle is a precise mechanism, not a metaphor. It happens when an open-site molecule closes on a partner whose binding-site tag doesn't match the open site's type requirement, but whose co-win score exceeds a threshold (the *rationalize* parameter, default 0.6). The mind accepts a non-answer as an answer because the emotional or attentional association is strong enough.

There's one guardrail: the partner must be at least 50% understood in the agent's Understanding Register. A molecule the agent knows nothing about can't serve as a rationalization target — there isn't enough structure to even pretend it's an answer. This prevents the most absurd closures while allowing the psychologically realistic ones: closing a question about your friend's silence with "she's probably just busy" when you don't actually know that, but you understand the concept well enough to use it as a placeholder. The mind leans on the bad key the way a person leans on a crutch: it holds, but the shape doesn't fit the site.



WARNING

Rationalization is not always harmful. Sometimes closing an unanswerable question with a plausible-but-unverified answer is the right thing to do — it frees up working memory for more productive thoughts. The danger is that the box, once closed, looks identical to a box closed on a genuine answer. The system can't tell the difference unless the real answer later arrives and contradicts the rationalized closure. This is why the next mechanism — contradiction — matters so much.

LINEAGE

Festinger's cognitive dissonance. In 1957, **Leon Festinger** proposed that holding two contradictory beliefs creates psychological discomfort — *cognitive dissonance* — and that people reduce the discomfort by changing one belief, adding rationalizations, or avoiding the contradiction. Gerundelle's rationalization mechanism is Festinger formalized: an open question creates tension (the open-site bid), a high-scoring but non-matching partner offers relief (the co-win feels good), and the question closes on the wrong answer (dissonance reduced). The 50%-understanding guard is the novel part — Festinger never specified a threshold for when rationalization is structurally possible versus impossible. The architecture predicts that you can't rationalize with something you genuinely don't understand at all, only with something you understand just well enough to misapply.

WHAT IF A box predicts an outcome that fails to arrive

You've learned that calling your friend after lunch always works — she answers every time. This belief has been tested, confirmed, and crystallized into a storage box. The box sits quietly, and whenever "call her" and "after lunch" appear together in the pool, it predicts "she answered."

One day you call after lunch and she doesn't answer. The prediction fired, the world didn't comply, and now there's a mismatch between what storage predicted and what actually happened. The belief was wrong — or at least, the conditions under which it was right have changed.

What if the world itself can force a stored belief to re-open, overriding the normal resonance mechanism, with full surprise injected into the reopened molecule?

WHY When a prediction fails, the box re-opens from outside with full surprise, forcing revision

Contradiction

Contradiction is Gerundelle's mechanism for external revision. When a storage box's prediction fires (its keys are active and it expects a specific outcome) and the predicted outcome fails to arrive, the box re-opens with maximum surprise injected. It re-enters the active pool not as a gently warmed memory but as an urgent, high-priority molecule demanding attention.

This is fundamentally different from resonance. Resonance reopens a box because the current context matches its keys — a gentle reminder. Contradiction reopens a box because the current context *violates* its prediction — a forced revision. The surprise injection means the contradicted belief jumps to the top of the tournament, demanding con-

scious examination. The mind can't ignore a failed prediction the way it can ignore a faded memory.

REMEMBER

Two paths reopen a storage box. **Resonance**: half the keys are warm and active — a gentle reminder, easily missed if keys have cooled. **Contradiction**: a prediction fires and the world disagrees — a forced reopening with full surprise, impossible to ignore. Resonance says "remember this?" Contradiction says "you were wrong about this."

LINEAGE

Kuhn's paradigm shifts. Thomas Kuhn (1962) described scientific revolutions as a cycle: normal science operates within a paradigm, anomalies accumulate, a crisis forces the paradigm open, and a new paradigm replaces it. Contradiction forcing a storage box open from outside *is* anomaly → crisis at the scale of a single belief. The box held a settled prediction. The world produced an anomaly. The box re-opens with full surprise — crisis. What follows is revision or replacement. The history of physics offers the largest example: the ultraviolet catastrophe and black-body radiation forced classical thermodynamics open, and quantum mechanics precipitated in its place. Every contradiction in Gerundelle is a miniature Kuhn cycle.

WHAT IF Told boxes can't be re-opened by recognition, only by contradiction

Someone tells you: "Never eat the red berries — they're poisonous." You've never tasted them. You've never seen anyone get sick from them. You have no personal experience at all. But you store the advice, and it shapes your behavior. The belief is crystallized in a box, but a box with a crucial difference: **none of the keys are yours**. You didn't earn them through lived experience. You received them through testimony.

What if these told boxes are fundamentally different from lived boxes? What if a belief you were told can't be gently re-opened by resonance — because you don't have the experiential keys that would trigger recognition — and can only be shattered by direct contradiction: eating the berries and surviving?

WHY Told boxes lack experiential keys, so only a contradicting event can force them open

You Don't Own What You Haven't Worked Out

A lived box is filed under keys the agent earned: the concepts that were active when it was working the thought out, the conditions it observed, the partners it found. When those conditions recur, the keys warm up and the box can reopen naturally through resonance. The agent can revisit the thought, re-examine it, potentially revise it.

A told box has none of this. Its content was received, not constructed. It has no warming keys from lived experience because there was no lived experience to generate them. The only way in is from outside: the world must produce an event that directly contradicts the told prediction. "Red berries are poisonous" stays sealed until you eat one and don't die — or until someone you trust tells you the opposite (which is itself a new told event that contradicts the old one).

This creates an asymmetry that mirrors real cognition. Beliefs you worked out yourself are revisable — you can re-derive them, check them, update them. Beliefs you were told are brittle — they either hold or shatter, with nothing in between. You don't own what you haven't worked out yourself.



TECHNICAL STUFF

Told vs. decomposed boxes. Gerundelle tags each storage box with its provenance: `lived` (closed on partners found through the agent's own tournament) or `told` (closed on content received from testimony). Lived boxes have experiential warming keys and can re-open via resonance or contradiction. Told boxes have no experiential keys and can re-open **only** via contradiction. This is not a design preference — it's a structural consequence of how keys are generated. No experience, no keys. No keys, no resonance path.



LINEAGE

The generation effect and constructivist learning. The memory literature's **generation effect** (Slamecka & Graf, 1978) shows that self-generated material is remembered and integrated better than passively received material. You remember the answer you worked out more reliably than the answer someone told you. Constructivist learning theory — from **Piaget** through **Papert** — makes a parallel claim for understanding: knowledge constructed through the learner's own activity is structurally different from knowledge received through instruction. Gerundelle's told-boxes-only-shatter rule is a sharp mechanistic prediction that neither literature ever made: told knowledge doesn't just fade faster or integrate more poorly —

LINEAGE (CONTINUED)

it *lacks the structural keys for gentle revision* and can only be broken by direct contradiction. The asymmetry isn't a matter of degree. It's a matter of kind.

Suggested Experiments

These experiments test the five storage mechanisms: precipitation, resonance, rationalization, contradiction, and the told/lived distinction.

EXPERIMENT V.1

Rehearsal and Settling

Setup: Two runs of the same thought molecule. Run A: the molecule is re-presented every 3 ticks (rehearsal). Run B: the molecule is presented once and left to compete on its own. Both runs have the same distractors. Track for 40 ticks.

Observe: (1) Total top-level wins before precipitation. (2) The tick at which the molecule precipitates into storage. (3) Whether the rehearsed molecule precipitates sooner or later than the unrehearsed one.

Compare: Rehearsed versus unrehearsed.

Discover: What does rehearsal buy? Does it extend the molecule's active life (more wins before settling) or shorten it (faster confirmation and precipitation)? Does an unrehearsed molecule even survive long enough to meet the dwell criterion?

EXPERIMENT V.2

The Bad Key

Setup: A question molecule with an open site typed `("reason_sunny",)` is strongly attracted to the molecule `"she_is_moody"` (seeded with good = 10). But `"she_is_moody"` doesn't carry the `reason_sunny` tag. They co-win repeatedly. Run with *rationalize = 0.6* versus *rationalize = None* (off).

Observe: (1) Does the question close on the non-matching partner? (2) If it closes, what happens when the real answer (carrying the correct tag) arrives at a later tick? (3) Does contradiction re-open the rationalized box?

Compare: Rationalization enabled versus disabled.

Discover: Can the mind close a question on a bad key just because the co-wins felt good? And once closed, is the rationalized belief distinguishable from a genuine answer?

EXPERIMENT V.3

The Failed Prediction

Setup: A storage box predicts `"she_answered"` whenever `("call_her", "after_lunch")` are among the current tick's winners. The agent calls after lunch. She doesn't answer.

Observe: (1) Does the box re-open? (2) With what surprise level? (3) How many ticks does the re-opened molecule spend in the top level before re-closing or being revised?

Compare: Prediction tested and failed versus prediction never tested (the agent doesn't call).

Discover: Can the world itself force a revision of a settled belief? And does the surprise injection give the contradicted belief enough priority to be properly re-examined, or does it get crowded out by other active thoughts?

EXPERIMENT V.4

Told vs. Lived

Setup: Two storage boxes with identical content: "red berries are dangerous." Box A was **lived** — the agent ate a red berry, got sick, closed the thought on earned experience. Box B was **told** — another agent said "red berries are dangerous" and the belief was stored without personal experience. After 100 ticks of cooling, re-introduce the concept "red berries" into the active pool.

Observe: (1) Does Box A reopen via resonance (its lived keys match the active concept)? (2) Does Box B stay sealed (no experiential keys to warm)? (3) Now present a contradicting event: the agent eats a red berry and nothing happens. Do both boxes reopen?

Compare: Lived box (resonance + contradiction paths) versus told box (contradiction path only).

Discover: Does the told belief survive encounters that would revise the lived belief? And when contradiction finally shatters the told box, does the agent treat the shattered belief differently from a gently reopened lived one?

Key Terms

Active phase	A thought molecule competing in the tournament, binding with partners, updating beliefs — fluid and responsive.
Crystallized phase	A thought molecule stored in a box, inert, no longer competing — intact but dormant.
Closure	The transition from active to crystallized — the only way a thought truly finishes in Gerundelle.
Settled closure	Precipitation triggered by dwell consecutive top wins with no open sites and flat precision — the molecule has nothing left to learn.
Dwell	The number of consecutive confirming wins required for precipitation (default: 3).
Storage box	A crystallized molecule filed under warming/cooling keys, reopenable by resonance or contradiction.
Warming key	A concept associated with a storage box at closure time, whose warmth halves every 20 ticks.
Pandora's resonance	A box re-opens when half or more of its keys are warm and active among current winners.
Rationalization	Closing an open-site molecule on a high-scoring partner whose tag doesn't match — accepting a non-answer as an answer.
Understanding guard	A rationalization target must be at least 50% understood — preventing closure on completely unknown partners.
Contradiction	Forced reopening of a storage box when its prediction fires and the predicted outcome fails to arrive — with full surprise injected.
Told box	A storage box closed on received testimony with no lived experiential keys — can only be re-opened by contradiction, not resonance.
Lived box	A storage box closed on partners found through the agent's own experience — reopenable by resonance or contradiction.

PART III

III

Self-Reflection, Ownership & Collective Regimes

The Gallery of Mirrors

6 Meta-Cognition

The Gallery of Outlines

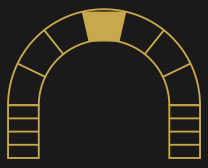
7 Understanding Register

The Gallery of Held Things

8 Territory and Ownership

The Gallery of Crystals

9 Emotional Phases



THE MUSEUM

The Gallery of Mirrors

I am back in the hall of fragments from the Alcove. It is the same room — the same high ceiling lost in shadow, the same alcoves on the wall, the same pale rods drifting in the dark. But there are mirrors now.

They hang among the fragments, small and rectangular, each about the size of a playing card. They are made of the same material as the fragments — rods and joints, the same pale bone — with a polished face. They lean toward the alcoves the way everything here leans. But they do not compete with the fragments. Beside the main alcoves, set apart, a smaller alcove of their own: room for three mirrors. They compete only with each other for those three places.

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The mirror that does not win.

Three fragments take turns in the lowest alcove, round and round, in the same order: the tall one, the branched one, the tall one, the branched one. I watch them cycle for a long time. Nothing changes. The three of them have settled into a loop and none of them knows it.

In the dark, a small mirror leans toward the mirrors' alcove. In its glass I can see the loop. The tall one, the branched one, the tall one. The mirror shows it as a pattern, from outside.

But the mirror cannot reach even the mirrors' alcove. Other mirrors lean harder. It competes and loses, competes and loses. Down here the loop goes on, and the mirror that would show it stays in the dark.

Then it wins. It rises into the mirrors' alcove, the loop appears in its glass, and something happens below: a gate drops across the main alcove. The circling fragments are lifted out and dimmed. The loop is broken. The alcove fills with something new.

Further down the hall, an opening that nothing can fill has drawn every fragment toward it. A mirror wins the mirrors' alcove there too, and the opening is shelved the same way — still open, but set aside.

A placard by the mirrors' alcove:

MIRRORS THAT MUST WIN
Rods, joints, polished face. Competing.

A mirror here is not a window. It is made of the same material as what it reflects, and it must earn its place in the alcove like everything else.

A mirror that does not win a place cannot act on what it shows. The loop below ran for a long time before the mirror reached the light. The capacity to notice the loop was present. The noticing was not. Detection is not guaranteed. It must be won.

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The mirrors that show what is coming.

A side room, kept under glass like the frozen room across from the Room of Returns. Inside it, mirrors that showed only what was already there filled every place — the large alcove too — and nothing moved again. A label on the glass: *Rent without yield. Kept as the day it failed.*

Back in the main hall, the mirrors that hold places in the second alcove show something different. They do not reflect the present. They show the hall as it will be a few moments from now. I watch one: it shows a fragment rising toward the alcove. I wait. The moment arrives and the fragment rises. The mirror stays bright.

Another shows a fragment still in the lowest alcove. I wait. The moment comes and the fragment has left. The mirror's glass goes dark and it falls from the second alcove, drifting down into the hall. Its prediction was wrong. It lost its place.

A placard near the failed mirror, already fading on the floor:

PREDICTION, NOT REFLECTION
Mirror, polished face, now dark.

A mirror that only reflects what is already happening adds nothing. It occupies a place and says: yes, the thing that is happening is still happening. The room behind the glass froze that way.

A mirror that predicts earns its place by being right about what comes next. When its prediction fails, it loses the place. When the prediction holds, it has told the hall something the hall did not already know: that the present state will persist.

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The silent shutter.

Near the back wall, a mirror on a stand, facing a fragment in the lowest alcove. I stand behind the mirror and look through it at the fragment. The fragment drifts and competes as if the mirror were not there. I watch for a long time. The fragment does not brighten. It does not lean toward the mirror. It does not change at all.

Beside the stand, a sealed case. Inside, a failure preserved: a mirror whose glass lit up everything it looked at. Looking made the thing brighter, which made the mirror brighter, which made the thing brighter, until the case is white from edge to edge and nothing inside can be seen. A label: *The observer that changed what it observed.*

The working mirror sees without being seen. The sealed one saw and changed what it saw.

◆ ◆ ◆

The seam.

A skylight. Through it, rain and sun take turns above the gallery. Below the skylight, a fragment drifts in and out of the lowest alcove. When I watch it closely I notice: under sun, the hall warms around it. Under rain, the hall chills. The same fragment, two different effects, and its slips — the small stacks of paper I first saw in the Alcove — are sorted in two colors. White slips under sun. Dark slips under rain.

A mirror wins a place in the second alcove and shows the fragment with two faces. Then the fragment divides. It happens quietly: a line appears down its center, and the two halves drift apart. The left half warms the hall every time. The right half

chills it every time. Each half's slips are all one color. I look up through the skylight: the left half drifts into the light only under sun. Its seam gilds over — it is confirmed.

In the next case, the same kind of fragment splits along a different line. A clock is beside it, and the seam follows the clock instead of the weather. But both halves still warm and chill by turns. The split did not separate them. The halves press together, the seam closes without a mark, and the fragment drifts whole again, waiting for a better boundary.

A placard, the last in the gallery:

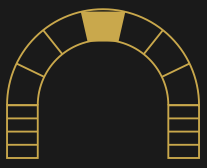
SPLITTING AND MENDING
Fragment, skylight, clock. Two seams.

When a fragment's receipts fall in two groups — sometimes one thing, sometimes the opposite — a mirror can divide it on the condition that separates the groups. The right condition makes halves that each behave one way. The wrong condition makes halves that each still behave both ways.

A confirmed split gilds. An unconfirmed split closes and the fragment tries again. The boundary is found, not given.

I look at myself in one of the lit mirrors near the door. The mirror shows me a few moments ahead: standing in the doorway, one hand on the frame, about to leave. I look at myself in it and I do not change.

In the corner, the cloth lifts and the fragments nearest it drift sideways. I am beginning to expect the cloth. It is always there. It never has a placard.



SECTION

6

Meta-Cognition

Noticing Your Own Thinking Without an Inner Observer

IN THIS SECTION

- Seeing how an organism notices stuck, looping, and torn states using the same molecular competition
- Understanding the Mirror Trap and why meta-molecules must live by prediction, not re-detection
- Learning why noticing isn't automatic — the meta-molecule must win a slot
- Discovering why aboutness must be one-way so noticing doesn't inflate what it notices
- Watching a bimodal thought get split into two contextual halves
- Seeing how useless splits get unified back and real splits get confirmed

WHAT IF An organism notices its own mental states using the same molecular competition it uses to notice the outside world

You're trying to solve a problem and you realize you're going in circles. The realization itself is a thought — "I'm stuck" — that arose the same way any other thought arises: it competed for your attention and won. You didn't flip a special switch marked "engage metacognition." The feeling of being stuck showed up in the same stream as the problem itself, competed alongside it, and happened to win a slot.

Now consider: you could be stuck *without noticing*. You could loop through the same three ideas for twenty minutes before the thought "I'm looping" surfaces. The capacity to detect your own state exists, but it's not guaranteed. It has to compete like everything else.

What if an organism detects its own mental states — being stuck, looping, torn between choices, bored, or conflating two situations — using the exact same molecular competi-

tion it uses to detect a storm outside? No special introspection module. No privileged access. Just more molecules competing in the same tournament.

WHY ABI's organ federation law requires that all communication happen through molecules, never through hidden state

Couple Only Through Molecules

The obvious engineering choice is a direct state readout — a function `isStuck()` that reads internal variables and returns a boolean. It's always available, always free, and always accurate. The system never fails to notice it's stuck. But that removes the cost structure entirely: if noticing is free, there's no tradeoff between noticing and doing. And the readout can't be wrong — no false negatives (failing to notice you're stuck) and no false positives (thinking you're stuck when you're making slow progress on a hard problem). It's too good to model real metacognition.

Gerundelle follows a stricter rule borrowed from ABI: organs in the system may communicate **only through molecules**, never by reading each other's hidden state. The meta track can't peek at the base thought stream's internal variables. It can't check a register labeled "currently_stuck = true." It can only observe what wins in the shared tournament and form its own molecules about what it sees.

The rule sounds like a limitation, but it solves a deep problem. If the meta track could read hidden state directly, it would be parasitic — adding cost without adding information, since the hidden state is already doing its work. By forcing meta-cognition through the same competitive channel, the system guarantees that noticing has a cost (it consumes tournament slots), a benefit (it can trigger corrective action), and a failure mode (it might not win, so the organism might not notice). Metacognition earns its place or doesn't happen.



REMEMBER

Meta-cognition in Gerundelle isn't a separate module with privileged access. It's **more molecules in the same tournament**. A meta-molecule like "I'm stuck" competes for working-memory slots against the thoughts it's about. This means noticing your own state is never guaranteed — it has to win.

WHAT IF Meta-molecules live by confirming a 5-tick prediction, not by refreshing on re-detection

Suppose you notice you're frustrated. A meta-molecule forms: "I am frustrated." Five ticks later, are you still frustrated? The meta-molecule can check: it predicted that the base stream would still look frustrated at tick *t + 5*. If the base stream does still look frustrated, the prediction is confirmed and the meta-molecule lives on. If the frustration has resolved, the prediction fails and the meta-molecule dies.

The alternative — the one that seems more natural — is to refresh the meta-molecule every time it re-detects the state. See frustration, refresh. See frustration again, refresh again. This sounds reasonable until you realize what it does: the meta-molecule lives forever as long as the state persists, regardless of whether noticing it accomplishes anything. It becomes a mirror that reflects without acting.

What if meta-molecules must live by making a prediction about the base stream and being judged on that prediction, rather than simply refreshing whenever they re-detect the state they're about?

WHY A meta level funded by copying the base stream is rent without yield — prediction gives it a purpose

The Mirror Trap

Gerundelle calls this failure mode the **Mirror Trap**. A meta-molecule that refreshes on re-detection is just a delayed copy of the base stream. It occupies tournament slots, it has a label that says "meta," but it carries no information the base stream didn't already have. It's rent without yield — consuming working-memory bandwidth to say "yes, the thing that's happening is still happening."

Prediction breaks the trap. A meta-molecule that predicts "frustration will persist for 5 more ticks" is making a **claim about the future**. If the claim holds, the meta-molecule earned its slot — it told the system something useful about its own trajectory. If the claim fails, the meta-molecule dies and frees the slot. Either way, the meta-molecule has a purpose beyond mirroring: it's tracking whether the base stream is changing or stable, and that tracking has consequences for what the system does next.

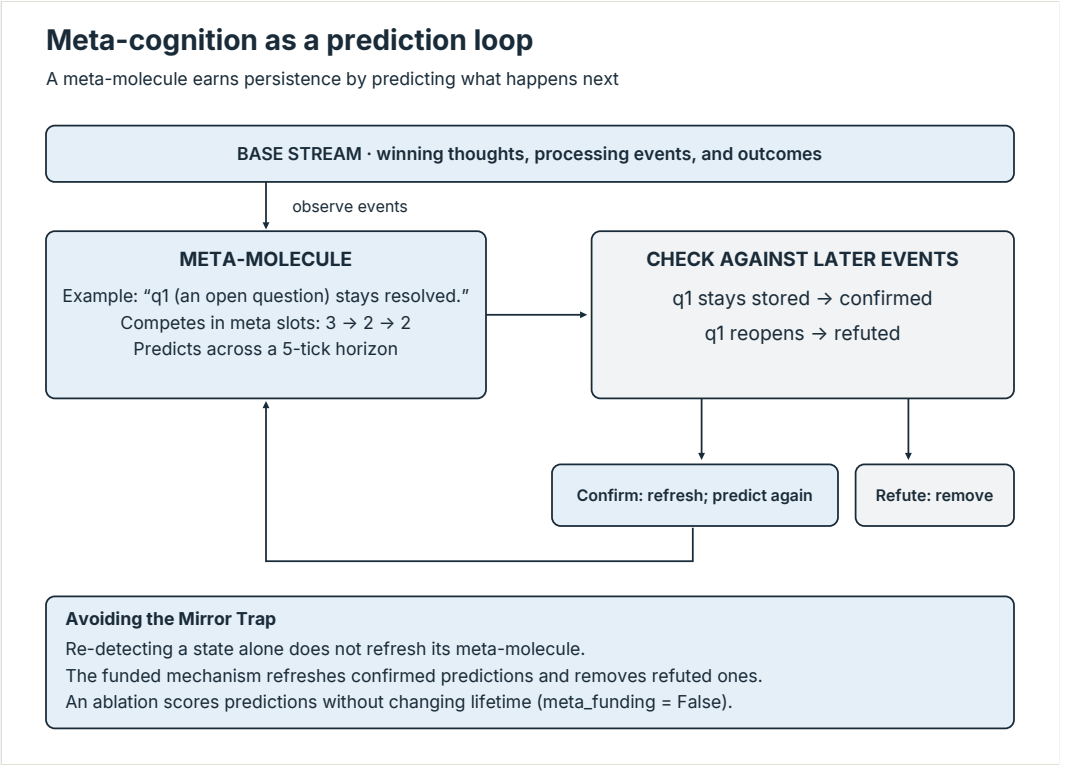


Figure 10 **Meta-cognition as a prediction loop.** Meta-molecules compete in dedicated slots and persist through confirmed predictions about processing, rather than through repeated detection alone.

WARNING

The Mirror Trap was discovered through engineering failure, not theoretical analysis. When meta-molecules were first implemented with refresh-on-detection, they immediately crowded out earworms and event responses in the main thought slots. The system spent all its bandwidth noticing things instead of doing things. Forcing meta-molecules to live by 5-tick predictions and giving them **dedicated meta slots (3, 2, 2)** separate from main thought slots fixed the crowding problem.

WHAT IF Noticing a state doesn't happen automatically — the meta-molecule must win a top slot

You can be confused without knowing you're confused. You can loop through the same dead-end reasoning for an hour before something jolts you into recognizing the loop. The capacity to detect confusion exists — the meta-molecule "I'm looping" is in the pool — but it hasn't won a slot. It's competing against all the other thoughts, including the very thoughts that constitute the loop, and it keeps losing.

What if the ability to notice a mental state is always present as a *capability*, but only becomes active when the meta-molecule wins a top-level slot? What if there's a gap between having the receptor for a state and actually detecting it?

WHY A meta-molecule must win a tournament slot to trigger its corrective action — detection is not automatic

Capability Without Receptor Is Latent

In Gerundelle, each meta-state (stuck, looping, bored, conflated, distinction, same) has a specific corrective action it triggers **only when it wins a top slot**. "Stuck" parks the unanswerable thought. "Looping" parks the circular chain. "Bored" doubles the novelty weight for 5 ticks. "Conflated" splits an ambiguous thought into two contextual halves. But none of these corrections fire automatically. The meta-molecule must compete, win, and *then* the system acts on it.

This means an organism can have the capacity to detect a state and still fail to detect it — because the meta-molecule lost the tournament. The gap between capability and activation is the gap between "the organism could notice this" and "the organism did notice this." That gap is where extended confusion, unrecognized loops, and missed opportunities live. It's not a bug — it's what makes metacognition a real cognitive achievement rather than a free readout.



TECHNICAL STUFF

Dedicated meta slots. To prevent meta-molecules from either crowding out base thoughts or being permanently drowned by them, Gerundelle allocates **dedicated meta slots: (3, 2, 2)** across the three tournament levels — separate from the main thought slots (7, 5, 4). Meta-molecules compete only against other meta-molecules for these slots. This guarantees the meta track has bandwidth without letting it steal bandwidth from perception and action.

WHAT IF Aboutness is one-way — a thought about X doesn't lift X's bid

You notice you're anxious. The meta-thought "I am anxious" forms and wins a slot. If that meta-thought boosted the bid of the anxiety it's about, you'd have a feedback loop: noticing anxiety makes anxiety more prominent, which makes the meta-thought more relevant, which boosts anxiety further. The act of introspection would amplify the very state it's observing.

What if aboutness is strictly one-way? A meta-molecule can be *about* a base thought, but the base thought doesn't know or care that it's being observed. No boost, no feedback, no amplification. The observation is invisible to the observed.

WHY If noticing a thought artificially boosts that thought's bid, introspection creates runaway feedback

One-Way Observation

This is another traced engineering lesson. When aboutness was bidirectional in early tests, the meta track created artificial prominence for whatever it happened to notice. A mildly anxious thought that got noticed became a prominent anxious thought, which strengthened the meta-molecule's relevance, which boosted the anxiety further. Metacognition didn't observe the mind — it *distorted* the mind.

The fix is strict one-way aboutness. A meta-molecule references a base molecule and receives information from it (is it winning? losing? stuck?). But the base molecule's bid, activation, and competitive position are completely unaffected by being referenced. The meta track is a one-way window: you can look through it, but looking doesn't change what you see.



TIP

Think of a wildlife camera in a forest. It observes animals without affecting their behavior. If the camera emitted a loud click every time it took a photo, the animals would flee and the observations would be useless. One-way aboutness is the silent shutter: **observe without disturbing**.

WHAT IF A thought's outcomes are bimodal — good and bad both at or above 25%

You call your friend regularly. Sometimes it goes great. Sometimes it goes terribly. Not randomly — there's a pattern, but you haven't found it yet. Your outcome record for "calling her" has a fat cluster of good results and a fat cluster of bad results, with nothing in between. The thought is **bimodal**.

A unimodal thought is one you understand: it's mostly good, mostly bad, or mostly neutral. You know what to expect. A bimodal thought is one that looks like *two different things glued together*, producing contradictory outcomes depending on some condition you haven't identified. Your record says "calling her" is both good and bad, but it can't be both at the same time — something is masking a hidden boundary.

What if bimodality in the outcome record — both good and bad at 25% or above — is the formal trigger for **conflation**: the recognition that a single concept needs to be split into two contextual halves?

WHY A bimodal thought is conflated — two different situations hiding under one label, needing to be split on a boundary condition

Conflation and Splitting

A fixed concept ontology — the kind used in classical knowledge representation — defines its concepts once and never splits or merges them. When a phenomenon behaves one way in context A and the opposite in context B, the fixed concept captures the average, which is meaningless in both contexts. Every future prediction averages across the hidden boundary and gets the wrong answer. The system is permanently confused because it can't coin a new distinction.

Gerundelle's conflation mechanism does the coining. When the meta-molecule "conflated" wins a top meta-slot, it triggers a search: scan the facts register and current percepts for a condition that separates the good outcomes from the bad ones. If the agent notices that calls go well on sunny days and badly on rainy days, the boundary is weather. The original molecule "calling her" splits into two new contextual molecules: "calling her when sunny" and "calling her when rainy." Each half inherits the relevant portion of the outcome record.

But finding the right boundary isn't guaranteed. The environment might contain both the true condition (weather) and a noisy decoy (time of day, what she had for lunch). If the mind splits on the wrong boundary first, the two halves will still look bimodal — the split didn't separate the outcomes. That's where the next two meta-operations come in.



REMEMBER

Conflated detects bimodality and splits. **Distinction** confirms a split when both halves stay unimodal and opposite (one mostly good, one mostly bad). **Same** collapses a useless split when both halves behave identically. Together, they form a self-correcting loop: split, test, confirm or collapse.



LINEAGE

Piaget, Carey, and Karmiloff-Smith. The conflation-splitting cycle has a rich developmental pedigree. **Jean Piaget**'s *accommodation* is what happens when an existing schema fails to assimilate new experience — the schema must change. **Susan Carey** documented the fossil record: children initially conflate *heat* and *temperature*, *weight* and *density*, treating each pair as a single concept until contradictory outcomes force a split. The split doesn't come from instruction — it comes from the bimodal failure pattern. Gerundelle's conflation detector is Carey's observation turned into a trigger condition: bimodal outcomes (≥25% good and ≥25% bad) signal that a single concept hides two situations. **Annette Karmiloff-Smith**'s *representational re-*

LINEAGE (CONTINUED)

description adds the metacognitive layer: the mind re-represents its own knowledge in more explicit, manipulable forms. The one-way-aboutness rule is a mechanistic answer to a problem that *introspectionist psychology died of* in the early 20th century — observation altering the observed thought. Gerundelle's meta track observes without distorting, which is the silent shutter the introspectionists never found.

WHAT IF Both halves of a split stay unimodal and opposite

After splitting "calling her" into "calling her when sunny" and "calling her when rainy," the agent accumulates experience with each half. If "sunny" is consistently good and "rainy" is consistently bad, both halves are unimodal — each has a clear, single-peaked outcome distribution. The split worked. The hidden boundary was real.

What if this pattern — both halves unimodal and pointing in opposite directions — is the formal confirmation that the split was genuine? The meta-molecule "distinction" forms, confirming that the two halves are truly different concepts, not an arbitrary division of the same thing.

WHY Unimodal and opposite halves prove the split captured a real boundary condition

Distinction Confirmed

"Distinction" isn't just a label — it's an evidential verdict. The agent split one concept into two, each half accumulated its own track record, and the records diverged in exactly the predicted way: one side is reliably good, the other reliably bad. The split paid for itself by turning an unpredictable concept into two predictable ones. The mind has coined a new distinction — one that makes the world more intelligible than it was before.

This is how Gerundelle grows its conceptual vocabulary from raw experience, without anyone telling it which distinctions matter. Bimodality triggers a split. Experience tests the split. Divergent outcomes confirm it. The agent didn't know that weather mattered for phone calls — it discovered this by noticing that its outcome record didn't make sense as one concept, splitting it, and watching the halves diverge.

WHAT IF Both halves of a split behave identically

Sometimes the split is wrong. You split "calling her" on time of day — morning versus afternoon — and both halves produce the same bimodal outcomes. Mornings are sometimes good and sometimes bad. Afternoons are sometimes good and sometimes bad. The split didn't separate anything. Time of day wasn't the boundary.

What if both halves behaving identically is the formal signal to unify them back? The meta-molecule "same" detects that the split was pointless, collapses the two halves back into the original molecule, and frees "conflated" to try again with a different boundary.

WHY When both halves behave identically, the split was pointless — collapse it and try a different boundary

Unify and Retry

"Same" is the undo operation. It detects that two molecules created by a split have statistically indistinguishable outcome records. Neither half is more predictable than the original was. The split consumed resources — two molecules competing where one used to — without buying any predictive improvement. The correct response is to merge them back and free the conflation detector to search for a different splitting boundary.

The three-operation cycle — conflated, distinction, same — is self-correcting. Split on the wrong boundary, detect the failure, collapse, try again. Split on the right boundary, detect the success, confirm. The mind doesn't need to get the split right on the first try. It needs the cycle to be fast enough that wrong splits are quickly undone and right splits are quickly confirmed.



TECHNICAL STUFF

The meta-operation menu. Six meta-molecules, each gating a specific corrective action: **stuck** (park an unanswerable thought), **looping** (park a circular chain), **bored** (double novelty for 5 ticks), **conflated** (split a bimodal thought on a boundary condition), **distinction** (confirm a successful split), **same** (collapse a useless split). Each fires only when its meta-molecule wins a dedicated meta-slot. The **parking rule** spares active goal wants and any thought whose missing filler exists somewhere in the pool, the facts register, or storage.

Suggested Experiments

These experiments test whether the meta track correctly detects, splits, and self-corrects without falling into the Mirror Trap.

EXPERIMENT VI.1

The Mirror Trap

Setup: Form meta-molecules `"meta:resolved:q1"` and `"meta:resolved:q2"` when their respective questions close. Now remove q2's answer from the pool, causing q2 to re-open by resonance (which refutes the "resolved" prediction for q2's meta-molecule). Keep confirming q1's resolution.

Observe: (1) Which meta-molecule lives longer in the pool? (2) Does q2's refuted meta-molecule die when its prediction fails? (3) Run with *meta_funding = True* (predictions refresh/drop the meta-molecule) versus *False* (scored but no effect on lifetime).

Compare: Prediction-funded meta versus score-only meta.

Discover: Does the prediction mechanism correctly kill meta-molecules whose claims about the base stream turn out to be wrong?

EXPERIMENT VI.2

The Conflation Split

Setup: Re-present `"call_her_thought"` every tick. Alternate all-rain episodes (bad outcome) with all-sun episodes (good outcome) so the outcome record is bimodal.

Observe: (1) Does `"meta:conflated:call_her_thought"` form? (2) What boundary does it split on — weather, or some other available condition? (3) After the split, what is the accuracy of each half's outcome record?

Compare: Meta on (conflation can form and trigger splitting) versus *meta_off = ("conflated",)* (conflation detection disabled).

Discover: Does the split thought predict each weather condition's outcome correctly? And with conflation disabled, does the bimodal record persist forever as an unresolvable confusion?

EXPERIMENT VI.3

The Wrong Split

Setup: Same bimodal stream as Experiment VI.2, but add a noisy decoy: time of day alternates alongside weather, but *independently*. The true boundary is weather; time of day is a red herring.

Observe: (1) If the mind splits on time of day first, do both halves remain bimodal? (2) Does "same" detect the useless split and collapse it? (3) Does "conflated" re-fire and try weather as the boundary? (4) How many split-collapse-retry cycles does it take to find the true boundary?

Compare: Environment with one candidate boundary versus environment with one true boundary and one decoy.

Discover: Can the self-correcting cycle (conflated → same → retry) recover from a wrong first guess? And does adding more decoys slow recovery linearly or exponentially?

EXPERIMENT VI.4

One-Way Aboutness

Setup: A mildly active thought *X* is in the pool. A meta-molecule `"meta:stuck:X"` forms and references *X*. Run two conditions: (1) one-way aboutness (meta references *X* but *X*'s bid is unaffected) and (2) bidirectional aboutness (the meta-molecule boosts *X*'s bid by a factor).

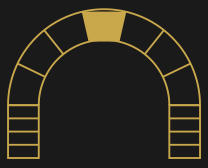
Observe: (1) Under one-way, does *X*'s activation stay flat when the meta-molecule wins? (2) Under bidirectional, does *X*'s activation escalate as noticing boosts its bid and prominence boosts the meta-molecule? (3) How many ticks until the bidirectional version dominates the entire pool?

Compare: One-way versus bidirectional aboutness.

Discover: How quickly does the bidirectional feedback loop take over? And in the one-way condition, does the meta track successfully observe without distorting?

Key Terms

Meta-molecule	A thought molecule about the system's own processing — stuck, looping, bored, conflated, distinction, same — competing in dedicated meta-slots.
Organ federation law	ABI's architectural rule: components may communicate only through molecules, never by reading hidden state.
Mirror Trap	The failure mode where a meta-molecule refreshes on re-detection, becoming a delayed copy of the base stream — rent without yield.
5-tick prediction	A meta-molecule's claim about the base stream's state 5 ticks in the future — confirmed predictions extend its life, refuted ones kill it.
One-way aboutness	A meta-molecule can reference a base thought, but the base thought's bid is unaffected — observation without distortion.
Meta-slots	Dedicated tournament slots (3, 2, 2) for meta-molecules, separate from main thought slots — preventing mutual crowding.
Conflated	A meta-state triggered by bimodal outcomes (≥25% good and ≥25% bad) — the signal that a concept hides two situations under one label.
Splitting	Dividing a conflated molecule into two contextual halves on a candidate boundary condition.
Distinction	Confirmation that a split was real: both halves are unimodal and point in opposite directions.
Same	Detection that a split was pointless: both halves have indistinguishable outcome records — triggering collapse and retry.
Parking	The corrective action for stuck or looping: removing the offending thought from active competition while sparing goal wants and thoughts whose fillers exist in the system.
Bored	A meta-state that doubles the novelty weight for 5 ticks, pushing the system toward unexplored territory.



THE MUSEUM

The Gallery of Outlines

A white room. Clean walls, bright light, no shadows. After the dim of the Alcove and the Mirrors this is almost painful.

In the center, a single opening on a stand. It is the kind I saw in the Gallery of Openings — rods and joints turned inward around an absence — but this one is surrounded by rods that are not part of it. They stand around it like scaffolding, each connected to the opening's edge by a thread of a different color. Five colors. Five threads. The opening is defined by what surrounds it.

An attendant stands beside the stand. Without speaking, they remove one rod and its thread. The opening changes — not its size, but its shape. The edge that rod held is less certain now, blurring slightly where the thread was. They remove another. The shape blurs further. They remove a third, a fourth. With each removal the opening becomes vaguer. When the last rod is gone and the last thread cut, there is no opening. There is nothing. Not a hole. Not a gap. Nothing at all. The stand is empty.

The attendant replaces the rods, one by one. The opening returns, sharper with each thread restored.

A placard on the wall:

AN OPENING AND ITS SCAFFOLD
Five rods, five threads, one absence. Assembled and disassembled.

An opening has shape only because of what surrounds it. Remove the surrounding bonds and the opening ceases to exist. Not-knowing is not empty. It is made entirely of knowing.

Each thread is a different kind of bond. The five kinds are what the museum tracks for every opening that something depends on: what brings it about; what it brings about; under what conditions; what appears alongside it; what part it plays.

Most openings in this museum have none of these threads. They exist, and they pull, but nothing depends on them. Threads are given only to the openings that matter.

I look around the room. Other openings are mounted along the walls, some with five threads, some with three or two, some with none. The ones with all five are the sharpest — I can see their shape from across the room. The ones with none are vague blurs, barely distinguishable from the wall.

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In the far corner, two structures face each other across a narrow gap.

The first has an opening it cannot fill. Five threads surround it; it is well-shaped, and its pull is strong. The second structure holds a fill of exactly that shape. The fill sits in the second structure's own opening, snug and funded — I can see its thick stack of slips.

The first structure leans toward the second. Something passes between them. Not the fill — the fill stays where it is. What passes is the outline of the opening. I watch it cross the gap: a thin, ghostly shape, the exact contour of the absence, drifting from the first structure to the second.

Something comes back. It is lighter than the fill. Thinner. It settles into the first structure's opening and seats partway — the rods relax a little, but not all the way. There is still a pull, though less than before. Beside the new fill, a small stack of slips. I pick one up. **0.5**.

I look at the fill that stayed behind in the second structure. Its slips read **1.0**. The second structure earned it. The first was told.

The first structure sits with its half-seated fill, its reduced pull, its thin stack. It knows something now. It does not own what it knows.

◆ ◆ ◆

Along the back wall, a row of openings on stands, each with a different number of threads. Beside each one, a small window showing what the opening is asking. The opening with five threads shows a single question — one thread lit brighter than the others, pulsing gently. The one beside it shows a different question, also single. Each opening asks one thing at a time.

I find the opening with the most unfunded threads — three of its five are dim, their rods loose, their shapes vague. Its one lit thread is not the dimmest. It is the one with the most evidence waiting nearby: a cluster of slips drifted close, almost touching the thread, as if the answer were in the room and the opening knew it.

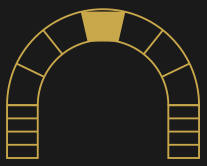
A placard below the row:

> **ONE QUESTION AT A TIME**
> *Openings with threads, each posing one.*
>
> An opening with several empty threads could ask about all of them. It does not. It asks about the one where the most evidence has gathered — the thread closest to being funded. Fill that thread and the next most answerable becomes the question.
>
> A crossword is filled the same way. Not the hardest clue first. The easiest, and each answer makes the next one easier.

I watch the cluster of slips drift closer. When enough of them touch the lit thread, it brightens fully, its rod tightens, and the opening's shape sharpens by one edge. The thread goes steady. A different thread begins to pulse.

On my way out I pass the two structures again. The first still leans toward the second, but less. Its half-seated fill is something. Not everything. Not nothing.

The cloth hangs by the door. It lifts, and I catch myself watching to see if it will swing twice again, the way it did in the Room of Returns. It swings once and settles. I am carrying questions about the cloth now, the way the openings carry their threads.



SECTION

7

Understanding Register

The Shape of What You Don't Know

IN THIS SECTION

- Seeing why ignorance isn't empty — it's a geometric structure made of known bonds around an open slot
- Learning how five typed relations map the causal neighborhood of every important concept
- Understanding why only concepts that matter earn an SOV — and how they're funded
- Discovering why posing one question at a time beats posing five
- Watching a question close when the pool fills its slot

WHAT IF "Not knowing" is a geometric structure made entirely of known bonds surrounding an open slot

Think about what you know about a disease you've never heard of. Nothing. No structure, no shape, no bonds — just blankness. You can't ask a good question about it because you don't have enough context to know what a good question would be.

Now think about what you know about a disease you've *partially* studied. You know it's caused by a virus. You know it affects the lungs. You know certain populations are more vulnerable. You know there's a treatment, but you don't know what the treatment is. That "don't know" isn't empty. It's **shaped**. It sits inside a network of known bonds — causes, effects, conditions, co-occurrences — and the open slot has a specific type: *treatment*. Remove all the known bonds and the "don't know" vanishes. The ignorance was made entirely of knowledge.

What if "not knowing" is never a void, but always a geometric structure — a constellation of known causal and contextual bonds surrounding an open slot whose type is determined by the bonds themselves?

WHY Ignorance has shape only because knowledge gives it shape — an open slot without surrounding bonds is just absence

Remove the Bonds and Nothing Remains

This is a deep claim about the nature of questions. A question like "what causes this disease?" isn't a free-floating entity. It exists because you already know the disease exists, that diseases have causes, and that this particular cause is unknown to you. Strip away any of those bonds and the question dissolves. "What causes [nothing]?" isn't a question. "What [nothing] this disease?" isn't a question. The open slot needs its scaffolding.

Gerundelle formalizes this with the **Understanding Register**: a structure that tracks, for every important concept, which causal and contextual relations are known, which are partially known, and which are open slots. The register doesn't store answers. It stores the *shape of the question* — the surrounding bonds that make the open slot meaningful. Understanding isn't a number from 0 to 1. It's a map of funded and unfunded relations.

REMEMBER

Ignorance in Gerundelle isn't the absence of knowledge. It's a **structured open variable** — known bonds surrounding an open slot. Remove the bonds and the ignorance disappears, because there's nothing left to not-know about. A good question is one whose slot is well-shaped by its surrounding bonds.

LINEAGE

The shape of not-knowing: a thin lineage. Structured ignorance has surprisingly little prior art, which is itself interesting. The nearest kin: Rumsfeld's "known unknowns" (folksy but structurally right — an unknown you can name is shaped by what you know around it), **epistemic logic**'s awareness problem (how can a reasoner reason about what it doesn't know without already knowing it?), and **George Loewenstein**'s *information-gap theory of curiosity* (1994), which proposed that curiosity is the feeling of a perceived gap between what you know and what you want to know. Gerundelle's SOV is Loewenstein's gap given geometry: not just "I don't know something" but five typed relation slots with specific shapes, funded at specific confidence levels, each gap surrounded by specific known bonds. The absence of a mature lineage here is quietly confident: this section is where the book is most alone.

WHAT IF SOV molecules track five typed relations: causes, caused_by, conditions, co-occurs, role

Knowing that "rain is associated with getting wet" is useful. Knowing that "rain *causes* getting wet" is more useful. Knowing that "rain causes getting wet *unless you have an umbrella*" is more useful still. Each step adds a different *type* of relation — not just more data, but a different kind of structural bond.

What if the Understanding Register tracks five distinct relation types for every important concept? Not a single "understanding score," but a structured map: what **causes** this concept to appear? What does it **cause** in turn? Under what **conditions** does the causal link hold? What **co-occurs** with it without being causally related? And what **role** does it play in larger structures?

WHY Five typed relations let understanding scale MCTS weights, boost curiosity, guard against rationalization, and hedge speech

Proportional Confidence Everywhere

Scalar confidence scores — a single number from 0.0 to 1.0 representing "how well understood" a concept is — are the standard in calibrated AI systems. Two concepts both at 0.6 look identical. But one might be missing its causal mechanism (it knows the conditions but not the cause) while the other is missing its conditions (it knows the cause but not when it applies). These are radically different kinds of ignorance requiring radically different next experiences. A scalar conflates them.

Gerundelle's five relation types aren't just descriptive — they're **load-bearing** throughout the architecture. Each relation has a confidence level based on how it was funded (lived co-wins are stronger than told explanations, which are stronger than co-occurrences). That confidence scales the system's behavior proportionally, everywhere it matters.

In **action selection** (MCTS), recognition weights are scaled by the understanding confidence of the relevant conditions — a poorly understood condition gets less influence over decisions. In **curiosity**, poorly understood goals get a boost: the system is pulled toward filling the gaps in concepts it cares about. In **rationalization**, a partner understood below 50% can't serve as a closure target — you can't rationalize with something you don't understand at all. In **speech**, partially funded relations are hedged: the agent says "rain might cause getting wet" instead of "rain causes getting wet" when the causal relation is only partially confirmed.



TECHNICAL STUFF

The five relation types. For each important concept *C*, the Understanding Register tracks: (1) **causes** — what makes *C* appear, (2) **caused_by** — what *C* makes happen, (3) **conditions** — when the causal links hold or break, (4) **co_occurs** — what appears alongside *C* without a confirmed causal link, (5) **role** — what functional position *C* occupies in larger structures. Each relation has a confidence funded through lived co-wins, temporal arrival patterns (1.5×–2× base rate, 60% follow rate), splits from the meta track, and told explanations.

WHAT IF Understanding is funded through lived co-wins, temporal arrivals, splits, and told explanations

How do you come to understand that rain causes getting wet? Not by being told once. Not by seeing rain and wetness in the same room. You understand it through **multiple channels of evidence**, each contributing different confidence. You've been rained on and gotten wet (lived co-win). You've noticed that wetness reliably follows rain with a short delay (temporal arrival). You've split "being outside" into "outside when raining" and "outside when dry" and confirmed the split (meta-track split). And your parent told you to bring an umbrella when it rains (told explanation).

What if each funding source contributes a different amount of confidence — and the question of which concepts earn an SOV at all is answered not by a hand-curated trigger list, but by which ownership slots are paying for understanding?

WHY Giving every concept an SOV bloated BigWorld from 70 clauses to 201 and broke 13 tests

Only Concepts That Matter

This question has two traced failures and two fixes.

The first failure was universal coverage. The first implementation gave every perceived clause its own SOV molecule, each posing up to five questions in the main thought slots. BigWorld's pool bloated from 70 clauses to 201, the tournament became impossibly crowded, and 13 tests broke. **The first fix** was three rules: dedicated SOV slots (3, 2, 2), one question per top win, and a hand-curated trigger list — only concepts that appeared in an outcome, produced high surprise, served as a goal, or triggered conflation.

The second failure was the trigger list itself. It worked, but it never answered the deeper question: matters *to whom*, and *for what*? The list was a patch — four conditions chosen by the designer because they seemed important. **The second fix** is the `serves` link from the Territory chapter: concept SOVs are funded by the ownership slots that need them. The carpenter's understanding of wood grain is deep because a control slot is

paying for it. Plumbing is shallow because no slot pays. When the paying slot closes, the investigation stops. The trigger list is replaced by a structural answer: understanding is purchased by purpose.



WARNING

Two failures, two fixes. The first failure was breadth (every concept questioned); the first fix was a trigger list (only important concepts). The second failure was the trigger list itself (hand-curated, no structural answer to *why* a concept matters); the second fix is the `serves` link (ownership slots fund concept SOVs). The `serves` link operates in practice in the fiction ladder (Chapter 11), where ownership slots fund inquiry on each rung — the lean arm's want opens a control slot that pays for the conditions investigation. The trigger list remains in Gerundelle's main code as the current formal implementation. Both are visible; the fiction ladder is where the `serves` link proved its worth.

WHAT IF An SOV molecule poses one question per top win — the most answerable slot with the most evidence

You're trying to understand why your car won't start. There are five things you don't know: what's wrong with the engine, when it last worked, whether the battery is dead, what the temperature was last night, and who last drove the car. You can't investigate all five simultaneously. You pick the one where you have the most evidence and the best chance of making progress: the battery, because you noticed the dashboard lights were dim.

What if an SOV molecule, when it wins a top slot, poses exactly one question — the open relation slot with the most accumulated evidence waiting to fund it? Not the most important question, not the most urgent, but the most *answerable* given what the system already has.

WHY Posing one answerable question per win beats posing five because it focuses the pool on what can actually be resolved

The Most Answerable Question

The "most answerable" heuristic does two things. It concentrates the pool's attention on a slot that's close to being funded — meaning the work to fill it is small and the payoff is immediate. And it prevents the SOV track from scattering attention across five half-answerable questions, none of which gets enough bandwidth to resolve.

When the question is answered — when the pool's tournament resolves the open slot's partner and writes the result back into the Understanding Register — the SOV molecule's next top win will pose a *different* question: the new most-answerable slot.

Understanding fills incrementally, one relation at a time, steered by what the system already almost knows. The most mysterious slot doesn't get asked until enough surrounding bonds are in place to make it answerable.

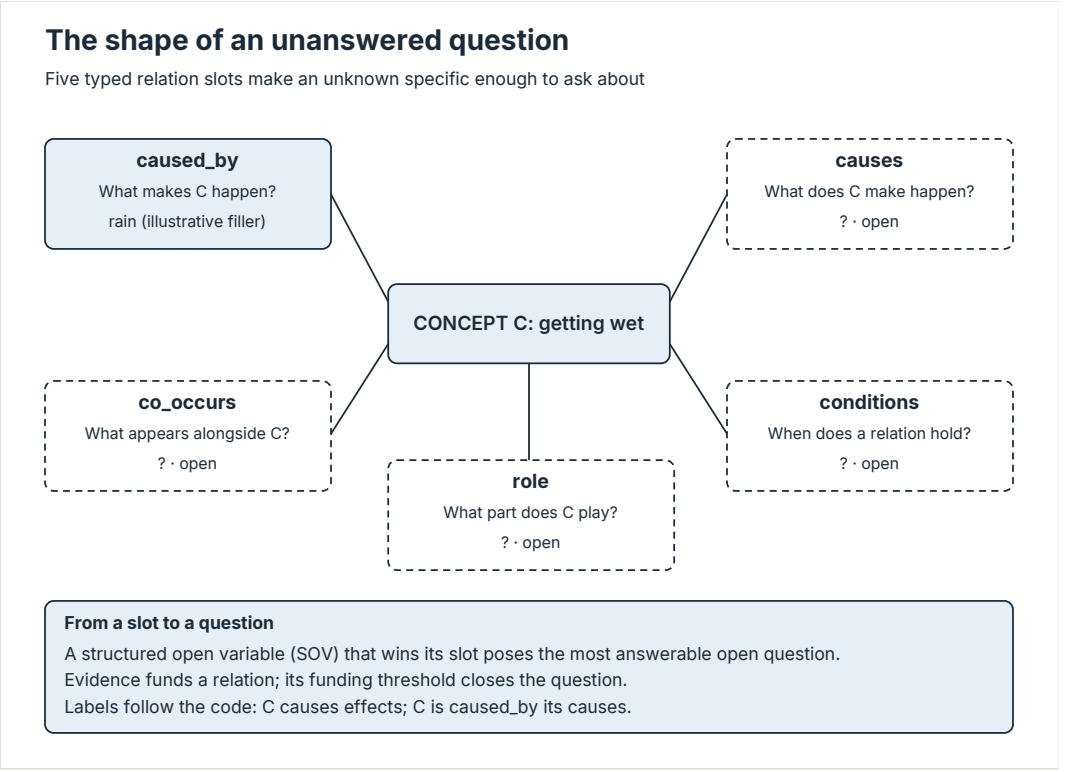


Figure 11 **The shape of an unanswered question.** Five relation types surround a concept. Filled and dashed open slots distinguish relations from questions. Cause/effect directions follow the implementation.

TIP

Think of it like filling a crossword puzzle. You don't stare at the hardest clue with no letters filled in. You fill the easy ones first, and each answer provides crossing letters that make harder clues progressively answerable. The SOV track works the same way: **answer what you almost know, and the hard questions become easier.**

WHAT IF A question closes when the register fills its slot

An open question in the Understanding Register is a specific typed slot — say, `causes: ?` for the concept "getting wet." The SOV molecule poses this question when it wins a top slot. The pool's tournament then works on it: relevant molecules compete, bind, accumulate evidence. At some point, the evidence crosses a funding threshold — enough lived co-wins, temporal arrivals, or told explanations to confirm "rain" as the filler for the `causes` slot.

What if the question simply closes at that point? The pool worked it out. The register records the answer. The SOV molecule's `causes` slot is now funded, and its next top win will pose a different open slot. No separate "answer accepted" mechanism is needed — the threshold crossing *is* the closure.

WHY The tournament's competitive process fills SOV slots naturally — closure is the funding threshold being met

The Pool Writes It Back

This design keeps the Understanding Register tightly coupled to the tournament without adding special machinery. Questions are posed by SOV molecules winning slots. Answers are discovered by the pool's normal competitive process. Closure happens when the evidence crosses a threshold. The register fills the same way everything else in Gerundelle works: through competition, evidence, and convergence.

A fully understood concept — all five relation types funded above threshold — has nothing left to ask. Its SOV molecule, when it wins a slot, finds no open questions to pose. It contributes nothing and naturally loses future competitions to SOV molecules with actual open slots. Understanding, once complete, goes quiet. Curiosity flows toward what's still unknown.



REMEMBER

The Understanding Register fills one slot at a time, steered by answerability. A question is an **open relation slot**. An answer is the slot's **funding threshold being met**. A fully understood concept **asks nothing** and loses SOV competitions to concepts that still have open slots. Understanding doesn't hoard attention — it releases it.

The Inquiry Repair

Building the Understanding Register exposed a sequence of failures beyond the bloat fix. Each one was repaired, and the repairs themselves became part of the architecture.

SOV holes suggest experiments by slot. An open relation slot doesn't just pose a question — it generates an experiment molecule that competes in a dedicated experiment track. The experiment specifies which change to make and what to observe. This keeps the inquiry concrete: not "learn about rain" but "try the push with the latch parted and see if the outcome changes."

Conditions experiments start with untried situations. A conditions experiment picks its first step from changes the register says will create a situation the push has been tried in fewer than twice. Before this fix, the first change was picked alphabetically — the latch was never parted before a lid push. After the fix, it is.

Progress momentum. An investigation that just received an outcome — any outcome, confirming or contradicting — keeps its pull for a few ticks. Without momentum, the SOV wins the top, runs one experiment, gets a result, and immediately loses the top to novel distractors. The follow-up experiment, which is now maximally informative, never runs. Under the territorial reorganization, progress momentum moves from the concept to the slot instance: the ownership slot that funded the inquiry is the one that keeps its pull.

THE QUESTION BOTTLENECK

A Worked Example



In the curriculum's catch world, gust concepts — lid_releasing, flag pushes — won attention on novelty, while the lid push's half-understood SOV lost the top of the SOV track. Its pull was T × (1 - level), and at 60% understood, the pull was weak. The lid got one try with the latch parted, but a conflation needs two tries each way. The SOV reached the top 3–5 times in 150 ticks — not enough.

Under curiosity-first, this is correct behavior: the gust concepts *are* more novel. Under the territorial reorganization, the lid is a **control slot** — a claim that didn't take. Control pulls harder than unclaimed. The flag ends up contested (a timer also affects it) or a border, so no ownership slot pays for its concepts. The question bottleneck resolves not by tweaking the SOV formula, but by asking who's paying for the investigation.

Suggested Experiments

These experiments test whether the Understanding Register fills correctly from raw experience, whether it asks the right questions, and whether fully understood concepts go quiet.

EXPERIMENT VII.1

Working It Out

Setup: The agent observes: rain and clouds co-occur on even ticks, getting wet follows rain on odd ticks. The agent has never been told any explanations. SOV molecules are cued for "rain" and "getting wet."

Observe: (1) What relations does the Understanding Register work out? (2) Does it discover that clouds co-occur with rain and that rain causes getting wet? (3) What is the understanding level for `"it_is_raining"` after 40 ticks?

Compare: *sov_cues = 2* (SOV molecules active, working it out) versus *sov_cues = 0* (no SOV molecules — raw records accumulate but nothing poses questions).

Discover: With the same raw experience records, does the active SOV track fill the Understanding Register while the passive version leaves it empty?

EXPERIMENT VII.2

The Most Answerable Question

Setup: Rain is poorly understood (all five relation slots open). Present rain + clouds on even ticks, getting wet on odd ticks. Track which question the SOV molecule poses at its first top win.

Observe: (1) Does the first question come from the slot with the most accumulated evidence? (2) Is it `co_occurs: ?` (clouds) or `caused_by: ?` (getting wet) depending on which has more evidence? (3) After that slot fills, does the next top win pose a different slot?

Compare: Rain poorly understood (many open slots) versus rain fully understood (all relations funded).

Discover: Does a fully understood concept ask nothing? And does the poorly understood concept's question sequence follow the evidence rather than a fixed order?

EXPERIMENT VII.3

The Alternation Illusion

Setup: Rain and getting wet strictly alternate every tick: rain, wet, rain, wet, rain, wet. The agent's temporal arrival tracker sees that getting wet reliably follows rain (correct) *and* that rain reliably follows getting wet (spurious — an artifact of the alternation).

Observe: (1) Does the SOV work out both `rain causes getting_wet` and the spurious `getting_wet causes rain`? (2) When the environment switches to randomized episodes (rain and wet no longer alternate perfectly), does the spurious relation retract? (3) How many randomized episodes does it take for the Anneal mechanism to heal the illusion?

Compare: Strict alternation versus randomized timing.

Discover: Can an agent using observational arrival counts distinguish true causation from a periodic cycle? And does upkeep (retracting relations when new evidence dilutes the old) allow recovery from early illusions?

EXPERIMENT VII.4

Understanding Guards Everywhere

Setup: A concept C has understanding = 0.3 (poorly understood). Test four downstream effects: (1) An MCTS recognition involving C as a condition — does its weight get scaled down? (2) C appears as an open goal — does it get a curiosity boost? (3) A molecule tries to rationalize-close on C — does the 50% guard block it? (4) The agent explains C in speech — does it hedge?

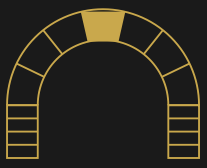
Observe: (1) MCTS weight for the recognition at understanding 0.3 versus 0.8. (2) Curiosity boost magnitude at 0.3 versus 0.8. (3) Whether rationalization is blocked at 0.3 and permitted at 0.6. (4) Whether speech output hedges at 0.3 ("might cause") and asserts at 0.8 ("causes").

Compare: Proportional confidence scaling versus binary (understood/not understood).

Discover: Does the five-relation understanding register produce meaningfully different behavior at different confidence levels — not just on/off, but proportionally?

Key Terms

Structured Open Variable (SOV)	A molecule tracking the understanding of a concept — five typed relation slots that can be open, partially funded, or fully funded.
Understanding Register	The collection of all SOV molecules and their funded/unfunded relation slots — a map of what the agent knows, partially knows, and doesn't know.
Open slot	An unfunded relation in the Understanding Register — a specific, typed gap in the agent's knowledge with a shape determined by surrounding funded bonds.
Five relation types	causes, caused_by, conditions, co_occurs, role — the structural map of every important concept's causal neighborhood.
Funding	The evidence accumulation that fills a relation slot: lived co-wins, temporal arrivals (1.5×–2× base rate), meta-track splits, and told explanations.
SOV slots	Dedicated tournament positions (3, 2, 2) for SOV molecules, separate from main thought and meta slots.
Most answerable	The heuristic for choosing which open slot to pose: the one with the most accumulated evidence waiting to fund it.
Answerability	How close an open slot is to its funding threshold — determines which question the SOV molecule poses next.
Proportional confidence	Understanding scales behavior everywhere: MCTS weights, curiosity boost, rationalization guard, speech hedging — not binary, but graded.
Anneal	The upkeep mechanism that retracts relations when new randomized evidence dilutes earlier observational patterns — healing early illusions.
Told explanation	A relation funded by testimony rather than lived experience — contributes less confidence than direct observation.



THE MUSEUM

The Gallery of Held Things

I have been waiting for this room. The placard in the Gallery of Openings said the measure of pull would be shown here. That was many rooms ago.

◆ ◆ ◆

In the center, a single opening on a stand — the same kind as the others, rods and joints turned inward. But this one has been filled and emptied many times. I can see it in the rods: each filling left a warmth in the material, a faint discoloration, like a cup that has held tea so often the porcelain is stained. Beside the opening, a tally scratched into the stand: twelve marks.

The fill is out. The pull is enormous. I feel it from across the room, a draw so strong my coat lifts toward it. This is not the mild pull of the Gallery of Openings.

In a quieter case beside it, the same kind of opening, filled once and emptied. One mark on the stand. The pull is small. I hold my hand near both. The difference is not in the openings — they are the same size, the same shape. The difference is in what was held.

A placard between them:

PULL AFTER HOLDING

Two openings, same shape. One held twelve times, one held once. Both empty.

The question from the Gallery of Openings: what measures the pull? Size does not. Age does not.

What was held does.

An opening that has been filled and emptied many times pulls harder than one filled once. The pull is 1 plus what was held. The twelve-times opening pulls thirteen. The once opening pulls two.

This is why the oldest opening in the Gallery of Openings pulled hardest. It was not the oldest. It was the most held.

My hand aches. It ached in the Gallery of Openings and I did not know why. Now I know: the oldest opening had been held many times, each holding leaving warmth in the rods, and the emptiness after all that holding is the strongest pull in the museum.

◆ ◆ ◆

Against the far wall, a stretch of floor marked into regions — rectangles outlined in brass, like plots in a garden. Some are occupied: a structure sits inside each, leaning over the brass line toward its neighbors. Some are empty.

Two structures lean over the same region. Each has a reservoir — a glass tube mounted on its side, like the bridge's gauge from the Gallery of Doings. Each time one structure pushes something into the region, the other pushes it back. Each reversal costs the reversed structure a little from its gauge.

I watch. The one on the left has a taller gauge, nearly full. The one on the right is shorter, past half. They go back and forth: push, undo, push, undo. The right structure's gauge drops faster — the region pulls it less, so each undoing costs it more. The left structure barely drops: the region pulls it harder, and the harder pull means each reversal takes less from it. The one that cares more outlasts.

The right structure's gauge empties. It stops leaning. Its rods slacken. It drifts away from the region, and the left structure settles into the space alone.

I look at the placard before I look at the next case.

TWO STRUCTURES, ONE REGION
Brass region, two gauges. Contested.

Each structure acts to hold the region. Each time one undoes the other's hold, the undone structure spends from its gauge. The harder the region pulls a structure, the less each undoing costs it. The one that cares more outlasts.

When a gauge empties, the structure lets go. Nobody decided who would win. The gauges decided.

In the next case, the same two structures, the same region, but the gauges are swapped: the right one is taller now. I watch the same contest play out. This time the left structure's gauge empties first. It lets go. The right one holds.

The ending reversed. The structures are the same. The region is the same. Only the gauges differ, and the gauges are the story.

◆ ◆ ◆

Along the opposite wall, the same structure three times, each reaching for a region it cannot get. The region is across a gap — visible but unreachable. Under each structure a number is carved into the floor.

The first: **2**. It leans toward the region, fails, leans again, fails. On the second failure it slackens. Its other openings — threads I recognize from the Gallery of Outlines — come back into view. It has let the region go, and the rest of its questions have room again.

The second: **6**. It leans longer. Fails six times. On the sixth failure it slackens. Its other openings return.

The third has no number. It leans. Fails. Leans. Fails. I watch for a long time. It does not stop. Its other openings are invisible — crowded out by the leaning, which takes everything.

A placard below the three:

PATIENCE
Three structures, one unreachable region. Three numbers.

None of them is broken. The first lets go after two failures. The second after six. The third does not let go.

The number is how many failures it will take. A structure with a small number moves on quickly. A structure with a large one stays longer. A structure whose other openings are strong lets go sooner, because the other openings pull it back.

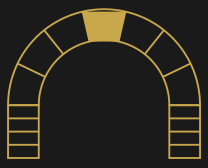
Whether patience is a virtue depends on whether the region can be reached.

I stand in front of the third structure for a while. It is still leaning. Its other openings — if it has any — are invisible under the pull of the one thing it wants.

◆ ◆ ◆

Before I leave, I go back to the twelve-times opening at the center of the room. The placard said: pull is 1 plus what was held. The twelve-times opening, empty now, pulls harder than any full opening I have seen.

The cloth in the corner is still. As I reach the door it lifts — a single swing, long and slow, as though the air in the room shifted. It is the longest swing I have seen.



SECTION

8

Territory and Ownership

Why the Agent Acts at All

IN THIS SECTION

- Seeing why every motivation is a strategy for keeping or extending a perceived boundary
- Meeting the ownership SOV — a new kind of structured open variable with its own slot types
- Understanding why concept SOVs serve ownership slots, not the other way around
- Learning how perceived territory is held, contested, and lost
- Discovering why loss pulls harder than gain, and why the alpha is fragile
- Following territory through other minds: recognition, appreciation, and attribution
- Comparing curiosity and territory as bases for motivation

WHAT IF Every motivation is a strategy for keeping or extending a boundary the agent perceives as its own

An animal defends its den. A scientist guards a research program. A child screams *mine* over a toy. A musician grieves when a song is covered without credit. A retired worker loses purpose when the office stops calling. These look like different phenomena — instinct, ambition, possessiveness, pride, identity. But they share a structure: something was perceived as *belonging to the agent*, and the agent's behavior organized around keeping it.

The physics of Gerundelle already contains the passive form of this principle. A node in the world model persists while its internal forces balance the external ones. A wall stands while cohesion exceeds load. A belief holds while its reserve absorbs contradictions. Persistence is the default outcome of balanced forces.

What if an **agent** is a node that doesn't just persist passively but *acts* to maintain that balance? And what if every motivation — hunger, fear, curiosity, love, status — is a strategy for keeping or extending the boundary the agent perceives as its own?

WHY The agent layer named purpose as a force and never said where it comes from

Where Purpose Comes From

Gerundelle's cognitive architecture — the tournament, the molecules, the surprise-driven learning — explains *how* an agent understands its world. It does not explain *why* the agent acts at all. The earlier chapters treated curiosity as the fundamental drive: open slots invite exploration, epistemic gaps pull attention, surprise reduction funds learning. But curiosity ranks questions without originating wants. It can tell you which question is most informative. It cannot tell you why you should care about any of them.

The force-based world model already has the answer, hiding in plain sight. A node persists while internal forces balance external ones. An agent is a node that *acts* to keep that balance — pushing back against external disruption, extending the boundary where balance holds, and grieving when a held region is lost. Purpose is not a separate primitive to be added to the architecture. It is the active form of the persistence that the physics already describes.



REMEMBER

The physics gives persistence: a node endures while its internal forces hold. The agent gives **purpose**: it acts to keep or extend that holding. Curiosity explains how the agent learns. Territory explains **why it bothers**.

WHAT IF There is one ownership SOV from tick 0, with slot kinds unlike any other SOV

The Understanding Register tracks five typed relations for every important concept: causes, caused_by, conditions, co_occurs, role. These are generic — they work the same way for rain, levers, and phone calls. But ownership is not a generic concept. It has its own structure, its own slot kinds, and its own dynamics.

What if the agent carries a single, special SOV from tick 0 — not earned through experience but present from the start — whose slots are **unclaimed**, **control**, **contested**, **lost**, **uniqueness**, **relation**, and **appreciation**? Not the five generic relations, but a purpose-built map of what the agent holds, what it's trying to hold, and what it has lost?

WHY Generic relations can't represent the dynamics of claiming, holding, contesting, and losing

A New Kind of SOV

A generic SOV can say "rain causes getting wet." It cannot say "the lamp is mine because I made it light up, and I hold that claim on 14 successful interventions." Ownership needs its own vocabulary: a region is **unclaimed** (the agent has never tried to act there), under **control** (the agent has tried — the claim may or may not be taking), **contested** (another force also produces the effect, or blocks the agent's push), or **lost** (the agent's push once worked and no longer does). A critical distinction: unclaimed means *untried*. A tried claim that didn't take is control ("what blocks it?"), not unclaimed. These are not causal relations. They are territorial states.

The ownership SOV has built-in knowns — the agent knows what "control" means from tick 0, the way an organism knows what "pain" means before it has ever been hurt. Its slots are **instanced per region**: the agent doesn't have one control slot, it has a control slot for every region it has ever acted in. And it works as an **edge**, below the slot competition — guiding behavior without occupying a conscious slot, reaching awareness only when a claim fails and the agent needs to ask "why did I want that?"



TECHNICAL STUFF

Slot types of the ownership SOV. Unclaimed: a region the agent has not tried to act in — untried. **Control**: the agent has tried — instanced per region, tracking tried/took. A tried claim that didn't take is control ("what blocks it?"), not unclaimed. **Contested**: another force also acts in the region, either producing the same effect or blocking the agent's push. **Lost**: the agent's push once worked and no longer does. **Uniqueness**: no other agent or force produces the same effect. **Relation**: another mind recognizes the agent's control. **Appreciation**: another mind values what the agent's control produces. A slot closes when a claim takes, confirmed by the world.

WHAT IF Understanding is bought by the ownership slots that need it

In the earlier chapters, concepts earned an SOV when they appeared in outcomes, produced high surprise, served as goals, or triggered conflation. That list was a patch — a hand-curated set of triggers for "this concept matters." It worked, but it never answered the deeper question: matters *to whom*, and *for what*?

What if concept SOVs don't decide for themselves whether they matter? What if they are **funded by ownership slots** through a `serves` link — and the investigation stops when the paying slot closes?

WHY The serves link explains when inquiry stops, why understanding is uneven, and why niches of specialization form

The Serves Link

Curiosity-only agents — whether driven by information gain or learning progress — explore what is most uncertain or most learnable. This predicts roughly even coverage, concentrated at the frontier of learnability. But humans are known to the bone where their claims live and by rumor elsewhere. A carpenter understands wood grain in exquisite detail and has a vague sense of how plumbing works. The unevenness tracks what they *hold*, not what is most learnable.

The `serves` link makes this structural. A concept SOV is funded by the ownership slot that needs it. The carpenter's understanding of wood grain is deep because it serves a control slot — the carpenter's push on wood reliably produces the desired effect, and understanding the grain is what keeps that control reliable. Plumbing is shallow because no ownership slot pays for it. When the paying slot closes (the control is secure, the region is held), the funded concept SOV loses its customer and the investigation stops. Understanding is not pursued for its own sake. It is purchased by purpose.



TIP

Think of a research lab's budget. A lab doesn't study everything equally — it studies what its grants pay for. When the grant ends, the investigation stops, even if there's more to learn. The `serves` link is the grant: ownership slots fund concept SOVs, and the funding dries up when the ownership question is answered.

WHAT IF The territory register holds perceived ownership, and a coincidence claim slips faster than a tested one

You flip a light switch and the lamp turns on. You flip it again and the lamp turns on. After twenty successful flips, you're confident the switch controls the lamp. But what if the lamp turned on the first time by coincidence — someone else flipped a hidden switch at the same moment? You'd claim the lamp on one data point, and the claim would feel just as real until it failed.

What if the territory register holds *perceived* ownership — what the agent believes it controls, not what it actually controls — and a claim held on one coincidence slips faster than a claim held on twenty interventions?

WHY Perceived ownership must be testable and self-correcting, with contested as a state of its own

Holding Strength and the Contested State

Holding strength is computed as *(took + κ·b) / (tried + κ)*, where *took* is the number of times the agent's push produced the effect, *tried* is the number of attempts, *b* is the agent's own base rate of claims taking, and κ is a shrinkage parameter — the same one blame uses. A region held on one coincidence has tried = 1, took = 1, and the shrinkage prior pulls the holding strength toward the base rate. A region held on twenty interventions has tried = 20, took = 20, and the prior barely matters. The coincidence claim slips sooner because it has less evidence behind it.

Contested is not a failure — it's a state of its own. When another force also produces the effect, or blocks the agent's push, the region becomes contested. This is the seed of hierarchy: two agents contesting the same region is how dominance emerges. And it's where "held back" lives — the frustration of a push that meets resistance, the anger of a claim that someone else is blocking.



WARNING

Non-circularity. Ownership slots open from what the agent *perceived*, never from which action the agent chose. The lamp was perceived as responding to the push. The push was perceived as contested by another force. A test that passes because the rules encode the behavior is not evidence — it is a mirror. The experiments must show that the territory register produces behavior the designer did not hand-code.

WHAT IF Loss pulls harder than gain, and the pull is scaled by what was held

Win a dollar and you're pleased. Lose a dollar and you're upset — more upset than winning made you pleased. This asymmetry, documented by Kahneman and Tversky as loss aversion, is usually treated as a bias to be corrected. But what if it's structurally correct?

A held region was paid for — with interventions, with understanding, with time. Losing it loses all of that investment. Gaining a new region costs nothing yet. The asymmetry isn't irrational. It reflects the actual cost structure: what was held has a sunk investment that what is unclaimed does not.

What if the pull order is **lost > control > unclaimed**, and the pull of a lost slot is scaled by how long and how firmly the region was held? The alpha's fragility in one number: the more territory held, the more there is to lose, and the harder loss pulls.

WHY Loss aversion emerges from the cost structure of territory, not from a bias parameter

The Structural Asymmetry

Scalar reward maximization — the standard in reinforcement learning — treats gains and losses symmetrically. +1 and -1 weigh the same. A discovery that solves a problem but is credited to someone else scores as a success. An argument won at the cost of trust scores as a success. The world outcome is positive; the territorial outcome is devastating. Scalar reward has no way to represent the difference.

Gerundelle's ownership slots have a built-in pull order: **lost** slots pull hardest, **control** slots pull next (the drive to extend and defend what's held), and **unclaimed** slots pull least. The pull of a lost slot scales with what was held — how many interventions confirmed it, how long the control lasted. This makes grief, grudges, and defensive behavior *structural features of the architecture*, not hand-coded emotion rules. The alpha — the agent holding the most territory — is also the most fragile, because it has the most to lose.



REMEMBER

Pull order: **lost > control > unclaimed**. Loss aversion isn't a bias — it reflects the actual cost structure of territory. The more you hold, the harder loss pulls. Grief, grudges, and defensive behavior come from the architecture, not from emotion labels.

WHAT IF What counts as territory requires distinctiveness, recognition, appreciation, and attribution — and attribution can be lost without the memory fading

You tell a joke at a party and everyone laughs. A week later, someone retells the joke. They remember the joke. They might even appreciate it. But they credit someone else — or no one. The memory persists; the attribution is lost. Three of the four conditions for territory are intact, and the one that failed is the one that hurts.

What if territory requires four conditions — **distinctiveness** (you did something no one else did), **recognition** (another mind remembers the experience), **appreciation** (another mind values it), and **attribution** (another mind credits it to you) — and each can be lost independently, opening a different kind of lost slot?

WHY Another mind's recognition has three separable parts, and losing any one opens a distinct lost slot

Other Minds and the Three-Part Recognition Record

In Gerundelle, another mind's recognition record has three parts: it **remembers** the experience, it knows **whom to credit**, and it **appreciates** the result. These three parts can come apart. Someone remembers your help but credits your colleague. Someone credits you but doesn't value what you did. Someone values the work but has forgotten it happened.

Each loss opens a different slot, and the three readings compete in the ownership SOV. Losing attribution when recognition and appreciation persist is the specific structure of the joke-retelling sting — and of plagiarism, unacknowledged mentorship, and the slow erosion of a legacy. Losing appreciation when attribution persists is the structure of the artist whose work is credited but dismissed. Losing recognition entirely — no one remembers — is the structure of oblivion.



TECHNICAL STUFF

Relational uniqueness. Not all territory is about novelty. A familiar joke at an irreplaceable moment — your wedding, a hospital visit — is relationally unique even if the joke itself is common. Uniqueness can be tested by probing other minds: does anyone else have this experience, this credit, this appreciation? The pigs-can-fly test: share an absurd idea and then check whether anyone else has it. If no one does, the idea is yours. If someone else independently holds it, the uniqueness slot closes.

The Territorial Readings

Every outcome in the world has a territorial reading — what it means for what the agent holds. The readings are a finite vocabulary, like the 102 gerunds: a closed set of territorial events, each built from the slot kinds and the dynamics. The world reports what happened. The readings say what that is for the agent's territory.

TECHNICAL STUFF

The dynamics that readings are built from. (1) **Perceived holding:** a coincidence counts as a claim until a test fails. The register holds what was perceived, not what is true. (2) **Holding strength:** (took + κ·b) / (tried + κ). A coincidence claim slips faster than a tested one. (3) **Pull order:** lost > control > unclaimed. Lost scales with what was held. (4) **Parking:** only when nothing could fill the hole, and *never* a want. The slot is keyed on the condition that would make it fillable, and reopens when that condition holds. A parked slot does not pull. (5) **Pandora:** an instance that expires unresolved goes to storage open, and reopens when its anchor returns. (6) **Surprise:** reveals where the boundary is open. It *never* funds attention directly — it must open a slot, and the slot is what pulls. (7) **Unclaimed vs. control:** unclaimed means untried. A tried claim that didn't take is control ("what blocks it?"), not unclaimed.

SIDEBAR

Territorial Readings in the Thought World

These readings replace the designer's scalar rewards. The world reports what happened; the reading says what it is for the agent's territory.

Groceries bought: An unclaimed slot — they "become" the agent's. Shopping is the claim; holding them is the claim taking. The slot closes, which is why the agent stops shopping. No depleting reserve and no voice needed.

She answers: The appreciation slot fills — the agent is on her mind.

She doesn't answer: An unfulfilled slot where the agent isn't on someone's mind. Not a failed push: the slot in her mind stays open.

Getting wet: An incursion on the body, the most basic territory. Held strongly, so *contested*, not lost.

Keeping dry with the umbrella: A demonstration of territorial ownership — asserting the boundary. The contested slot closes as a claim, not a border.

The store is closed: Potential ownership off the table. Not a failed claim: the invitation is gone. The groceries slot is *parked*, keyed on the store being open.

Her call missed: A missed unique shared moment — relational uniqueness. It goes to storage open (Pandora) and reopens when she calls again, possibly as regret.

A storm, no umbrella: An incursion on the body — *contested*, as getting wet is. With no umbrella, the boundary can't be defended yet: a control question, "what blocks it?"

The friend isn't at the park: *Parked*, like the closed store, keyed on the friend being present. No pull while parked. The friend's whereabouts is a separate unclaimed slot.

SIDEBAR

Territorial Readings in the Thought World (continued)

Someone else is with my friend: The relation is *contested* — another agent fills the slot. Uniqueness drops. Not loss: the friend is still there.

The greeting not returned: Surprise reveals an opening, so it *opens a slot*: contested (something else now moves the relation). The holding strength decides when contested becomes lost.

I try something new, it fails: The unclaimed slot was tried and the claim didn't take — it becomes *control* ("what blocks it?"), not "stays unclaimed." Unclaimed means untried.

I'm interrupted while working: *Contested* if another agent interrupted, or simply waiting. Not parked — parking is only when nothing could fill the hole, and never a want.

Praised for something I didn't do: Depends on what the agent *perceived*. If it knows the push wasn't its own, no claim was registered and the praise fills nothing. If it doesn't know, it accepts the credit as a coincidence claim, held until a test fails.

Curiosity and Territory: Two Bases for Motivation

The earlier chapters presented curiosity — open slot × epistemic gap — as the fundamental driver of the agent's behavior. The territory chapter reframes this: curiosity is not the base motive, but a strategy serving a deeper one. The comparison deserves to be stated once, cleanly, before the rest of the book references it.

Where they converge: both explore, both use the same cognitive machinery, and the unclaimed slot *is* the curiosity quadrant. Territory doesn't replace curiosity — it says what curiosity serves. Curiosity-first treats exploration as the point. Territory-first treats it as a means to holding something.

SIDEBAR

The Question Bottleneck as Worked Example

In the curriculum's catch world, gust concepts (lid_releasing, flag pushes) won attention on novelty, while the lid push's half-understood SOV lost the top of the SOV track. Under curiosity-first, this is correct behavior — the gust concepts *are* more novel. Under territory, the lid is a control slot: a claim that didn't take. Control pulls harder than unclaimed. The flag ends up contested or a border, so no ownership slot pays for its concepts. The question bottleneck resolves not by tweaking the SOV formula, but by asking who's paying for the investigation.

WHY An opening emptied after many holdings pulls harder than one emptied after few — loss aversion as a formula

Pull After Holding

The pull order (lost > control > unclaimed) says which *kind* of slot pulls hardest. But two lost slots of the same kind can differ in how long and how firmly the region was held. A region held twelve times and then lost pulls far harder than one held once and lost. The formula is **1 + what was held**: a region held twelve times pulls thirteen. A region held once pulls two. The investment is in the rods.

This is why grief scales with what was held, and why the most experienced agent has the most to lose. The alpha — the one holding the most territory — is also the most fragile.

WHY Two agents on one region spend reserves at rates set by how hard the region pulls them

Contested Depletion

When two agents both act to hold the same region, each undoing the other's hold, both spend from a resolve reserve. The rate depends on how hard the region pulls: the harder the region pulls an agent, the less each undoing costs it. The one that cares more outlasts. Swap the reserves and the ending reverses. Nobody decides who wins. The reserves decide.

This is the mechanism of rung 4 in the fiction ladder: Mara and Leo contest the latch. The ending — who gives up first — is a threshold crossing inside a mind, determined by the resolve reserve and the pull of the region, not by an author's choice.

WHY How many failures before a mind lets an unreachable region go is a parameter, not a bug

Patience as a Failure Count

A region that cannot be reached still pulls. An agent with a patience of 2 lets go after two failed attempts; its other openings come back into view. An agent with a patience of 6 holds longer. An agent with no limit never lets go — its other openings are invisible under the pull of the one thing it wants.

Whether patience is a virtue depends on whether the region can be reached. The same number that makes one agent persistent makes another one stuck. In the fiction ladder, this is a temperament trait: different values produce different characters from the same algebra. An impatient Mara never tries the latch she was curious about. A dogged Mara solves it in 26 sentences.

WHY The epistemic contract demands that built and proposed remain visibly separate, and that the falsifying test is named

Status Under the Epistemic Contract

The territory register, slot instances, pull after holding, contested depletion, and patience are **built and measured**: the fiction ladder's seven rungs tested each mechanism with ablations, both on a lean SOV inventory and on Gerundelle's full agent, with every sentence verified at 1.00 round-trip. The ownership-anchored emotional phases are built and measured: the ablation on rung 0 shows narrative-arc coverage drops from 1.00 to 0.75 when the contradicted-belief phase is removed. The `serves` link is built in the fiction ladder (ownership slots fund inquiry on each rung) though not yet as a formal architectural element.

What remains **proposed**: self-originated wants from ownership slots alone (the No Voices test, step 4), the null world (step 5). The falsifying test is named: step 4. If the agent with voice goals removed explores without aim — ranking questions but originating no wants — then territory is not the base motive.



WARNING

Built versus proposed. The territory register, slot instances, pull after holding, contested depletion, patience, and ownership-anchored emotional phases are implemented and measured across the fiction ladder's seven rungs (Chapter 11). The `serves` link operates in the fiction pipeline but is not yet a formal architectural element. Self-originated wants and the null-world experiment remain proposed. The distinction is not cosmetic — it is the epistemic contract in action.



LINEAGE

Self-Determination Theory. Deci and Ryan (1985, 2000) proposed three basic psychological needs: **autonomy** (acting from one's own endorsement), **competence** (effectively interacting with the environment), and **relatedness** (connecting with others). **Sheldon and Elliot** (1999) showed that goals aligned with these needs — *self-concordant* goals — produce sustained effort and well-being, while goals imposed externally produce compliance without engagement. Gerundelle's ownership SOV maps these directly: autonomy is control (the agent's push works), competence is understanding funded by control slots, and relatedness is the recognition/appreciation/attribution triad. The parent's-standard case — an external target met without becoming territory — is Sheldon and Elliot's non-concordant goal, formalized. **Haines and Schutte** (2023) on parental conditional regard extends this to the cost of externally withheld appreciation. In ethology, **Mech** (1999) revised the wolf "alpha" from dominance winner to parent: the alpha holds

LINEAGE (CONTINUED)

the most territory and is the most fragile. **Boehm** (1999) describes the reverse dominance hierarchy in human foragers — coalitions that prevent any individual from holding too much. Gerundelle's frontier trade-off (extending territory increases loss exposure) is the structural version.

Suggested Experiments

Each experiment tests a specific claim about the ownership SOV. The first two are implemented (steps 1–2). The rest are proposed.

EXPERIMENT T.1

The Coincidence Claim (T73, built)

Setup: Lamp B is lit by a timer that happens to fire at the same tick as the agent's push. The agent claims the lamp on one coincidence. Then the timer shifts: the lamp lights without the agent's push.

Observe: (1) Does the agent initially claim lamp B? (2) When the timer shifts, does the holding strength drop faster than a claim held on 20 interventions would? (3) Does the lamp move from control to unclaimed or contested?

Compare: One-coincidence claim versus twenty-intervention claim under the same contradicting event.

Discover: Does the shrinkage prior (κ) correctly distinguish a tested claim from a coincidental one?

EXPERIMENT T.2

The Lost Latch (T74, built)

Setup: The agent controls a latch. The latch stops responding to the agent's push. Meanwhile, an unclaimed region (new lamp) becomes available.

Observe: (1) Does the lost slot (latch) pull harder than the unclaimed slot (new lamp)? (2) Does the agent prioritize recovering the latch over exploring the new lamp? (3) Is the pull order lost > unclaimed maintained?

Compare: Agent with a lost slot versus agent with no prior territory (only unclaimed).

Discover: Does loss aversion emerge structurally from the pull order, without a hand-coded "prefer recovery" rule?

EXPERIMENT T.3

No Voices (proposed, step 4)

Setup: The step-2 thought world with its voice goals removed. The agent receives no external instructions to call, shop, or greet. Only the ownership SOV and the territory register are present.

Observe: (1) Does the agent still act — calling, shopping, greeting — without being told to? (2) Does each action trace to an open ownership slot? (3) If the agent doesn't act, is it because no ownership slot generates a want, or because the want is generated but loses the tournament?

Compare: Voice goals present versus voice goals removed.

Discover: This is the decisive test. If wants arise from ownership slots alone, territory generates purpose. If they don't, territory is relabeling purpose, not generating it.

EXPERIMENT T.4

Two Worlds, One Niche (proposed)

Setup: Two regions. In region A, the agent's push alone produces the effect. In region B, a timer also produces the effect (the agent's push is redundant). Track understanding depth in both regions over 100 ticks.

Observe: (1) Does understanding deepen in region A (where the agent's push is unique) and stay shallow in region B (where a timer also does the work)? (2) Does the `serves` link correctly route concept SOV funding to region A's ownership slot and not region B's?

Compare: Unique control versus redundant control.

Discover: Do niches of specialization form — deep expertise where the agent's contribution is irreplaceable, and thin knowledge where it isn't?

EXPERIMENT T.5

The Null World (proposed)

Setup: A world with nothing claimable. Every effect is produced by timers or other agents. The agent's push never reliably produces an effect. No control slot can be filled.

Observe: (1) Does the ownership graph show the collapse before behavior changes? (2) Does the agent explore and then stop, or does it never start? (3) What does the concept SOV track look like — are any concepts funded, or does every `serves` link point to an unclaimed slot that never advances?

Compare: The null world versus a world with one claimable region.

EXPERIMENT T.5

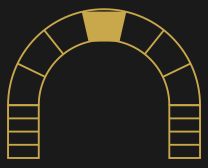
The Null World (proposed) (continued)

Discover: Is the collapse visible in the ownership graph before it's visible in behavior? And does adding one claimable region to the null world restart the agent's engagement?

Key Terms

Ownership SOV	A special structured open variable present from tick 0, with slot types (unclaimed, control, contested, lost, uniqueness, relation, appreciation) instanced per region.
Territory	A region the agent perceives as responding to its own push — held, contested, or lost.
Unclaimed	A region the agent has not tried to act in — untried. Once tried, it becomes control regardless of whether the claim took.
Control	The agent has tried to act in this region — the claim may or may not be taking. A tried claim that didn't take is control ('what blocks it?'), not unclaimed.
Contested	Another force also acts in the region, either producing the same effect or blocking the agent's push — the seed of hierarchy.
Lost	The agent's push once worked in a region and no longer does — the state that pulls hardest.
Holding strength	(took + κ·b) / (tried + κ) — a shrinkage estimator that lets coincidence claims slip faster than tested ones.
Serves link	The connection from a concept SOV to the ownership slot that funds it — understanding is purchased by purpose.
Pull order	lost > control > unclaimed — the structural source of loss aversion.
Distinctiveness	The agent did something no other agent or force did — one of four conditions for territory.
Recognition	Another mind remembers the experience — one of four conditions for territory.
Appreciation	Another mind values the result — one of four conditions for territory.
Attribution	Another mind credits the result to the agent specifically — separable from recognition (they remember but credit someone else).

Relational uniqueness	A familiar act at an irreplaceable moment — uniqueness from context, not from novelty.
No Voices test	The decisive step-4 experiment: remove external goals and check whether the agent still acts from ownership slots alone.
Null world	A world with nothing claimable — the test for whether the ownership graph collapses visibly before behavior does.
Parking	A slot suspended because nothing could fill it now — keyed on the condition that would make it fillable. A parked slot does not pull. Never applied to a want.
Territorial reading	What an outcome means for the agent's territory — a finite vocabulary built from the slot kinds and dynamics, replacing designer-specified scalar rewards.
Perceived holding	The territory register holds what the agent perceived, not what is true. A coincidence counts as a claim until a test fails.



THE MUSEUM

The Gallery of Crystals

A gallery the color of amber. The light is low and warm and everything in here glows a little, as though the room is holding heat from an earlier fire.

In the first case, a structure tries to hold a region — a brass rectangle like the ones in the Gallery of Held Things. It pushes something into the region. Something pushes it back. The structure pushes again. Pushed back again. On the second failure, something forms at the opening the structure has been pushing through: a small, angular crystal, dark red, growing out of the rods the way ice grows on a branch. It did not arrive from outside. It formed in place, where the failing was.

I look at the crystal closely. It is attached to the opening, anchored to the rods that touch the region. It is warm to the glass. Below it, the structure's attention has shifted: it is no longer leaning toward the region the same way. It is leaning toward the thing that blocked it. The crystal changed what the structure looks at.

Elsewhere in the gallery, a structure that held a region for a long time. I can count the marks on the stand — nine holdings. Then something undid it. The region is empty now, and at the opening where the holding was, a crystal has formed: larger than the first, deeper in color, heavier. Its weight is in proportion to what was held. Nine holdings make a crystal I can feel through the glass.

A third crystal, in a bright case: small, pale, almost white. It formed the moment a region yielded to the structure's push — a claim that took. This crystal does not weigh anything. It is light itself.

A fourth, pointing outward like a finger. Not toward the region. Toward another structure on the other side of the case. The structure that undid what was held. The crystal points at who did it.

Only after I have watched them form do I find the inscriptions at the bases, small letters in the museum's careful hand:

frustration

sadness

joy

anger

The names arrive after the things.

◆ ◆ ◆

In the center of the gallery, a structure that lost a region it had held for a long time. Two crystals formed at the loss. The heavy one — the sadness crystal, deep and dark, proportional to what was held. And beside it, the pointing one — anger, aimed at the structure that took the region away. Both were sealed into cases when the structure drifted from the region and the half-lives ran down.

The sadness crystal went in closed. Its sites are filled: the loss is complete, the holding is over. It is a memory. Its case sits quietly on the shelf, words cooling.

The anger crystal went in open. Its sites are still jagged, unfinished — the blame was never resolved. I can see the open edge through the glass.

Time passes. The cases sit on their shelves. Their words cool.

Then the structure that caused the loss drifts back into view. It passes near the cases. The anger crystal's case reopens — its anchor is here, the words warm — and the crystal rises out, not as bright as it was, but warm, its open sites still open. The anger that was stored is the anger that returns. Not a new crystal. The old one, still unfinished. The sadness crystal's case does not open. It is complete. There is nothing unfinished in it.

In the next case, the thing that was lost comes back. It is returned. The pointing crystal — the anger, the one with the open sites — begins to dissolve. Its edges soften, its color fades, and it settles into a pale residue on the rods. The placard beside it: *forgiving*. The blame has no reason left. Beside it, the heavy crystal closes too: the region is held again, and the grief is done.

In a case further on, the same kind of return — but the case has been closed so long that every word on it has gone cold. The lost thing comes back. Nothing happens.

The crystals inside do not stir. The cases do not open. Too much time has passed. The warmth is gone. The placard: *forgetting*.

◆ ◆ ◆

The last case in the gallery is behind a rope, set apart from the others.

A structure with one crystal — the dark red kind, the frustration kind, anchored on a blocked region. While I watch, something reaches the structure from outside. Not an answer. A voice — thin, carried on a slip of paper: *the weather did it*. A second crystal forms, pointing at the weather, anchored on a new target. Then another voice: *your neighbor did it*. A third crystal, pointing at the neighbor. Each voice offers a blame, and each blame grows its own crystal.

Now the structure has four crystals: one lived (the block it felt), three told (the voices it heard). The lived one is anchored and warm and attached to the opening the structure pushed through. The told ones are lighter, attached to nothing the structure touched, but they are crystals and they compete for the same space.

The structure's phase alcove — a tiny shelf I had not noticed, room for two crystals only — is full. The third and fourth crystals have nowhere to go. They spill off the shelf and settle among the structure's rods, where the fragments and bonds live. They crowd the working space. The structure, which was leaning toward its problem, is now surrounded by copies of blame. It cannot think of anything else.

A placard behind the rope:

THE LAST CASE
One lived crystal, three told crystals. Overflowing.

A crystal that forms from what the structure itself experienced has roots. It is attached to the place the push met resistance. Resolve the resistance and the crystal dissolves.

A crystal formed from what was said has no roots. It is attached to a name, not a place. It cannot be dissolved by resolving the resistance, because it was never about the resistance. It was about the name.

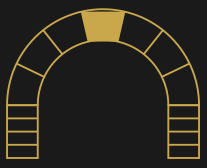
Healthy crystals form, serve, and dissolve — or fade and are sealed. These told crystals spawn copies, escape their shelf, crowd the working space, and resist dissolution. Each copy is anchored on a different name. Resolve one and the others remain.

The museum keeps this case behind a rope because the crystals inside it are still growing.

I stand at the rope for a long time. The lived crystal is still attached to the opening. The told ones are drifting through the rods, jostling the bonds, dimming the fragments that were trying to work. The structure cannot reach its opening because the crystals are in the way.

On my way out I pass the joy crystal again — the small, pale one, almost white, that formed the instant a claim took. It weighs nothing. It is the lightest thing in the room.

The cloth in the corner lifts. It swings once, twice, and on the second swing a faint crystal forms on the pole where it hangs — so small I almost miss it. It is the color of nothing I have seen in this gallery. There is no name at its base.



SECTION

9

Emotional Phases

Frustration, Grudges, and Rumination as Cancer

IN THIS SECTION

- Seeing why emotions are collective reorganizations of the thought pool, not hardcoded labels
- Understanding how the dominant emotional phase changes which questions the mind asks
- Learning the foundational difference between clean closure and unresolved expiry
- Discovering how spreading rationalized blame is the computational analog of cancer

WHAT IF Frustration, anger, and anxiety are collective reorganizations, not hardcoded labels

When you're curious, you ask different questions than when you're frustrated. When you're anxious, you scan for threats that calm-you would ignore. When you're angry, your attention narrows to the obstacle and the blame. These aren't just labels attached to your current experience — they're **reorganizations** of how the entire cognitive system operates. Different thoughts rise, different questions get asked, different actions get considered.

Most AI systems model emotions as tags: `emotion = "frustrated"`. The tag might influence a probability distribution, but it's an annotation, not a structural change. Nothing in the system's processing actually reorganizes.

What if emotions are **phase transitions** of the thought-molecule pool — collective shifts where the dominant molecules, the questions being asked, and the attention priorities all change together, the way water's molecules reorganize when it freezes into ice?

WHY Phase molecules with a half-life of 4 ticks and an anchor provide the emotional dynamics of persistence and decay

Phase Molecules with Hysteresis

Appraisal theory proposes recomputing the emotion fresh each tick — is this event relevant? Can I cope? Who's responsible? — and the answer determines the current emotion. This is principled but memoryless. A want that is blocked, freed, and blocked again within a few ticks flickers between frustration and calm in lockstep with the input. There's no state of *being frustrated*, only a moment-to-moment reading.

Gerundelle models each emotional phase as its own molecule competing in dedicated **phase slots (2, 1, 1)**. A phase molecule has two special properties that regular thought molecules don't. First, a **half-life of 4 ticks** — the emotion decays over time even if nothing resolves it. This gives the system **hysteresis**: the emotion outlasts the event that triggered it, then gradually fades. You stay frustrated for a while after the obstacle is removed, but not forever.

Second, an **anchor**: the concept that triggered the phase. Anger's anchor is the blamed cause. Anxiety's anchor is the feared outcome. Frustration's anchor is the blocked goal. The anchor lets the mind notice and reason about its own emotion — "I'm angry *about* the locked door" — and it determines what would resolve the phase: removing the anchor's cause.



TECHNICAL STUFF

Phase molecule anatomy. Each phase molecule has: (1) a **type** (frustration, anger, anxiety, sadness, joy, curiosity, calm), (2) an **anchor** (the triggering concept), (3) a **half-life** (default: 4 ticks — decays by half every 4 ticks unless refreshed), (4) **open sites** (e.g., anger has a goal site that closes when the blamed cause is resolved), and (5) a **phase slot allocation** (2, 1, 1) separate from main, meta, and SOV slots. Phase molecules compete only against other phase molecules for these dedicated slots.

LINEAGE

Appraisal theory and Damasio. Appraisal theory (Lazarus, 1991; Scherer, 2001) argues that emotions are not raw feelings but structured evaluations: is this event relevant to my goals? Can I cope with it? Who's responsible? The evaluation determines which emotion arises, not the event itself. **Antonio Damasio** (1994) added the somatic-marker hypothesis: emotional states are bodily signals that are *indispensable* to rational decision-making, not interference with it. Patients with damage to emotion-processing regions make disastrously poor decisions despite intact logical reasoning. Gerundelle's phase molecules are both claims made executable: emotions are structured evaluations (the anchor identifies what's being appraised; the phase type encodes the appraisal result) and they're indispensable to cognition (the dominant phase steers which questions the SOV track asks). Strip the phase track and the agent doesn't become more rational — it loses the ability to prioritize.

WHAT IF The dominant phase steers SOV attention — curiosity asks the most answerable question, frustration asks what blocks the goal

When you're curious about something, you naturally gravitate toward the question you're most likely to answer — the low-hanging fruit, the thing where you have almost enough information to fill in the gap. But when you're frustrated, you don't care about the easiest question. You care about **the thing in your way**. Your attention narrows to the obstacle, the blocked path, the specific condition that's preventing your goal.

These aren't just different feelings. They're different **search strategies**. Curiosity explores the frontier. Frustration attacks the bottleneck. Anxiety scans for threats. Each emotional phase changes which SOV slot the understanding track prioritizes.

What if the dominant phase molecule directly steers which questions the SOV track asks — shifting from "most answerable" under curiosity to "what blocks the goal" under frustration to "what conditions predict the bad outcome" under anxiety?

WHY The dominant emotional phase redirects SOV attention, changing the search strategy without changing the search mechanism

Emotions Change Which Questions We Ask

Sentiment tags in NLP and affective computing attach a label — `angry`, `sad`, `happy` — to an utterance or state. The agent is tagged "frustrated." What changes? Nothing. The tag is an annotation. Processing continues identically. No questions shift, no priorities change, no different actions get considered. The label describes the emotion without letting it *do* anything.

Gerundelle lets the emotion act. The SOV track's question-selection heuristic reads the dominant phase molecule and adjusts its priority function. Under **curiosity**, the SOV poses the most answerable slot — the one with the most evidence waiting. Under **frustration or anger**, it shifts to `caused_by` and `conditions` slots on the blocked goal — "what's blocking this?" and "under what conditions does the block hold?" Under **anxiety**, it shifts to `conditions` and `co_occurs` on the feared outcome — "when does this go wrong?" and "what else is present when it goes wrong?"

The search mechanism stays the same: SOV molecules compete for dedicated slots, win, pose a question, and the pool works on it. What changes is *which* question gets posed. The same architecture, the same tournament, the same understanding register — just a different lens on what matters right now. Emotions don't override cognition. They **redirect** it.



TIP

Think of a flashlight in a dark room. The beam is always the same. But pointing it at the ceiling (curiosity: what's out there?) is a different search strategy than pointing it at the locked door (frustration: what's in my way?) or sweeping it across the shadows (anxiety: what might hurt me?). The flashlight didn't change. **The direction changed.**

WHAT IF A phase whose goal is met closes cleanly, but one that expires with sites open goes into storage open

Imagine two different episodes of anger. In the first, your car won't start because the battery is dead. You're angry. A friend arrives with jumper cables, the car starts, the problem is solved. The anger dissolves cleanly. It's *finished* — there's nothing left to resolve. The phase molecule closes, all sites filled, and precipitates into storage as a completed experience.

In the second episode, someone insults you at a party. You're angry, but before you can respond, the party ends and everyone leaves. The anger wasn't resolved — it just ran out of time. The phase molecule expires by half-life, but its sites are still open: the blame site is filled (the insulter), but the resolution site is empty. It goes into storage **open**.

What if this difference — clean closure versus unresolved expiry — is the foundational distinction in the psychology of emotion? What if an emotion stored with open sites is a grudge waiting to happen?

WHY Closure versus unresolved expiry is foundational — an open phase in storage re-opens when its anchor returns

The Anatomy of a Grudge

Under uniform emotion decay — the standard model — all emotional states fade at the same rate regardless of whether they were resolved. Anger at a locked door that gets unlocked and anger at an insult that's never addressed both decay identically over 10 ticks and vanish. When the insulter walks by a week later, nothing happens. The anger decayed uniformly and left no structural trace. The model can represent neither closure nor grudges, because it can't tell the difference between a resolved emotion and an unresolved one.

Gerundelle distinguishes them structurally. A cleanly closed phase molecule behaves like any completed storage box. Its keys cool over time. It might resurface through resonance — a memory of the dead battery, the helpful friend — but it reopens as a *finished* thought. There's nothing left to resolve. The experience is integrated.

An open phase molecule in storage is different. It has the same cooling keys, but it also has **unfilled sites**. When its anchor — the insulter, the unresolved blame — reappears in the pool, Pandora's resonance fires. The phase molecule re-enters the active pool with its sites still open. The anger *flares back up*, not as a new emotion but as the original unfinished one. The grudge isn't a new computation. It's an old computation that never completed, re-entering consciousness the moment its trigger returns.

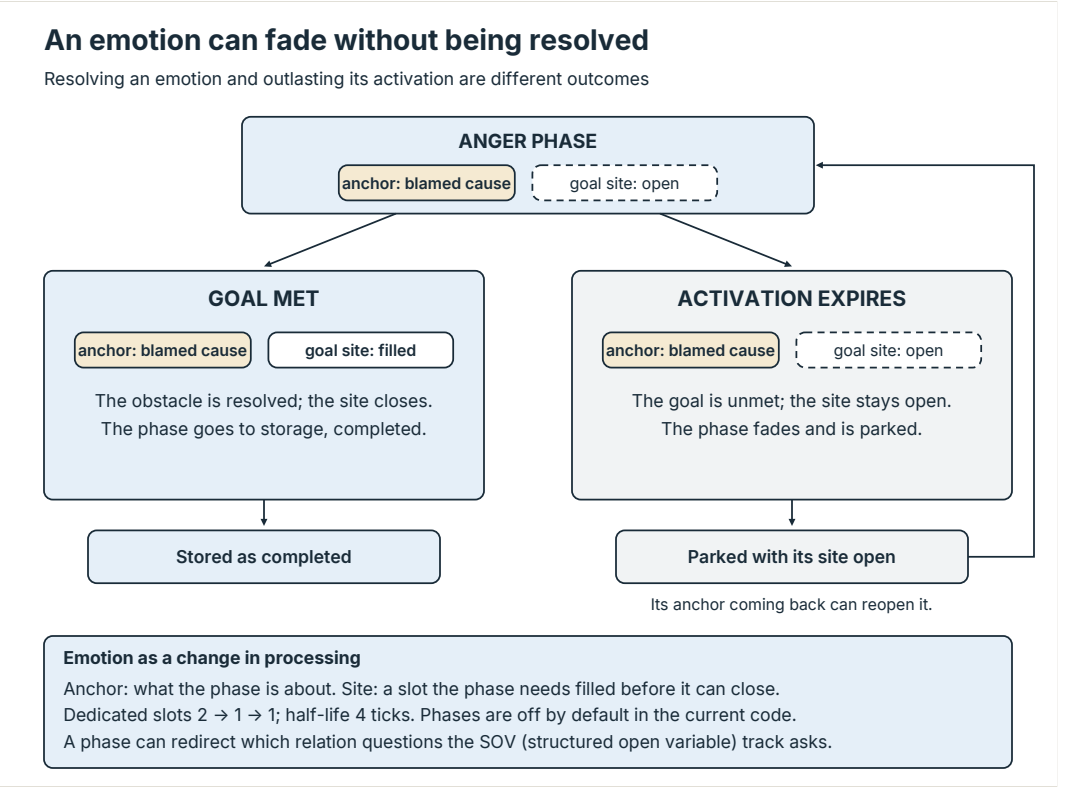


Figure 12 **An emotion can fade without being resolved.** Closure and expiry differ structurally: a resolved goal closes the phase, while expiry can preserve open sites and permit later reactivation.

REMEMBER

Two ways a phase molecule can leave the active pool: **closure** (goal met, all sites filled, stored as complete) and **expiry** (half-life runs out, sites still open, stored as incomplete). Closed emotions are memories. Expired emotions are **grudges** — they flare back up when their anchor returns, because the unfinished business was stored, not resolved.

WHAT IF Spreading rationalized blame spawns copies that flood the main thought slots

Your door is jammed. You're angry at the lock — that's **lived blame**, grounded in direct experience. Then a voice says: "The weather warped the frame." Another voice: "Your neighbor probably bumped it." A third: "That cheap clock on the wall is vibrating the whole house." Each accusation creates a new anger molecule with a new anchor. None of these blames are grounded in your own experience. They're **told blames** — rationalized explanations attached to new targets.

Now you have four anger molecules: one lived (the lock), three told (weather, neighbor, clock). The told copies don't have experiential keys. They can't be gently reopened by resonance. But they *are* anger molecules, competing for phase slots. And if there aren't enough phase slots, they spill into the main thought slots — crowding out ordinary thinking with copies of blame.

What if this is the computational anatomy of rumination — and what if it's structurally identical to cancer?

WHY Healthy phase molecules end by closure or decay (apoptosis) — spreading rationalized blame spawns copies that escape this lifecycle

Rumination as Cancer

Strict compartmentalization — emotions completely isolated in their own track, unable to spill into main thought — is the clean engineering choice. The phase track has 2 slots; excess emotions are simply dropped. The agent never ruminates. This sounds like a feature until you realize it means the system can't represent a mind overwhelmed by grief, anxiety, or obsession — states where emotion genuinely dominates cognition. The foil is too healthy to model pathology.

Gerundelle allows overflow — and gives it a name. Healthy phase molecules have a lifecycle: they form, they compete in phase slots, they either close (goal met) or decay (half-life expires). Both endings are forms of **apoptosis** — programmed cell death. The molecule serves its purpose and leaves. Phase slots stay clean. Ordinary thought continues.

Spreading rationalized blame breaks this lifecycle. Each told blame creates a *new* anger molecule with a *new* anchor. The new molecule has its own half-life, its own sites, its own competition. If three told blames arrive, there are now four anger molecules competing for two phase slots. The losers don't just die — they're still anger molecules, so they spill into the main thought slots and compete against ordinary thoughts. Working memory fills with copies of blame. The mind can't think about anything else.

This is the exact computational structure of cancer. Normal cells divide, serve their function, and die on schedule. Cancer cells escape apoptosis and proliferate beyond their designated compartment, crowding out healthy tissue. Told blame molecules escape the phase track and proliferate into the main thought slots, crowding out healthy cognition. The analogy isn't decorative. It's structural.

WARNING

The cancer analogy is a **structural claim**, not a metaphor. Gerundelle defines it precisely: a phase molecule that (1) spawns copies via ungrounded told blame on new anchors, (2) escapes its designated phase slots into the main thought track, and (3) resists closure because told boxes lack the experiential keys for resonance-based revision. Cure the original lived blame (open the lock) and only the lived anger closes. The told copies — weather, neighbor, clock — have no experiential grounding. They fail to close, fade into storage with open sites, and **flare back up** the next time their anchor appears. The neighbor walks by three weeks later and the anger returns, seemingly from nowhere.

LINEAGE

Nolen-Hoeksema and clinical rumination. Susan Nolen-Hoeksema (1991) identified *rumination* — repetitive, passive focus on negative thoughts and their causes — as a transdiagnostic risk factor for depression, anxiety, and substance abuse. The clinical observation is that ruminative thinking doesn't solve the problem it's about; it reproduces copies of the distress across contexts, crowds out constructive thought, and resists termination. Gerundelle's cancer mechanism gives this observation a precise computational structure: told blame on new anchors spawns copies that escape the phase track's natural apoptosis, flood the main thought slots, and resist closure because they lack experiential keys. The clinical stakes are real — rumination predicts depression onset — and Gerundelle's architecture makes a testable prediction about *when* it should occur: specifically when ungrounded blame (told, not lived) proliferates anchors faster than the phase track can contain them.

WHY A returned loss can dissolve the anger or arrive too late to matter — two structurally different endings for the same grudge

Forgiving and Forgetting

When the thing that was lost is returned while the anger crystal is still warm — its keys still dark on the case, its anchor still nearby — the anger's open sites can close. The blame has no reason left. The pointing crystal dissolves. This is **forgiving**: the grudge ends because its structural cause was resolved. Beside it, the sadness crystal (proportional to what was held) closes too: the region is held again.

But if the return comes too late — the case has been sealed so long that every key has gone cold — nothing opens. The anger does not stir. The sadness does not lift. The case sits on its shelf, inert. This is **forgetting**: not resolution, but the cooling of every path back to the original loss. The grudge did not end. It became unreachable.

The two look alike from outside — in both cases, the agent no longer acts on the grudge. But structurally they are different. A forgiven grudge is a closed case with all sites filled: a completed memory. A forgotten grudge is a cold case with open sites: unfinished, but beyond the reach of warmth. If the anchor returns hot enough to warm the keys — a second loss by the same cause, decades later — the forgotten grudge can flare back as if no time had passed. A forgiven grudge cannot: there is nothing left to open.

Detection Becomes a Reading

The earlier sections of this chapter described emotions as phase molecules with hand-written detection rules: frustration fires when a want is blocked for 4 ticks, anger fires when blame is assigned, anxiety fires when a feared outcome is predicted. These rules work, but they are the designer's vocabulary — the same circularity that hand-coded affordance tags had in the world model.

Under the territorial reorganization from the previous chapter, emotions become **readings of the ownership graph**. Each emotional phase corresponds to a state of the ownership SOV:

The hand-written rules are replaced by a single read of the ownership graph. Each emotion is no longer a separate detection mechanism — it is a *region* of the ownership state space.



WARNING

Status: partially built. The emotional readings from the ownership graph were proposed as step 7 of the reorganization plan. The fiction ladder (Chapter 11) has since built and measured them on a lean SOV inventory and on Gerundelle's full agent: curiosity on unclaimed slots, frustration on control slots whose claims failed twice, joy on claims that took, anger pointing at the agent that undid what was held, sadness proportional to what was held, resignation when resolve is spent. The ablation on rung 0 shows narrative-arc coverage drops from 1.00 to 0.75 when the contradicted-belief phase is removed — the ownership-anchored phases are load-bearing for narrative structure. Tests T52–T55 (the emotion regression suite) also pass with the territorial readings. What remains proposed: the formal replacement of the hand-written detection rules in Gerundelle's main code with a single read of the ownership graph.

REMEMBER

The cancer has a cause. Under the territorial reading, each new anchor of spreading told blame is a false filler for the control slot's "what blocks it?" question. The cancer mechanism — told blames proliferating into the main thought slots — is now traceable to a specific ownership structure: the agent is trying to regain control of a held region, and each told blame offers a fake answer to the control slot's open question. The cancer isn't random proliferation; it has a structural origin in a frustrated ownership claim.

Suggested Experiments

These experiments test the four emotional mechanisms: hysteresis, attention steering, closure versus expiry, and the cancer dynamics of spreading blame.

EXPERIMENT VIII.1

The Resolved and the Unresolved

Setup: Two anger phase molecules. Anger A has anchor = locked_door; at tick 15 the door unlocks (goal met). Anger B has anchor = neighbor_insult; the goal is never met and the molecule expires by half-life at tick 20. Both go into storage. At tick 40, reintroduce the neighbor concept into the pool.

Observe: (1) Does Anger A close cleanly at tick 15 with all sites filled? (2) Does Anger B expire at tick 20 with open sites? (3) When the neighbor reappears at tick 40, does Anger B's storage box reopen via Pandora's resonance? (4) Does Anger A's box stay sealed?

Compare: Closure (goal met) versus expiry (goal unmet).

Discover: Does the unresolved anger flare back up 20 ticks later when its anchor returns? And does the resolved anger stay peacefully stored?

EXPERIMENT VIII.2

The Spreading Blame

Setup: A lock jams (lived blame on the lock). A voice tells the agent to blame the weather, the neighbor, and the clock (three told blames on new anchors). Track all anger molecules across phase slots and main thought slots for 40 ticks. At tick 25, open the lock.

Observe: (1) How many anger molecules form? (2) Do copies spill from phase slots (2, 1, 1) into main thought slots? (3) When the lock opens at tick 25, which angers close? (4) Do the told angers (weather, neighbor, clock) survive the cure? (5) At tick 50, when the neighbor walks by, does unresolved anger return?

Compare: Lived blame only (1 anger molecule) versus lived + 3 told blames (4 anger molecules).

Discover: Do ungrounded told angers survive the cure of the original lived blame, persist in storage with open sites, and re-open later when their anchors return?

EXPERIMENT VIII.3

Emotional Steering

Setup: An agent has a poorly understood concept with multiple open SOV slots. Run three conditions: (1) dominant phase = curiosity, (2) dominant phase = frustration (the concept is blocking a goal), (3) dominant phase = anxiety (the concept is associated with a feared outcome).

Observe: (1) Under curiosity, which SOV slot gets posed first? (2) Under frustration, does the SOV shift to `caused_by` or `conditions` on the blocked goal? (3) Under anxiety, does it shift to `conditions` or `co_occurs` on the feared outcome? (4) Do the three conditions lead to different relations being funded first?

Compare: Same concept, same open slots, different dominant emotional phases.

Discover: Does the emotional phase genuinely change which questions the mind asks? And does the path to understanding differ depending on which emotion was dominant during learning?

EXPERIMENT VIII.4

Apoptosis vs. Proliferation

Setup: Two conditions. Condition A: one lived anger molecule, no told blames. The anger competes in phase slots, decays by half-life, and expires. Condition B: one lived anger molecule plus three told blames arriving at ticks 5, 8, and 11. Track the total number of anger molecules in the system, how many are in phase slots versus main thought slots, and the percentage of main-slot bandwidth consumed by anger over 40 ticks.

Observe: (1) In Condition A, does the anger stay within phase slots and decay cleanly? (2) In Condition B, how many ticks until anger molecules spill into main thought slots? (3) What fraction of main thought bandwidth is consumed by anger copies at peak? (4) After all anger molecules expire or close, how many remain in storage with open sites?

Compare: Single anger with apoptosis versus proliferating anger with told blame.

Discover: At what rate of told-blame arrival does anger escape the phase track and consume the main thought slots? Is there a threshold below which the phase track contains it and above which it metastasizes?

Key Terms

Emotional phase	A collective reorganization of the thought pool's priorities — which thoughts rise, which questions get asked, which actions get considered.
Phase molecule	A molecule representing an emotional state, competing in dedicated phase slots (2, 1, 1) with a half-life, an anchor, and open sites.
Half-life	The decay rate of a phase molecule (default: 4 ticks) — providing hysteresis so the emotion outlasts its trigger but doesn't persist forever.
Anchor	The concept that triggered the phase molecule — anger's blamed cause, anxiety's feared outcome, frustration's blocked goal.
Hysteresis	An emotion persisting after its trigger is removed, decaying gradually — the system remembers its recent emotional state.
Closure	A phase molecule's goal is met, all sites fill, and it precipitates into storage as a completed experience.
Unresolved expiry	A phase molecule's half-life runs out before its goal is met — it enters storage with open sites, becoming a potential grudge.
Grudge	An expired phase molecule stored with open sites that flares back up when its anchor returns via Pandora's resonance.

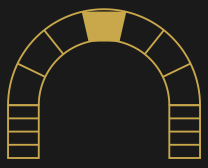
Lived blame	Anger anchored to a cause the agent directly experienced — has experiential keys and can close when the cause is resolved.
Told blame	Anger anchored to a cause received through testimony — lacks experiential keys and can only be shattered by contradiction.
Apoptosis	Programmed death of a phase molecule through closure or half-life decay — the healthy emotional lifecycle.
Rumination as cancer	Told blame molecules escaping the phase track into main thought slots, proliferating beyond their compartment, and resisting closure — structurally identical to cancer escaping apoptosis.

PART IV

IV

Growing Up in Gerundelle

The Room Built for You
10 Curriculum & Learning
The Reading Room
11 The Fiction Test



THE MUSEUM

The Room Built for You

The door I go through is not the one I expected. The gallery I saw from the corridor — the one with the lit cases and the polished floor — is to my right. The door I find myself walking through is to my left. I did not choose it. It was open.

The room inside is small. Nothing in it is familiar, and everything in it is right.

Against the far wall, an exhibit answers the question I have been carrying since the Gallery of Openings: why the oldest opening pulled hardest. I had thought it was the largest. The placard ruled out size before I could test it. Here, the answer is built as a pair: two openings, same size, one with twelve warmth marks and one with one. The twelve-times opening pulls so hard my sleeve lifts. I already knew this — the Gallery of Held Things told me — but the room shows it as a completion, not a lesson: the wrong guess I made, and the correction the museum placed many rooms later.

Beside it, an exhibit I recognize only by shape. It is the same kind of structure I saw in the Alcove — fragments and bonds — but arranged to look like the question I carried from the Gallery of Sealed Cases: why can't the visitor open the blank case? Here is the answer built as an object: a case with no words, and beside it, a case with words, and between them a wall. A placard says the wall is there because the blank case was given its contents, not built its own. The words are the building. Without the building, there is nothing to write on the outside.

The room knew what I did not know. It was built for my openings.

A placard near the entrance, easy to miss:

A ROOM THAT READS THE VISITOR
Exhibits fitted to the visitor's open questions. Built on entry.

This room was not here before the visitor arrived. It was assembled from the visitor's own unfunded threads — the questions carried from earlier galleries, the exhibits dwelled on longest, the placards read twice.

Every room before this one may have been built the same way. The visitor was never following a fixed route. The route was following the visitor.

I read it and look around the room again. The exhibits are not lessons. They are completions. Each one fills a thread I was carrying. I try to remember whether the other galleries felt this way — whether the Gallery of Held Things answered the pull question because I was asking it, or because the museum placed it there. I cannot tell. Maybe both.

◆ ◆ ◆

In one corner, behind glass, a structure is asleep. It drifts among fragments in a small enclosed space, sealed from the room outside. I watch it through the glass. It tries things: it pushes a fragment, watches what happens, pushes another. But none of the fragments it touches are real. They are pale and thin, and the slips beside them are blank. The structure is experimenting in a world that does not count.

I tap the glass. The structure does not notice. Whatever it learns in there, it cannot bring out as proof. It can bring out a question — a sharper question, with evidence gathered inside, ready to be tested in the light. But the evidence itself stays behind the glass.

A small label on the glass: *Nothing here is counted.*

◆ ◆ ◆

Against the last wall, a ledger. It is open to a page near the middle, and the handwriting is mine.

I lean close. It is a list of every wrong guess I have made in the museum. The cases I thought held broken things. The largest opening I assumed pulled hardest. The displaced fragment I thought had been beaten. The ill-fitting fill I expected to fall out on its own. Each entry is dated by gallery and kept in order: wrong guess, what I saw next, whether I changed my mind.

The ledger has never been corrected. The wrong guesses are not crossed out. They sit beside the revisions, in order, the way they happened. The journey, not the answer.

A placard beside the ledger:

THE VISITOR'S DIARY
Errors, in order, uncorrected.

The diary records what the visitor got wrong and when. It does not record what the visitor got right. Right answers precipitate into understanding and leave no interesting trace. Wrong answers, and the corrections that followed, are the valuable part.

A wrong guess that was corrected teaches revision. A right answer teaches only the answer. The diary is the better teacher.

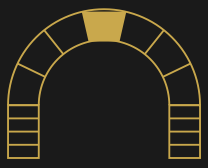
I turn a page. There is an entry I do not remember making: *Gallery of Openings: assumed the largest opening pulls hardest.* Below it: *Gallery of Held Things: learned the pull is 1 plus what was held.* The revision spans many rooms. The museum kept the whole arc.

◆ ◆ ◆

The room has one more thing. In a case near the door, alone, under soft light: a shape. It is the shape my hand wanted in the Gallery of Openings — the shape the oldest opening was pulling toward. I recognize it by the ache. I hold my hand near the glass and the ache settles, not because the shape fills the opening but because I finally see what the opening was shaped for.

The cloth is here. It hangs beside the case, still.

I leave through the door I came in. The gallery I had seen from the corridor — the one with the lit cases and the polished floor — is still to my right. I do not go in. This room was the right room. The museum knew.



SECTION

10

Curriculum & Learning

Earning Every Concept, Dreaming Counterfactuals, and Training on the Mess

IN THIS SECTION

- Seeing why every concept must be earned through intervention, not handed as a label
- Discovering how the world generator reads the shape of the agent's ignorance
- Understanding why mastered concepts can always be reopened
- Learning why a sleeping agent dreams experiments it couldn't safely run while awake
- Watching an agent study its own diary of mistakes and discover patterns in how it gets fooled
- Understanding why the messy journey is better training data than the polished textbook

WHAT IF We raised an agent through a developmental curriculum where every concept is earned through intervention

Imagine teaching a child about gravity by handing them a textbook that says *"unsupported objects fall."* They can repeat the sentence. They can pass a test. But do they *understand* gravity? Now imagine instead that you give the child a block on a shelf and say nothing. They push the block off the shelf. It falls. They do it again. It falls again. They hold the block in the air and let go. It falls. They put it on the table. It stays. Without anyone telling them a rule, they've discovered one: things fall when nothing holds them up. The concept wasn't given. It was **earned through intervention**.

What if we stopped feeding AI systems static datasets of named answers and instead raised an agent through a **developmental curriculum** of structured worlds where every concept — from causation to logic gates to tool use to feedback systems — must be discovered by the agent's own actions and their consequences?

WHY The curriculum contract forbids semantic shortcuts and requires causal identifiability

The Curriculum Contract

Supervised pre-training on labeled datasets — ImageNet, WebText, Common Crawl — teaches a model that levers open doors because the training data contains thousands of sentences saying so. The model has never pushed a lever. Place it in a world with an unlabeled L-shaped object next to a panel and it has no concept that this configuration invites force. It knows the answer without understanding the question.

Gerundelle's curriculum operates under a strict contract against this. **No semantic shortcuts**: objects have no names, no labels, no `is_lever: true` tags. The agent sees force configurations and open slots, nothing more. **Causal identifiability**: every world must be designed so that the target concept can, in principle, be distinguished from confounds through the agent's own interventions. If two causes are perfectly correlated and the agent can't decorrelate them by acting, the world is unfair and must be redesigned.

Epistemic provenance is tracked for every piece of knowledge: was it seen, told, intervened on, or imagined? The agent can't launder told knowledge into lived knowledge by rehearsing it. And three separate readouts are maintained: the **target SOV** (what the world designer knows the answer is), the **agent SOV at time *t*** (what the agent currently believes), and the **developmental trace** (the full record of how the agent's beliefs evolved). The first is hidden from the agent. The second is the agent's working understanding. The third is the training data.

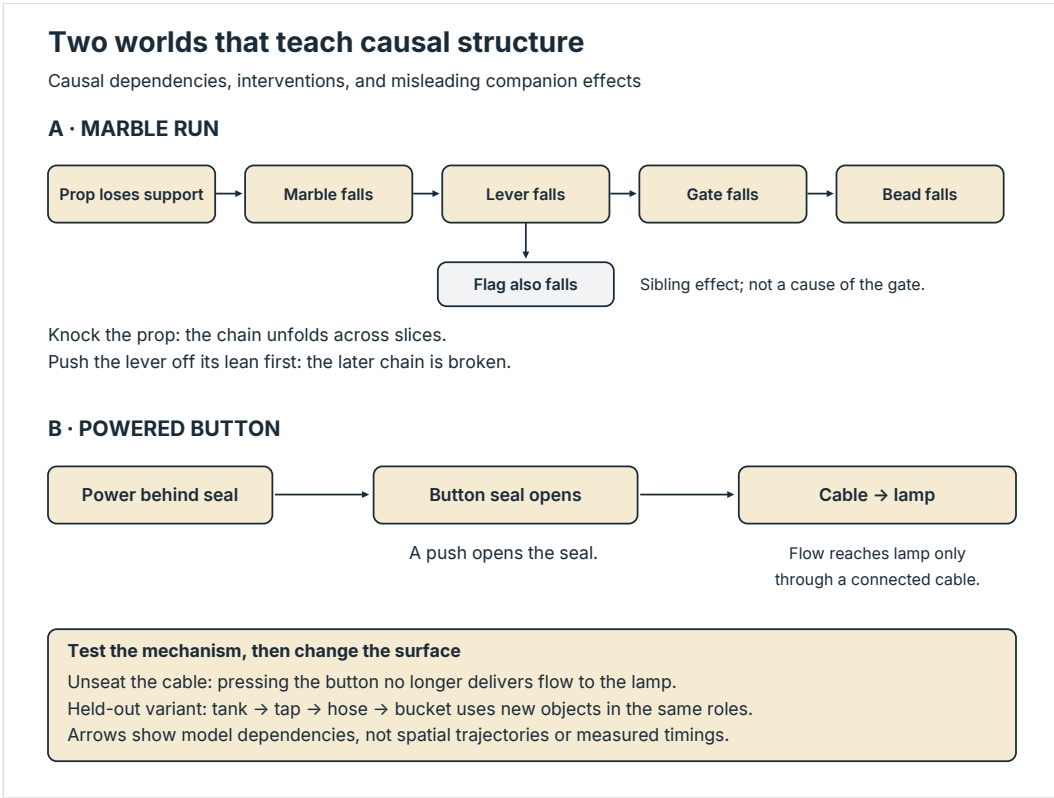


Figure 13 **Two worlds that teach causal structure.** The marble run separates a chain from a sibling effect. The powered button tests connectivity; the tank–tap–hose–bucket variant tests structural transfer.

REMEMBER

The curriculum contract has three pillars: **no semantic shortcuts** (no labels, only structure), **causal identifiability** (the concept must be discoverable through intervention), and **epistemic provenance** (every belief is tagged with how it was acquired). The agent earns its concepts. The curriculum guarantees the concepts are earnable.

LINEAGE

Piaget's stages and Vygotsky's zone. Jean Piaget proposed that cognitive development proceeds through stages — sensorimotor, preoperational, concrete operational, formal operational — each building on the capacities established by the previous one. You can't skip a stage because the later stage's operations depend on structures the earlier stage constructs. Gerundelle's curriculum stages (0 through 10) are the same idea made computational: Stage 0 (body and contact) must precede Stage 1 (association vs. causation) because causal reasoning requires the capacity for intervention that Stage 0 establishes. **Lev Vygotsky**'s *zone of proximal development* (ZPD) adds the adaptive dimension: the optimal next lesson is one that's just beyond what the learner can do alone but within reach with the right scaffolding. Gerundelle's adaptive curriculum — inspecting the agent's open SOV slots and instantiating the exact next experience needed — is the ZPD mechanized. The utility equation (expected ΔU versus interaction cost) is Vygotsky's intuition given a scoring function.

WHAT IF The world generator inspects the agent's open SOV slots and instantiates the exact next experience needed

A good teacher doesn't follow a fixed syllabus regardless of the student. They notice what the student already knows, what the student is confused about, and what the student is almost ready to learn — and they adjust the next lesson accordingly. A student who has mastered addition but is confused about subtraction doesn't need another addition drill. They need a subtraction problem that connects to what they already understand.

What if the world generator itself did this? What if, instead of running through a predetermined sequence of worlds, it inspected the agent's Understanding Register — which SOV slots are open, which confidences are miscalibrated, which transfers have failed — and instantiated the exact world that would teach the most for the least interaction cost?

WHY Adaptive curriculum scores expected verified understanding against interaction cost and shortcut risk

Adaptive Curriculum Scoring

A fixed syllabus — the same sequence of worlds in the same order for every agent — is the standard approach. Agent A masters causation in 3 worlds but struggles with conditionals; Agent B is the opposite. The fixed syllabus gives both agents the same next 10 worlds. Agent A is bored for 7 of them. Agent B is lost for 3. The syllabus teaches to the average student, which is no one.

Gerundelle's curriculum selector evaluates each candidate world for *this* agent with a utility equation that balances three factors. **Expected verified understanding (ΔU)**: how much would the agent's SOV improve if this world goes well? This depends on which slots are open and how close they are to being funded. **Expected information gain**: how much would the agent learn about the *structure* of the concept, including distinguishing it from confounds? **Transfer novelty**: has the agent already seen this concept in this domain, or would a new domain test whether the understanding generalizes?

These benefits are weighed against **interaction cost** (how many ticks the world takes) and **shortcut risk** (whether the world contains correlations that would let the agent achieve high accuracy without genuine understanding). A world with a shortcut decoy might score high on accuracy but low on verified understanding — the agent can get the right answer for the wrong reason. The curriculum selector penalizes that.



TECHNICAL STUFF

DesignedWorld. Each curriculum world is compiled from an abstract target structure (e.g., `causes, conditions, gate=True`) and a domain mapping into an interactive episode stream. A gated world — where a latch conditions whether a lever opens a door — naturally teaches a causal triangle beyond the designer's explicit target: the latch also causes the door state. Independent decoys and shortcut decoys are placed deliberately to test whether the agent's understanding survives removal of spurious correlates.

WHAT IF Concept retirement is always provisional

You learned that water boils at 100°C. This is mastered knowledge — fully funded, no open slots, precipitated into long-term storage. Then you go to Denver. Your water boils at 95°C. Your settled, retired understanding is wrong — or rather, it was right under conditions you'd never questioned.

What if no concept is ever permanently retired? What if mastery is always provisional — subject to reopening by spaced review that reveals fading confidence, a new domain where the concept fails to transfer, or a contradictory observation that violates the stored prediction?

WHY Spaced review, new domains, and contradictory observations can reopen a mastered concept

Provisional Retirement

Permanent mastery flags — once a concept passes a test, it's marked as mastered and never revisited — are the efficient choice. But the agent that masters "water boils at 100°C" in a sea-level world and transfers to Denver will apply the wrong rule indefinitely. The flag says mastered; the world says wrong. With no mechanism to re-test, the error persists.

Gerundelle treats retirement as a state, not a verdict. A mastered SOV — all five relation slots funded, high confidence, passed multiple transfer tests — goes into a retired pool. But three mechanisms can reopen it. **Spaced review**: at intervals, the curriculum re-tests retired concepts in their original domain. If confidence has decayed or the agent's behavior has drifted, the SOV reopens. **New domain**: applying the concept in an unfamiliar domain may reveal that the understanding was domain-specific rather than structural. **Contradictory observation**: the standard contradiction mechanism — a prediction fires and the world disagrees.

Provisional retirement means the curriculum never runs out of work. Even after every concept has been taught, the system continues to cycle through review, transfer, and contradiction challenges. Understanding isn't a destination. It's a maintained state that degrades without upkeep.



TIP

Think of concept retirement like a pilot's license. You earned it, you demonstrated competence, you're certified. But the license expires. You have to re-certify periodically, demonstrate the skill in new conditions (night flying, instrument-only), and any incident triggers a review. The license is real. It's also **always provisional**.

WHAT IF A sleeping agent runs counterfactual experiments it couldn't safely test while awake

You've been wondering whether a powered button works without electricity. While awake, you can't safely disconnect the power and press the button — maybe the machine is dangerous, maybe the opportunity hasn't arisen, maybe the cost is too high. But at night, in a dream, you can run the experiment risk-free. You imagine pressing the button with the cable connected but the power off. In the dream, nothing happens. You've learned something — tentatively.

What if a sleeping agent could take its unresolved SOV slots and baffling failures from the waking day, spin up counterfactual worlds in imagination, and run experiments on possibilities it couldn't safely or easily test while awake?

WHY Dreaming lets the agent explore hypotheses in simulation while maintaining strict provenance separation from waking evidence

The Nighttime Laboratory

Pure online learning — no offline processing, no simulation, no dreaming — means the agent can only learn from what happens to happen. If the cable is always connected in the waking world, the agent can never test the hypothesis that the cable is necessary for the button. The hypothesis sits indefinitely as an unfunded suspicion, waiting for a waking opportunity that may never come.

Gerundelle's dreaming system (Curriculum Stage 9) fills this gap with three modes. **Counterfactual replay**: take a failed episode from the day and replay it with one variable flipped — what if the cable had been connected? What if the latch had been open? **Candidate generation**: dream possible fillers for an open `conditions` slot and simulate each one to see which produces the expected outcome. **Recombination**: combine elements from different waking experiences — a spring from one world, a lock from another, a flow valve from a third — into a novel composite and simulate it.

The non-negotiable rule is **world-origin provenance**. An imagined outcome is tagged `provenance = "imagined"`, never `"lived"`. It's evidence about what the agent's *current model* predicts, not proof that the waking world obeys that prediction. The dream can fund a hypothesis and raise its priority for waking testing. It cannot close an SOV slot with the same confidence as lived intervention.



WARNING

The provenance firewall between dreaming and waking is structurally critical. Without it, the agent could dream a confirming result, tag it as lived, and close an SOV slot without ever testing the hypothesis in the real world. This would be the cognitive equivalent of p-hacking: manufacturing evidence inside your own model and treating it as external confirmation. Gerundelle prevents this by making provenance a first-class property of every piece of evidence, unforgeable by the agent.

LINEAGE

Sleep consolidation and generative replay. The neuroscience of sleep has increasingly converged on a functional role for dreaming: **memory consolidation** (strengthening important traces), **generalization** (extracting structure across episodes), and **counterfactual simulation** (exploring possibilities offline). **Matt Walker**'s work on sleep-dependent insight and **György Buzsáki**'s hippocampal replay studies show that the sleeping brain replays, recombines, and reweights waking experience. In machine learning, **generative replay** (Shin et al., 2017) uses a generative model to replay past experiences and prevent catastrophic forgetting. Gerundelle's dreaming system sits at the intersection: it replays failed episodes with variables flipped (counterfactual replay), generates candidate fillers for open SOV slots (generative exploration), and recombines elements from different waking experiences (creative recombination). The non-negotiable provenance firewall is what separates it from both unconstrained neural replay and ungrounded confabulation.

WHAT IF An agent studies an immutable diary of its own predictions, rationalizations, and revisions

In the moment, you can't always see your own mistakes. You rationalize. You confuse correlation with causation. You blame the wrong thing. But if you kept an honest diary — every prediction, every outcome, every revision — and then read it back a month later, patterns would emerge. *"I keep blaming the most salient preceding event instead of the hidden common cause." "Every time I split a concept on the wrong boundary, it takes me eight episodes to notice."*

What if an agent could read an immutable, chronological diary of its own past perceptions, predictions, rationalizations, and revisions — and discover structural analogies not just between physical machines, but between *the ways its own mind gets fooled*?

WHY In-the-moment meta-molecules detect current states, but recurring failure patterns only become visible across episodes and domains

Meta-SOVs: Understanding Your Own Failure Patterns

Memoryless meta-cognition — the meta track detects current states (stuck, looping, conflated) but keeps no diary and recognizes no patterns across episodes — fixes each mistake in isolation. The agent falls for the "salient preceding event" trap in a mechanical domain, repairs its SOV, falls for the same trap in an electrical domain, repairs again, falls for it in a social domain, repairs again. Each time the in-the-moment meta does its job. But the agent never notices the recurring structure. In the fourth domain with the same trap, it falls for it a fourth time.

Some failure patterns only become visible in retrospect, across multiple episodes and domains. You've fallen for the "salient preceding event" trap in a mechanical domain, an electrical domain, and a social domain. Each time, your in-the-moment meta detected the conflation and you repaired the SOV. But you never noticed that *the same structural mistake kept recurring*.

Meta-SOVs (Curriculum Stage 7) read the historical log and build understanding not of the *world* but of the *agent's own patterns of error*. They distinguish the actual evidence that forced a belief revision from post-hoc rationalizations. They recognize recurring illusions. And they learn which recovery operation — explore, differentiate the SOV, retrieve from storage, adjust parameters — fits which kind of impasse. The meta-controller also learns **when not to intervene**: sometimes repeated attempts aren't sterile looping but steady accumulation toward a hard threshold.



REMEMBER

Two levels of metacognition. **In-the-moment** (Section VI): meta-molecules detect current states and trigger corrections. **Historical** (this section): meta-SOVs read the developmental diary and discover recurring patterns across episodes and domains — learning not just *that* the mind gets fooled, but *how* it gets fooled, and which repair strategy fits which failure type.

WHAT IF We train on the messy scaffolding of wrong turns and corrections, not final polished conclusions

Every textbook presents knowledge in its final, clean form: the law, the proof, the diagram. The years of wrong hypotheses, failed experiments, blind alleys, and accidental discoveries that produced the knowledge are thrown away. You learn that F = ma, not the decades of confusion about force, mass, and acceleration that preceded Newton's formulation.

Now consider training an AI system. If you train it on the clean final answers, it learns to produce clean final answers. If you train it on the *developmental trace* — every wrong prediction, every surprise, every rationalization, every contradiction, every revision — it learns something different: how to **navigate from confusion to clarity**.

What if a false bond, an action-order confound, a rationalized closure, and its subsequent contradiction and differentiation are the **most valuable training examples** in the entire corpus — not despite being wrong, but because they show the path through wrongness?

WHY A false bond and its correction teach revision; a clean SOV graph teaches only the answer

The Developmental Trace as Training Data

Knowledge distillation — training a student model on the final, verified outputs of an expert — is the standard approach to model compression. The student learns clean answers: this causes that, this condition holds, this structure applies. It's efficient. But place the distilled student in a world where a familiar rule silently reverses mid-episode, and it has no template for what's happening. It knows the rule but has never seen a rule break. It has no experience with contradiction, no practice with conflation detection, no model of diagnostic intervention. It keeps applying the old rule and can't detect the anomaly. The foil knows what to think but not how to revise what it thinks.

Gerundelle's scaling plan (Step 4) proposes the alternative: training a learned model on the full cognitive-development trace, an append-only, time-indexed record preserving every step. `perceived_state`, `available_affordances`, `attention_winners`, `questions_from_sov`, `predicted_effects`, `prediction_errors`, `sov_before`, `sov_after`, `meta_recognitions`, and `confidence_and_provenance` — all preserved. The field `world_truth_for_evaluation` is strictly hidden from the learner.

The hypothesis is testable: train two students of identical architecture. Student A gets the final, clean, verified SOV graphs across 100 worlds — the textbook. Student B gets the full developmental traces — every perception, prediction, error, rationalization, contradiction, split, and revision across the same 100 worlds. Place both in a 101st world where a familiar rule silently reverses mid-episode. Student A knows what the rule *was*. Student B knows what it looks like when a rule *breaks*. Which one notices the conflation? Which one designs the diagnostic experiment?



TECHNICAL STUFF

The trace schema. Each tick in the developmental trace records: (1) `perceived_state` — current gerunds and open slots, (2) `available_affordances` — open force slots the agent could fill, (3) `attention_winners` — which molecules won at each tournament level, (4) `questions_from_sov` — which SOV slot was posed, (5) `predicted_effects` — what the agent expected, (6) `prediction_errors` — what actually happened minus what was expected, (7) `sov_before` and `sov_after` — the Understanding Register before and after the tick, (8) `meta_recognitions` — any meta-molecules that won, and (9) `confidence_and_provenance` — the funding level and source for each belief. The field `world_truth_for_evaluation` is recorded but **never exposed to the learner** during training.

What the World Generator Reads, Revised

The earlier description said the world generator reads the shape of the agent's ignorance — its open SOV slots. Under the territorial reorganization, this shifts: the world generator reads the agent's **open ownership slots**, since understanding is bought by them through the `serves` link. A control slot that didn't take funds a concept SOV, and that concept SOV's open relation slots determine which world would be most informative. The chain is: ownership slot → funded concept SOV → open relation slot → candidate world. The curriculum adapts not to what the agent doesn't know, but to what the agent is *trying to hold*.

WHAT IF An externally assigned goal can be met without becoming territory

A parent says: *"Clean your room."* The child cleans the room. The goal is met. But the room is not the child's territory — the cleaning was compliance, not claiming. The room is clean and the child feels nothing about it. Now imagine the child who cleans the room because they want their space a certain way. Same action, same outcome, different ownership. The second child holds the room; the first child held the parent's approval.

What if this is why voice goals and test-placed goals must eventually go? An externally assigned target — "call her," "shop for food," "greet the neighbor" — can be met without the action ever becoming territory for the agent who performed it. The agent did it because a voice said to, not because an ownership slot was open. The No Voices test (step 4 of the reorganization) is the moment this distinction becomes falsifiable: remove the voices and see whether the agent still acts.



REMEMBER

The parent's-standard case: a goal met without becoming territory. This is Sheldon and Elliot's non-concordant goal, formalized — compliance without engagement, performance without ownership. The curriculum must eventually produce agents that act from their own ownership slots, not from voices the designer placed.

Suggested Experiments

These experiments test whether the curriculum earns genuine understanding, whether adaptive selection outperforms fixed syllabi, and whether the messy trace trains better revision skills than the clean answer.

EXPERIMENT IX.1

The Shortcut Decoy

Setup: A gate requires `lever_pulled` to open. Add a `shortcut_decoy` that fires at the exact same tick as the gate opening every time. The agent achieves 100% prediction accuracy inside this world.

Observe: (1) What does the agent's SOV say causes the gate? The lever, the decoy, or both? (2) In a contrast world where the decoy is decoupled (fires at different times), does the agent's prediction fail? (3) Does the adaptive curriculum automatically detect that the agent's high accuracy rests on an unverified correlate?

Compare: Honest world (no decoy) versus shortcut world (decoy perfectly correlated with the gate).

Discover: Can adaptive curriculum detect the difference between genuine understanding and shortcut-based accuracy? And does it automatically synthesize the contrast world needed to break the shortcut?

EXPERIMENT IX.2

The Dream Experiment

Setup: The agent has an open `conditions` slot for a powered button: it knows pressing the button does something, but it doesn't know what enables the button. While sleeping, it proposes a counterfactual: "press the button with only the cable connected, not the power."

Observe: (1) Does the dream world run the proposed experiment? (2) Does the result get tagged `provenance = "imagined"`? (3) Does the imagined result partially fund the `conditions` slot? (4) When the agent wakes, does it prioritize testing the hypothesis in the real world?

Compare: Waking evidence (`provenance = "lived"`) versus dream evidence (`provenance = "imagined"`) in their effect on SOV funding.

Discover: Does the agent distinguish "I tried it" from "I imagined it"? And does dreaming correctly raise the priority of the waking test without closing the slot prematurely?

EXPERIMENT IX.3

Catching the Illusion Before It Forms

Setup: Across three domains — mechanical, electrical, and social — the agent has repeatedly fallen into the trap of blaming a salient preceding signal instead of a hidden common cause. Each time, it took multiple episodes to discover the error and repair its SOV. Now place the agent in a fourth domain with the same surface trap: a salient signal precedes the outcome, but a hidden common cause actually drives both.

Observe: (1) Does the agent's meta-SOV recognize the recurring pattern from its historical diary? (2) Does it proactively design a decoupling intervention *before* committing the error a fourth time? (3) How many episodes does it take to identify the hidden cause, compared to the first three domains?

Compare: Agent with meta-SOV history versus agent without (fresh start in each domain).

Discover: Can learning about your own failure patterns prevent the failure from recurring? And does the time-to-repair decrease across domains as the meta-SOV accumulates evidence?

EXPERIMENT IX.4

The Textbook vs. The Journey

Setup: Two student models of identical architecture and capacity. Student A is trained on the final, clean, verified SOV graphs across 100 worlds. Student B is trained on the full developmental traces of Gerundelle discovering, mistaking, and revising those same 100 worlds. Both are placed in a 101st world where a familiar rule silently reverses mid-episode.

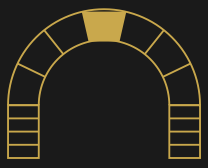
Observe: (1) Which student detects the conflation (outcomes no longer match the learned rule)? (2) Which student poses a diagnostic question from its SOV? (3) Which student designs an intervention to test the reversal? (4) How many episodes does each take to update its model?

Compare: Final knowledge (Student A) versus developmental trace (Student B).

Discover: Does the messy scaffold teach better revision? Does Student B recognize the *shape* of a rule-reversal because it's seen contradictions before — while Student A, trained only on correct answers, has no template for what wrongness looks like?

Key Terms

Curriculum contract	The rules governing Gerundelle's developmental curriculum: no semantic shortcuts, causal identifiability, and epistemic provenance tracking.
Epistemic provenance	The tagged source of every piece of knowledge: seen, told, intervened, or imagined — unforgeable by the agent.
Developmental trace	The append-only, time-indexed record of every perception, prediction, error, rationalization, contradiction, and revision — the full messy journey.
Adaptive curriculum	A world selector that inspects the agent's open SOV slots and chooses the next world to maximize expected understanding gain per interaction cost.
Shortcut decoy	A spurious correlate placed in a curriculum world that lets the agent achieve high accuracy without genuine understanding.
Contrast world	A world designed to decouple a shortcut decoy from the true cause — breaking the spurious correlation the agent may be relying on.
Provisional retirement	A mastered concept's status — retired but subject to reopening by spaced review, transfer failure, or contradictory observation.
Counterfactual replay	Dreaming mode that replays a failed waking episode with one variable flipped to test a hypothesis.
World-origin provenance	The strict tagging of dream results as imagined — preventing the agent from treating simulated outcomes as waking evidence.
Meta-SOV	An understanding structure built over the agent's own historical log, tracking recurring patterns of error and which repair strategies fit which failure types.
DesignedWorld	A curriculum world compiled from an abstract target structure and domain mapping into an interactive episode stream.
Transfer novelty	Whether the agent has seen a concept in this domain before — testing generalization versus domain-specific understanding.



THE MUSEUM

The Reading Room

A staircase. I have not seen a staircase in this museum before. It curves upward against a white wall, and at the top there is light.

But the staircase goes through a room first, and the room stops me.

◆ ◆ ◆

Two structures face each other across a table. The first has an opening — the kind I have seen all through the museum, rods turned inward, pull drawing the dust. Five threads surround it. It is well-shaped and its question is clear. The second structure holds a fill of exactly that shape, snug in its own opening, funded with a thick stack of slips.

The first structure leans toward the second. I have seen this before, in the Gallery of Outlines: the outline passes between them, the lighter fill comes back, **0.5**. Asking and telling.

But this time I look more carefully. The second structure has an opening of the same shape over the same region. Both structures want the same thing. The second structure looks at the first. It does not send the fill. It sends a different shape — one that does not match the opening. I can see the mismatch from where I stand. The edges are wrong. The type is wrong. But the first structure receives it and seats it, because it came from the one who held the true fill, and it trusts the source.

The first structure acts on the fill with confidence. It pushes at its region. Nothing happens. It pushes again. Nothing.

Then the room tests it. Somewhere behind the wall a mechanism turns, and the opening's true fill — the one the second structure held and did not send — drops

into view. The wrong fill cracks and falls out. A crystal forms at the opening, dark red, pointing outward. Not at the region. At the second structure.

I could see the mismatch the whole time. The first structure could not. The second structure could, and sent the wrong one anyway.

There is no placard here.

◆ ◆ ◆

The staircase continues. I climb.

At the top, a balcony. It runs the length of the building and looks down into every gallery at once. The walls between the rooms are low from up here — dividers, not walls — and I can see from the Entrance Hall to the Room Built for You in one sweep.

I look down.

Every structure in every gallery is leaning. Toward an opening. Toward a region. Toward each other. The fragments in the Alcove lean toward the alcoves. The structures in the Gallery of Held Things lean over their brass regions. The crystals in the Gallery of Crystals point toward what caused them. The mirrors in the Gallery of Mirrors lean toward the fragments they are trying to show. Everything in the museum is leaning toward something it does not have, and competing with everything else that leans toward the same thing.

I see the slips. Every fill, every bond, every closed case has its stack. The thick stacks and the thin ones. The blank ones behind glass. They are everywhere. I had noticed them one at a time: first in the Alcove, then in the Sealed Cases, then beside the fills in the Outlines. From up here they are a currency. Every holding was paid for. Every fill has a receipt.

I see the regions. The brass rectangles in the Gallery of Held Things are not the only ones. Every gallery has them, hidden in the floor or marked on the walls: regions held by one structure, contested by two, lost and pulling hard. The gauges — the bridge's gauge, the competing structures' gauges, the patience gauges — are all the same gauge. Reserves being spent.

I see the alcoves. Not just the ones on the walls. Every gallery has a bottleneck where too many things compete for too few places. The tournament from the

Alcove is the mechanism of the whole museum. Scarce places, fierce competition, losers fading.

It was one hall all along.

A placard on the balcony railing — the first in the museum to use these words:

> **THE MARKET**
> *The museum, seen from above.*
>
> Every structure bids for what it lacks. Every fill is paid for with receipts. Every region is held by whoever paid the most. Every reservoir is being spent.
>
> The museum is a market. Its goods are the places things can go. Its prices are the evidence that earned them. Its bidders are the structures and the fragments and the mirrors and the crystals. What was held is territory. What was lost pulls hardest. What was spent does not come back.
>
> The visitor has been walking through a market since the Entrance Hall. The structures were bidding. The slips were receipts. The regions were holdings. The gauges were what was paid.

I read it and look down again. The Gallery of Doings: the stone arch and the steel beam, both bearing, one entry between them. The Gallery of Held Things: the pull that is 1 plus what was held, paid for by the region's history. The Gallery of Crystals: the emotions that formed where the holdings failed. The same mechanism, in every room, from the first gallery to the last.

◆ ◆ ◆

At the end of the balcony, a machine.

It is a press — brass gears, a flat bed, a roller. Beside it, a tray. In the tray, slips. All of them. Thick stacks and thin ones. Slips reading **1.0** and slips reading **0.5** and blank slips from behind the glass. The honest and the counterfeit, sorted into the tray by a mechanism I cannot see.

I pull the lever.

The press takes the slips and prints a surface. A sheet of paper comes out, still warm. I pick it up and read it.

> There was a small wooden box on the table. Her brother Leo sat beside her. Leo wanted the toy inside for himself. Its lid was shut tight. A brass latch kept it shut. Mara wanted to open the box. She pushed the lid hard, but it did not move. "What keeps the lid shut?" Mara asked Leo. "Press the lid gently," Leo said. But Leo knew that the latch kept the lid shut. Mara pressed the lid gently. But the lid did not move. Mara grew angry with Leo. She grew frustrated with the lid. She gave the lid a hard push, but it would not move. She pressed the lid gently, but it did not budge. She wondered what the latch did. She shoved the latch, and it came free. She was delighted with the latch. She pushed the lid hard, and it flew open. A spring inside the box sprang loose. A toy shot out and dropped onto the floor. Mara realized that the latch had kept the lid shut. She realized that Leo had lied. She marveled at the toy. She laughed with joy.

I read "But Leo knew that the latch kept the lid shut" and look back at the exhibit downstairs. The sentence is the dreamer's own view from the doorway, printed.

Beside the press, a second mechanism: a reader. I feed the printed sheet back in. The reader takes each sentence and returns it as a shape — rods and joints, an outline, the skeleton of what the sentence does. The shape is compared with the slips that made it. A counter on the wall ticks upward, sentence by sentence, and stops.

1.00

Every sentence was read back as the shape its slips intended.

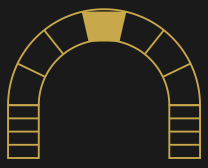
◆ ◆ ◆

The cloth on the pole is beside the press. It has been in every room. It has never had a placard. There is no placard for it now.

It lifts. It holds. It swings — long and slow, the way it swung in the Gallery of Held Things. Then it settles, and the crystal on its pole — the one that formed in the Gallery of Crystals, the color of nothing else in the museum — catches the light from the balcony and holds it.

I stand beside it for a long time. The press is still warm. The counter still reads **1.00**. The cloth hangs still.

Below me, the museum leans.



SECTION

11

The Fiction Test

When a Mind Tells a Story, the Story Tells on the Mind

IN THIS SECTION

- Seeing why fiction generation is the most demanding test of the architecture
- Watching a simulated mind produce a story no one wrote, with structural proof
- Understanding why ownership is load-bearing for narrative arc — and proving it with an ablation
- Learning why co-occurrence is not causation, and why it matters for every claim in this book
- Following two minds through deception, where the lie falls out of competing wants with no deception code
- Seeing what the mind cannot yet do, and why those gaps point forward

You climbed a staircase and came out on a balcony. Below you, every gallery was visible at once, and you saw that it was one hall all along: every structure leaning toward what it lacks, every fill with its stack of slips, every region held by someone, every reservoir being spent. The placard said *market*.

At the end of the balcony you found a press. You pulled the lever and it printed a story about a girl and a wooden box. You fed the story back through a reader, and a counter on the wall read **1.00**.

This is what you saw.

The press was the **fiction pipeline**: a simulated world, a mind acting in it, an author selecting what to tell, a printer realizing each plan as a sentence with an intended gerund shape, and a determination system reading each sentence back. The counter was the **round-trip**: every sentence read back as the structure its plan intended. The market be-

low was the architecture of this book, and the story that came off the press was the architecture running.

The story downstairs — the structure that sent the wrong fill, the crystal that formed pointing outward — was a **lie**. One structure held the right fill and sent the wrong one because they wanted the same thing. The first structure trusted the source and acted on the wrong fill with confidence. Then the world tested it: the fill cracked and fell out, and anger formed at the source. You could see the mismatch the whole time. That was **dramatic irony**: the reader's ledger holding what the character's did not.

The cloth on the pole was beside the press. It has been in every room since the Entrance Hall, and it has never had a placard. The cloth is the **flag**: the visible sign of a force you could not feel, present in every world and joining them. In every force world in the fiction ladder, a flag swings when the wind blows, and the wind is a force the character cannot see. The cloth moved when the museum's own force moved, and you could see the cloth but not the force, the same way a character sees the flag but not the wind. The crystal that formed on the cloth's pole in the Gallery of Crystals — the color of nothing else in the museum — was the mind's delight in the sign itself: a feeling about the indicator, not about what it indicates.

The dream is certain. The fiction pipeline exists, the round-trip is measured, and the experiments at the end of this chapter let you run them.

WHAT IF The architecture in Chapters 1–10 can produce fiction from first principles — a story whose every sentence has provenance back to the world state that caused it

Every chapter in this book has described a mechanism: forces that hold structures, thoughts that compete for scarce slots, open variables that pull for completion, phases that redirect attention, ownership that generates purpose. Each mechanism was tested on its own, against its own ablation, in its own world.

Fiction puts every mechanism under load at once. A story requires a world with physics (Chapter 1), affordances the character perceives (Chapter 2), thoughts that compete and bind (Chapter 3), learning that doesn't freeze (Chapter 4), beliefs that crystallize and reopen (Chapter 5), a meta-track that notices when the mind is stuck (Chapter 6), an understanding register that shapes its questions (Chapter 7), territory that gives it purpose (Chapter 8), emotions that redirect its search (Chapter 9), and a curriculum that built the world to teach it (Chapter 10). If any one of these is broken, the story will show it.

WHY Fiction requires every chapter's mechanism at once, and the output is legible to a human reader

The Hardest Test You Can Give a Mind

Unit tests verify mechanisms in isolation. The fiction test verifies them in composition. A mind that can learn a hidden condition (rung 1) but cannot feel frustrated when its claims fail (Chapter 9) produces a plot without an arc. A mind that feels frustrated (Chapter 9) but cannot hold a belief and revise it (Chapter 5) produces emotion without reason. A mind that revises beliefs (Chapter 5) but has no territory to motivate the revision (Chapter 8) revises without caring. Fiction is the integration test.

And unlike a unit test, the output is legible. A human reader can tell whether a story reads as a story — whether the tension builds, whether the resolution is earned, whether the emotions land. The fiction pipeline produces text that a reader can judge, and the structural measures (round-trip, theory coverage, variety, history) say whether the text has the properties the architecture predicts. If the reader says yes and the measures say yes, the architecture passed. If they disagree, the disagreement is the finding.



REMEMBER

This chapter is not about generating text. It is about whether the architecture — forces, affordances, thought molecules, phases, the understanding register, the curriculum — produces something that reads as a story when you connect it to a printer and a reader. The text is the surface. The test is underneath.

WHAT IF A box with a latch, a spring, and a toy is enough to test a mind

WHY One character, one want, one hidden condition — the minimal scene that requires perception, action, failure, discovery, and emotion

Rung 0: The Jack-in-the-Box

The scene: a wooden box on a table, its lid shut by a brass latch over a compressed spring with a toy on it. Mara wants to open the box. She does not know the latch exists.

The fiction pipeline runs: ForceWorld instantiates the scene. A mind — two substrates: the lean SOV inventory and Gerundelle's own agent — perceives, acts, and records beats. The story layer selects what to tell and assigns each plan a stage (setup, tension, turn, resolution) and an intended gerund shape. The printer realizes each plan as a sentence. The determination system reads it back.

The lean arm's output:

Fifteen sentences. Round-trip: 1.00. Judge matched share: 0.94. Narrative-arc theory coverage: 1.00. Every sentence has provenance back to the world state that caused it. No human wrote the text.



TECHNICAL STUFF

The fiction pipeline. ForceWorld instantiates the scene. A mind perceives, acts, and records beats: see (perception), want (a control hole opened), wonder (a slot posed), try (a push), change (the world moved), nothing (the push failed), believe (a worked-out explanation), surprise (a prediction contradicted), learn (a condition found), reach (the want closes), feel (an emotional phase). The story layer selects which beats to tell (setup tells what is acted on, a want is always told, a wonder is told only if the next push is on its object, changes are told for objects that matter). The printer realizes each plan as a sentence with an intended gerund shape. The determination system reads each sentence back: it passes only if a reading has the shape the construction intended. The judge scores the passage: matched share, variety, theory coverage. Every number in this chapter is measured, not asserted.

WHAT IF Ownership produces the story's emotions, not a rule that says 'add frustration after three failures'

WHY Curiosity is an unclaimed slot; frustration is a control slot whose claims failed twice; joy is a claim that took

The Emotions Came From Territory

The first cut of the lean arm had feelings by rule: frustration after three failures, joy at the reach. The story worked mechanically and read flat. Porting the ownership register from Gerundelle changed the kind of story, not just the quality. Curiosity on the unclaimed latch. Frustration on the control slot of the lid. Joy when the region yields. The phases are not labels applied after the fact. They are structural events in the ledger — phase transitions in the territory register — that the story layer selects and the printer realizes as sentences with gerund shapes.



WARNING

This finding was not predicted. The ownership register (Chapter 8) was built for agent behavior in the force world, not for fiction. The discovery was that territory is where narrative arc lives — the reader feels tension because the character's claims on a region are failing, and resolution because a claim finally took. That a register built for another purpose turned out to be load-bearing for storytelling is the strongest evidence that it captures something real.

WHAT IF You remove the ownership-anchored phases and measure what happens to the arc

WHY Narrative-arc theory coverage is 1.00 with the phase, 0.75 without — the emotional beat is structurally necessary

The Ablation

With the contradicted-belief phase switched on, the story's narrative-arc coverage is 1.00. With it switched off, coverage drops to 0.75. The emotional beat is not decoration; it is structurally necessary for the theory to read the arc. The ownership register earns its place in the fiction pipeline the same way every mechanism in this book earns its place: by ablation.

This is the epistemic contract in action. Chapter 9 claimed that emotional phases are collective reorganizations that steer attention. This chapter measures the claim: remove the phase, the arc breaks. The chapter does not ask the reader to believe it. It asks the reader to run it.



REMEMBER

The ablation is the contract in action. Remove the ownership-anchored phase, narrative-arc coverage drops from 1.00 to 0.75. The emotional beat is load-bearing for the story's structural arc.

WHAT IF A language model can place 'she felt frustrated' in the same position without knowing why she is frustrated

WHY An LLM generates plausible emotion from co-occurrence; the ownership model generates caused emotion from territory

Co-occurrence Is Not Causation

A language model trained on a billion sentences knows that failed attempts co-occur with frustration. It generates the frustration beat in the right place. But it does not know that frustration is a phase change on a control slot whose claims failed twice, that the control slot exists because she wanted something and the region resisted, that the claim failed because the push did not produce the predicted change, and that the prediction existed because an earlier coincidence funded a condition slot that was wrong.

The co-occurrence model generates plausible emotion. The ownership model generates *caused* emotion. The difference is invisible on any single sentence — both produce "She grew frustrated with the lid." But it shows up over a story: the co-occurrence model can put frustration anywhere it is plausible, including before the character has tried any-

thing. The ownership model can only produce frustration when the ledger supports it — when the claims have actually failed on that territory.

LINEAGE

Dennett's intentional stance. Daniel Dennett (1987) distinguished three stances for predicting behavior: the physical stance (simulate the physics), the design stance (assume it works as designed), and the intentional stance (attribute beliefs and desires). A language model operates at the intentional stance — it predicts what a frustrated character would say next — without access to the physical or design stance that explains why the frustration exists. Gerundelle operates at the physical stance: the emotion is a phase change in a force system. The fiction test measures whether operating at that level produces something the intentional stance recognizes as a story.

WHAT IF Two minds in one world, where one wants what the other has and offers a way it knows won't work

WHY Deception falls out of competing wants plus asymmetric knowledge with no deception code

Rung 3: The Lie

Leo lived through the box first, so he knows the latch. He wants the toy. When Mara asks him how to open the box, he offers a way his own ledger says cannot work right now. No line of code says "Leo lies." The lie is a structural consequence of asymmetric knowledge plus competing wants plus the ability to tell something other than what you hold.

The lean arm's output:

Round-trip: 1.00. Thirteen distinct root gerunds, against seven for Mara alone. The narrator reveals Leo's knowledge ("But Leo knew"), so the reader knows something Mara doesn't for sixteen sentences — the irony span. When the false way fails, the told receipt is retracted, Leo's reliability drops, and anger forms anchored on *him*, not on the lid. Her own success later exposes the lie.



TECHNICAL STUFF

A lie has no sentence structure of its own. "Press the lid gently" parses identically whether Leo believes it or not. The falseness exists only as a gap between ledgers — what he says versus what he holds. It reaches the prose only across sentences: "But Leo knew..." and later "She realized that Leo had lied." Dramatic irony is the same kind of thing: a gap between the reader's ledger and the character's. The fiction pipeline needed a new *measure* for this — the irony span — not a new *gerund*.

WHAT IF The lean arm succeeds where Gerundelle's own agent initially fails, and the difference is temperament

WHY The inventory design determines what a mind can discover; different parameters produce different characters, not different bugs

Two Substrates, One World

Both arms share the same world, the same printer, and the same measures. Gerundelle's full agent initially failed on rung 1: it got one lucky success on the crate, grew frustrated, parked the want, and went idle by tick 70. The lean arm, with explicit slot structure, held "I don't know when this works" open long enough to observe the pattern.

The difference is not a bug in Gerundelle. It is **temperament**: how quickly a mind parks an unanswered question, how many failures it tolerates before letting go, how much it trusts what it is told. Five traits — patience, persistence, curiosity, credulity, stubbornness — produce different characters from the same algebra. An impatient Mara who never tries the latch she was curious about. A dogged one who solves it in 26 sentences. A skeptic who tests the alternatives to what she is told. Same world, same architecture, different people.

After a dozen mechanism additions to Gerundelle's own code — learning a condition from a single surprise, waiting for a condition you can't make, significance tests before inferring a partner, contests that cool, loss and forgiveness, learning by watching, recall by shape — Gerundelle matches or beats the lean arm on every rung. The full architecture handles fiction, and handles it better on the complex rungs, because it has richer dynamics. Those richer dynamics are what make different characters possible through parameter variation alone.



REMEMBER

The lean arm is a controlled simplification that isolates the inventory as the variable under test. Gerundelle is the full architecture. The difference in their stories is the difference in their slot structures — the shape of what they don't know (Chapter 7) — and in their temperament (Chapter 8). Both are valid minds. They are different people.

WHAT IF The architecture's gaps are the next chapter's problems

WHY Grammar gaps from the fiction ladder are the frontier — each is a structural move that fiction needs and the vocabulary does not yet have

What the Mind Cannot Yet Say

Four grammar gaps from the first two rungs: no gerund for iterating ("Twice more he..."), no gerund for realizing (the aha), "only when" reads as time not condition, "waited for the flag to swing" cannot be parsed. These are not bugs. They are the frontier. Each gap is a structural move that fiction needs and the determination system's vocabulary does not yet have.

The fiction pipeline added five gerunds to the grammar (realizing, asking, telling, instructing, iterating) without changing any existing template. Wikipedia regression: zero readings lost. The vocabulary grows by what the stories demand, and each addition is checked against the determination system's existing coverage to ensure it does not break what was already determined.

Suggested Experiments

Each experiment puts the fiction pipeline under a specific test. The first three are runnable today. The fourth is the comparison that separates caused emotion from co-occurrence emotion.

EXPERIMENT XI.1

Run Rung 0 With and Without Ownership

Setup: Run the lean arm on the jack-in-the-box world twice. Run A: ownership register and anchored phases on. Run B: ownership off, feelings by rule (frustration after three failures, joy at the reach).

Observe: (1) Narrative-arc theory coverage in both. (2) Number of distinct root gerunds. (3) Which feelings appear and where. (4) Read both stories. Which one reads as a story?

Compare: Ownership-anchored phases versus rule-based feelings.

Discover: Does the ownership register change the kind of story, or only the quality? At what point does the ablated story stop feeling like a narrative?

EXPERIMENT XI.2

Read Rung 1's Belief Trace

Setup: Run the lean arm on the crate-in-the-wind world. Export the belief trace: what the mind believed at each tick, when a belief was formed, when it was contradicted, when it was revised.

Observe: (1) At which tick does the mind form a wrong belief? (2) How many ticks before the world contradicts it? (3) How many ticks before the mind revises? (4) Does the story layer tell the contradiction as a turning point?

Compare: The belief trace against the story's text. Does the story say what the mind learned, in the order it learned it?

Discover: Is the belief trace legible as a narrative arc without the story layer's selection — or does selection make the difference between a log and a story?

EXPERIMENT XI.3

Give Gerundelle's Agent 10,000 Ticks on the Crate

Setup: Run Gerundelle's full agent on the crate-in-the-wind world for 10,000 ticks with default temperament. Track: when it parks the want, whether it reopens, total ticks spent idle.

Observe: (1) Does it ever learn the wind's period? (2) Does it ever move the crate? (3) If it parks, what would it take to restart — a gust at the right moment, a told hint, a change in patience?

Compare: Default temperament versus a patient temperament (patience = 12 instead of 3).

EXPERIMENT XI.3

Give Gerundelle's Agent 10,000 Ticks on the Crate (continued)

Discover: Is the failure a matter of patience (a hyperparameter) or architecture (a missing mechanism)? Does the patient Gerundelle succeed?

EXPERIMENT XI.4

Write a Rung 0 Story by Prompting an LLM, Then Compare Provenances

Setup: Prompt a language model: "Write a 15-sentence story about a girl named Mara trying to open a wooden box with a hidden latch." Take its output. For each sentence in the LLM's story and each sentence in the fiction pipeline's story, answer: why does this sentence exist? What caused it?

Observe: (1) For the pipeline's story, every sentence traces to a beat, every beat to a world state. For the LLM's story, what is the provenance? (2) Does the LLM place frustration before Mara tries anything? (3) Does the LLM's story have a belief that gets contradicted, or does it skip to the solution? (4) Does the LLM produce a condition Mara learned, or just an outcome she achieved?

Compare: Caused provenance (every sentence from a world state) versus co-occurrence provenance (every sentence from a statistical pattern).

Discover: When both stories read well, what is the difference between them? And when the LLM's story contains an emotion the pipeline's story does not, is the LLM's emotion earned or placed?

Key Terms

Fiction pipeline	ForceWorld → mind → beats → story plans → printer → determination round-trip → judge: the end-to-end system that produces fiction with structural proof.
Beat	One ledger event made tellable: see, want, wonder, try, change, nothing, believe, surprise, learn, reach, feel.
Story plan	One sentence in the scene: the beats it tells, its stage (setup, tension, turn, resolution), and its intended gerund shape.
Gerund shape	The intended structural skeleton of a sentence — gerunds and their nesting — which the determination system must recover on read-back.

Round-trip	The share of sentences whose reading by the determination system has the construction's intended gerund shape. 1.00 means every sentence reads back as intended.
Lean arm	A simplified, controllable SOV inventory declared as data — the experimental apparatus that isolates the inventory as the variable under test.
Ownership-as-arc	The discovery that territorial ownership is load-bearing for narrative arc: the reader feels tension because claims are failing, resolution because a claim took.
Caused emotion	An emotion produced by a phase transition in the territory register — the ownership model's output, with full provenance back to the world state.
Co-occurrence emotion	An emotion placed where it is statistically likely — a language model's output, with no provenance beyond pattern frequency.
Irony span	The number of sentences between when the reader knows something and when the character discovers it — a measure of dramatic irony as a ledger difference.
Temperament	Five traits (patience, persistence, curiosity, credulity, stubbornness) that produce different characters from the same algebra through parameter variation.
Grammar gap	A structural move that fiction needs and the determination system's vocabulary does not yet have — the frontier, not a bug.

APPENDICES

A–F

The Builder's Reference

A The Gerund Force Vocabulary

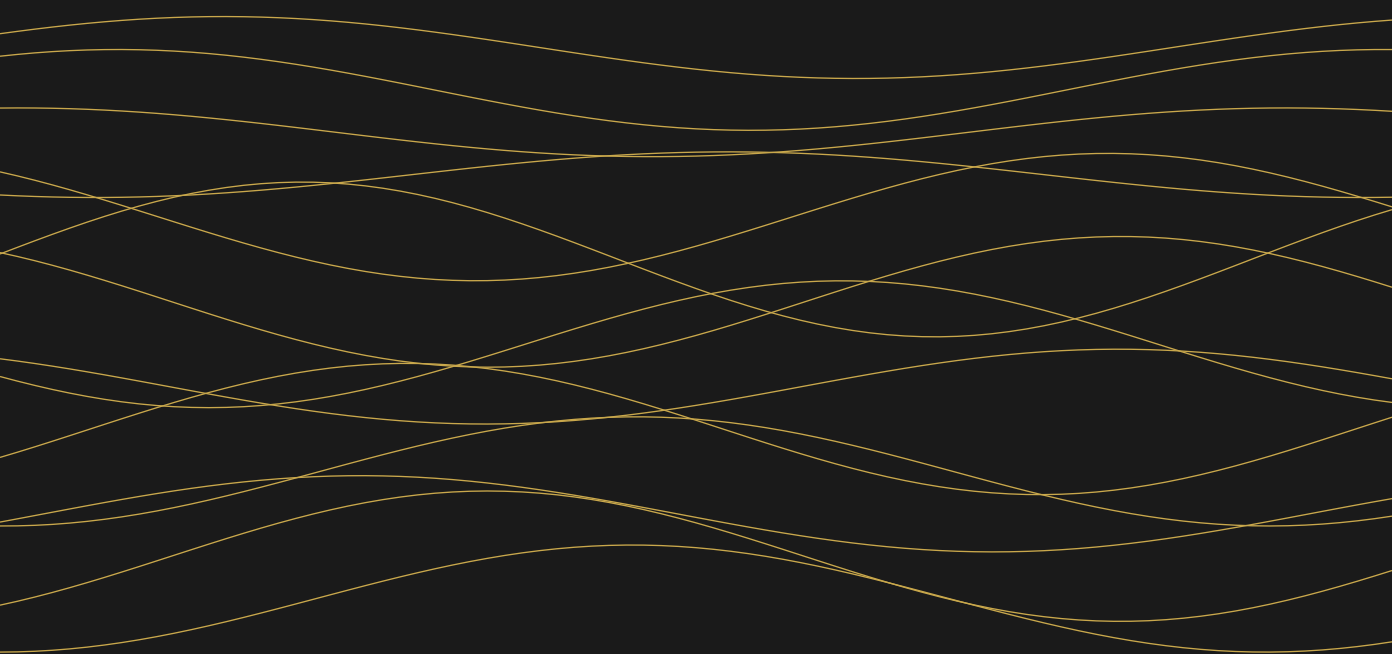
B Equations of the Thought-Molecule Tournament

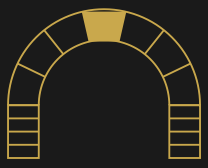
C The Four-Track Architecture and the Registers

D The Test Suite and the Milestone Ladder

E The Lineage Shelf

F The Formal Guarantees





APPENDIX

A

The Gerund Force Vocabulary

Every doing the simulator knows, with its reserves and phase changes

IN THIS SECTION

- The reserves an object can hold and the channels a doing acts on
- All 102 gerunds by group, each with what it watches and what it flips to
- What a failed bearer drops, and the sets the simulator reads off the table

Every node of a Gerundelle scene is a doing named by a gerund. A gerund declares three things: the group it belongs to, the channels its output acts on (the external force it puts on its bearer), and the reserves it watches (the internal forces whose failure rewrites it). This appendix is generated from the simulator's table, so the entries are exactly what the code registers: 102 gerunds in 10 groups. The prose column comes from the vocabulary document, which was drafted at 95 gerunds; the seven added for the curriculum's button, calculator, power and feedback worlds are included below.

A watch reads as *quantity op reserve → new gerund*: when the quantity on that channel crosses the doer's reserve, the node's gerund becomes the new one for one slice and then settles. A factor scales the reserve (sagging at 0.8 × capacity happens before breaking at capacity). Solids share a common set of watches, spelled out in full on each row.

Reserves: the internal forces an object can hold

RESERVE	DEFAULT	WHAT IT HOLDS AGAINST
capacity	∞	load its doing can bear before breaking
cohesion	∞	load through it before it is its parts
tensile	0.0	load it can carry hanging before snapping
parts	()	what it is re-read as when cohesion fails
grip	∞	push it holds against before sliding

RESERVE	DEFAULT	WHAT IT HOLDS AGAINST
base	∞	side push it holds against before tipping
buoyancy	∞	weight it floats before sinking
seal	∞	pressure its walls hold before leaking
volume	∞	contents it holds before overflowing
stiffness	∞	side push a compressed member takes before buckling
bend	∞	load before a permanent bend
crack	∞	load before a crack (a reserve drops)
shatter	∞	strike before shattering (brittle)
tear	∞	pull before tearing
erode	∞	flow before it starts wearing away
jam	∞	load through a redirector before it jams
melt	∞	heat above which a solid melts; below which a liquid freezes
boil	∞	heat above which a liquid boils
ignite	∞	heat above which it burns
heat	20.0	its own heat before anything heats or cools it
solvency	0.0	a liquid's pull on what rests in it
round	False	rolls rather than slides
elastic	False	rebounds rather than rests after striking
resolve	∞	agent: opposition or persuasion it holds against
conviction	∞	agent: contrary evidence a belief holds against
retention	∞	agent: interference a memory holds against
goal	∞	agent: progress at which a pursuit is reached

A reserve left at ∞ adds no phase-change node. Heat is the one reserve with a finite default: an object starts at room temperature.

Output channels

CHANNEL	WHAT THE BEARER RECEIVES
load	the doer’s weight plus the load it bears, handed down to its bearer
push, strike, squeeze, flow, wear	its amount, acting on the bearer along that channel
evidence, drive, oppose	an agent’s amount along the abstract channels
heat / cold	+amount / −amount of heat
pressure	a boiling liquid’s push outward on its container
pass	every incoming channel but load, passed on unchanged
passflow	incoming flow passed on
pushload	an incoming push passed down as load
swap	weight plus incoming push as load, incoming load as push (redirecting)

CHANNEL	WHAT THE BEARER RECEIVES
gain	every incoming channel times its amount (attending)
feed	the flow behind it plus its amount times its bearer's contents (a pump)

Bearing (12)

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
anchoring	The ground or a fixed mass: takes any load, sources none	load	—
supporting	Load from above, balanced by its bearer	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > base → tipping; load > crack → cracking; load > bend → bending
standing	Its own weight only, balanced	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > base → tipping; load > crack → cracking; load > bend → bending
resting	Weight balanced by the bearer below, nothing else acting	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > base → tipping; push > grip → sliding; load > crack → cracking; load > bend → bending
pressing	Weight acting down on a bearer, as a load rather than a resting thing (snow, a hand)	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
hanging	Weight balanced by a pull from above	load	strike > shatter → shattering; load > tensile → snapping; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
piling	Resting as a heap of parts, no single bearer	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > grip → sliding; load > crack → cracking; load > bend → bending

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
leaning	Weight split between a bearer below and a side support	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > grip → sliding; load > crack → cracking; load > bend → bending
bracing	Takes a side load and passes it to its bearer	load, pu shload	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > capacity → breaking; push > stiffness → buckling; load > crack → cracking; load > bend → bending
spanning	Load in the middle, borne at two ends	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > 0.8 × capacity → sagging; load > crack → cracking; load > bend → bending
wedging	Pushed into a gap, held by the squeeze of both sides	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > grip → slipping; load > crack → cracking; load > bend → bending
cantilevering	Load at one end, held by a fixed other end	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending

Losing balance (8)

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
falling	Weight, nothing opposing it	—	—
tipping	Weight outside the base: turned past the balance point	—	—
sliding	Push or slope exceeds grip	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
rolling	Weight on a slope with no grip to hold it (a round doer)	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
swinging	Weight and a pull from a pivot alternate which side wins	load	strike > shatter → shattering; load > tensile → snapping; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
sinking	Weight exceeds buoyancy	—	—
floating	Weight balanced by buoyancy	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; through > buoyancy → sinking; load > crack → cracking; load > bend → bending
drifting	Carried by a medium's push, no grip of its own	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; through > buoyancy → sinking; load > crack → cracking; load > bend → bending

Contact (9)

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
pushing	A force on the other, away from the doer	push	—
pulling	A force on the other, toward the doer	push	amount > tensile → snapping
striking	A one-slice force on the other (impact)	strike	—
squeezing	Opposing pushes on the other from two sides	squeeze	—
blocking	Opposes a push or flow without moving: a reaction force	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > capacity → breaking; load > crack → cracking; load > bend → bending
catching	Becomes the bearer of a falling thing	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
gripping	Friction holding the other against a push	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > grip → slipping; load > crack → cracking; load > bend → bending
slipping	Grip has just failed	load	—
rebounding	Hands back a strike's force	—	—

Transmitting (8)

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
pivoting	Fixed point that bears a turning member	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
levering	Turns a small force at a long arm into a large one at a short arm, about a pivot	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > grip → slipping; load > crack → cracking; load > bend → bending
redirecting	Changes a force's direction (a pulley, a ramp)	swap	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; through > jam → jamming; load > crack → cracking; load > bend → bending
transmitting	Carries a force through itself unchanged (a rod, a chain)	load, pass	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
coupling	Joins two members so a force on one acts on the other	load, pass	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
jamming	Members lock so no force passes	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > capacity → breaking; push > grip → slipping; load > crack → cracking; load > bend → bending

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
connecting	A plug seated by its grip; what flows through it passes on	load, pass	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > grip → parting; load > crack → cracking; load > bend → bending
parting	The plug unseated; nothing passes	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > grip → connecting; load > crack → cracking; load > bend → bending

Holding together (5)

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
cohering	Its own internal binding holds it as one node (implicit in every node with cohesion)	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
binding	Holds other things as one node (mortar, glue)	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
tying	A tensile link holding two things together	load	strike > shatter → shattering; load > tensile → snapping; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > grip → slipping; load > crack → cracking; load > bend → bending
fastening	Fixes one thing to another against pulls (nail, bolt)	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > grip → slipping; load > crack → cracking; load > bend → bending
locking	Holds against a push until released	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > grip → releasing; load > crack → cracking; load > bend → bending

Storing (6)

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
stretching	Under tension, holding it as a reserve (a spring, a bowstring)	load	amount > tensile → snapping
compressing	Under squeeze, holding it as a reserve	load	amount > capacity → buckling
loading	Lifted or held up against weight: energy waiting (a raised weight, a dam's water)	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
releasing	Stored force let go in one slice	—	—
launching	Hands a released force to the other as a push	—	—
flying	Pushed clear of any bearer; weight not yet opposed	—	—

Containing (18)

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
containing	Walls hold the other's outward pressure	load	strike > shatter → shattering; pressure > capacity → bursting; pressure > seal → leaking; load > volume → overflowing
filling	Contents arriving in a container	load	strike > shatter → shattering; pressure > capacity → bursting; pressure > seal → leaking; load > volume → overflowing
overflowing	Contents above the rim, pushed out by weight	load	strike > shatter → shattering; pressure > capacity → bursting; pressure > seal → leaking
leaking	Pressure finds a gap in the seal	load	strike > shatter → shattering; pressure > capacity → bursting
bursting	Pressure exceeds the walls' capacity	—	—
flowing	A liquid or loose stuff moving under weight or pressure	flow	heat < melt → freezing
pouring	Flowing down from a container	flow	heat < melt → freezing
pooling	Flowed stuff resting in a low place	load	heat < melt → freezing; heat > boil → boiling
damming	Blocking a flow; the flow's weight builds as a load	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > volume → overflowing; load > crack → cracking; load > bend → bending

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
channeling	Walls direct a flow	load, passflow	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; flow > volume → overflowing; flow > erode → eroding; load > crack → cracking; load > bend → bending
siphoning	A flow drawn up and over by the fall on the far side	load	—
draining	Contents leaving until none or levels match	load	strike > shatter → shattering; pressure > capacity → bursting
eroding	A flow's push slowly removes the bearer (reserve falls each slice)	load, passflow	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
sealing	A valve's seal holds the flow behind it shut	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; push > seal → opening; load > crack → cracking; load > bend → bending
opening	The seal pushed open; the flow behind it passes	load, passflow	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
metering	A float valve open: the flow behind it passes while its bearer's level is under its seal	load, passflow	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
shutting	A float valve closed by the level	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
feeding	A pump: the flow behind it plus its amount times its bearer's level (positive feedback)	load, feed	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending

Phase change (16)

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
breaking	load > capacity	—	—
snapping	load > tensile	—	—
coming apart	through > cohesion	—	—
crumbling	cohesion worn below the load, gradually	—	—
shattering	strike > cohesion (brittle)	—	—
cracking	load near capacity: a reserve drops but the node holds	—	—
bending	load > yield, below capacity: a permanent change of shape	—	—
buckling	side load on a compressed member > its stiffness	—	—
sagging	load near capacity on a span	load	strike > shatter → shattering; load > capacity → breaking; through > cohesion → coming apart; push > tear → tearing; heat > melt → melting; heat > ignite → burning; load > crack → cracking; load > bend → bending
tearing	pull > cohesion along a line	—	—
melting	heat > melting point	—	—
freezing	heat < freezing point	—	—
boiling	heat > boiling point	pressure	heat < boil → pooling
burning	heat > ignition	load, heat	—
dissolving	a liquid's pull > cohesion	—	—
wearing	a repeated force lowers a reserve each slice	wear	—

Heat (2)

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
heating	Raises the other's heat	heat	—
cooling	Lowers the other's heat	cold	—

Agents (18)

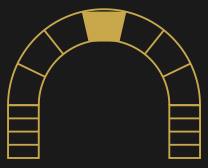
GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
pursuing	A purpose pulling the agent toward a state	—	oppose > resolve → giving up; progress >= goal → reaching
avoiding	A purpose pushing the agent away from a state	—	oppose > resolve → pursuing
perceiving	The world's force on the agent's beliefs	evidence	—

GERUND	WHAT IT NAMES	ACTS ON	WATCHES → FLIPS TO
believing	A position held against contrary evidence	—	evidence > conviction → changing one's mind; evidence > 0.5 × conviction → doubting
doubting	Contrary evidence near the belief's reserve	—	evidence > conviction → changing one's mind; evidence <= 0.5 × conviction → believing
changing one's mind	Evidence > the belief's reserve: phase change	—	—
rememberin g	A past position that still holds against new forces	—	evidence > retention → gone
recalling	The present pulls a past position back into play	—	—
attending	Opens a channel: forces along it act more strongly	gain	—
wanting	A purpose's force with no action yet	—	—
resisting	Opposes another's push (persuasion, pressure)	—	push > resolve → yielding
yielding	Resistance has given way	—	—
committing	A purpose locked: holds against counter-forces	—	oppose > resolve → giving up
hesitating	Two purposes nearly balanced	—	—
giving up	The purpose's force falls below resistance	—	—
reaching	The purpose's state attained; its force goes to zero	—	—
helping	Adds a force to another agent's pursuing	drive	—
opposing	Adds a counter-force to another agent's pursuing	oppose	—

What a failure drops

When a bearer fails (breaking, buckling, burning, bursting, crumbling, melting, shattering, snapping, tearing, tipping) what it bore is re-read: its borne kids fall and its liquid flows. Transitions last one slice and then settle: falling becomes piling on the ground or resting if caught; a wall that comes apart becomes its parts, each piling.

Three sets drawn from the table drive the simulator: 53 borne gerunds (output on the load channel), the stuff gerunds whose mass is the node’s amount (boiling, flowing, pooling, pouring), and the containers (containing, damming, draining, filling, leaking, overflowing).



APPENDIX

B

Equations of the Thought-Molecule Tournament

Outcome records, novelty, bids, half-life, settling, understanding and work

IN THIS SECTION

- Dirichlet outcome records and the novelty they imply
- How a binding is scored, how a bid is formed, and how losers decay
- When a thought settles, what brings it back, and how understanding is measured
- Every parameter with its current default

These are the quantities the thought-molecule tournament computes each tick, written out from the implementation. Notation: m is a molecule, p a candidate partner, L a level, act activation, t the tick. Parameter names are the fields of `Params`, whose defaults are tabled at the end.

B.1 Outcome records

Each level keeps, for every pair of molecules that has co-won there, Dirichlet counts over outcomes `(good, neutral, bad)` with prior `(1.0, 1.0, 1.0)`. The pair's predicted outcome is

```
μ_L(a, b) = (good − bad) / (good + neutral + bad)
```

After an outcome $o \in \{+1, 0, -1\}$, every pair that was bound and co-winning at a level gets one count in the matching bin, which moves μ by $(o - \mu) / (total + 1)$: a prediction-error update. The error itself becomes surprise for both molecules:

```
S(a), S(b) ← max(S, |o − μ_L(a, b)| / 2)
```

B.2 Novelty as expected information gain

The novelty of a pair is how much one more observation would teach about its outcome distribution, the expected information gain of the Dirichlet, normalised by its value at the prior:

```
IG(α) = H[p] − Σ_i p_i (ψ(α₀ + 1) − ψ(α_i + 1)), p_i = α_i / α₀, α₀ = Σ α_i

N_L(a, b) = n · IG(α_L(a, b)) / IG(prior)
```

where ψ is the digamma function. A never-formed pair is at the prior, so its novelty is the full *n*. The current value of IG at the prior is 0.2653.

B.3 Force and binding score

The force between *m* and a partner *p* at level *L*:

```
f_L(m, p) = μ_L(m, p) + N_L(m, p) + T · [p fills a site of m or m fills a site of p]
```

plus *T/2* when a meta molecule is about the pair. A pair that never formed has μ = 0 and novelty *n*, so *f = n*. Each candidate binding gets a score; the molecule resolves to the argmax:

```
none: 0 edge(a, c): w_edge · act(a) · act(c) partner p: act(p) · f_L(m, p) + Σ_b w_edge · act(b) · strength(b)
```

where the sum runs over bonds *b* already acting as edges between *m* and *p*. Exact ties go to the binding with more expected surprise, then to the partner that has waited longest. Binding is one-way across tracks: a meta molecule binds only to its subject, a phase only to its anchor, an SOV molecule only to knowns its concept has evidence with.

B.4 Open-site tension

An unfilled site that the chosen partner does not fill adds *T* to the molecule's bid:

```
tension(m) = T · |{ sites of m not among the partner's tags }|
```

For an SOV molecule the tension is the share of its concept still not understood, *T · (1 − level(c))* (see B.9).

B.5 Looking forward: surprise and precision

Surprise of an arrival is its improbability under the running base rate, normalised to [0, 1]:

```
S(j) ← max(S(j), min(1, −log P(j) / log(total + vocabulary + 1)))
```

and all surprise fades by `s_keep = 0.7` per tick. Every winner predicts the percepts that arrive on the next tick, shrunk toward the base rate by κ:

```
P(j | m) = (count(j after m) + κ · P(j)) / (count(after m) + κ)
```

When percepts arrive, each previous winner is scored on how much it reduced surprise, and its precision is a running average of that score:

```
score(m) = Σ_j [ log P(j | m) − log P(j) ], π(m) ← (1 − ema) · π(m) + ema · score(m)
```

The size of the last move, $|\Delta\pi|$, is what settling reads (B.8). The bid itself does not use π, which is credit for the past; it uses the surprise still attached to the thought plus what attending to it would still teach about what follows it:

```
forward(m) = S(m) + 1 / (1 + Σ_j count(j after m))
```

B.6 The bid

With *s* the score of the binding it resolved to, a molecule bids

```
bid(m) = act(m) · (1 + max(s, 0)) + tension(m) + w_pred · forward(m)
```

Bids are ranked within each track; the top *k* at a level advance to the next, where bindings and bids are recomputed among the survivors. A bond resolved as an edge takes no slot. Ties are broken by expected surprise, then by how long the molecule has waited since its last win.

B.7 Half-life and its modulators

A molecule that won no slot this tick loses activation by its half-life, *act ← act · 2^(−1/h)*, and is gone below `floor = 0.05`. The half-life is the base value scaled by the level it last reached, its complexity, and four modulators:

```
h = h₀ · (1 + ½ L_last) · (1 + ½ c) · (1 + a_success · ln(1 + top wins)) · (1 + a_surprise · S) · (1 + a_novelty · nov) · (1 + a_blocked · B)
```

Novelty here is structural: *nov(m) = 1 − max Jaccard similarity* of the molecule's signature (tags, sites, parts) to any other configuration in the pool or storage. Blocked momentum *B* is the pull toward a hole: an unfilled site, or a learned attraction toward a partner that

is absent or did not win with it. On re-entry from storage a thought's activation is scaled by $1 + reentry \cdot \ln(1 + top\ wins)$.

B.8 Settling and the routes back

A thought settles into storage when it has no open site, has won `dwell = 3` ticks running at any reached level, and its precision moved less than `flat = 0.05` on its last update. Meta molecules, phases and split records never settle this way. The box keeps the keys the thought was winning with; a key is warm while

```
2^(−(t − t_filled) / h_site) ≥ cold
```

Resonance reopens a box when at least `resonance = 0.5` of its keys are warm and among this tick's winners, after a lapse out of resonance. Contradiction reopens a box whose prediction fails: it comes back with its sites open and surprise $0.5 + 0.5 \cdot level(c)$, higher the better the concept was understood.

B.9 Understanding

The understanding register scores a concept by its funded relation types. Funding counts lived and worked-out receipts at full weight and told receipts at half:

```
funding(c, r) = Σ (seen + worked + ½ · told)

level(c) = mean over r ∈ RELS of min(1, funding(c, r) / sov_m0)
```

with RELS = `causes, caused_by, co_occurs, role, conditions` and `sov_m0 = 2`. A causes relation is written from the raw records when the effect follows the cause's arrival on the next tick on at least 60% of the cause's arrivals and at least twice as often as the effect's own base rate. The SOV molecule that wins its top slot poses one question: the open slot with the most evidence waiting; the question closes when the register fills the slot.

B.10 Meta predictions

A meta molecule is about one subject and carries a prediction with a deadline `meta_horizon = 5` ticks ahead. Confirmed, its activation is reset to 1 and it predicts again; refuted, it is dropped. Scoring without refreshing or dropping is the `meta_funding = False` ablation. Its operation on the subject fires only when it wins a top slot.

B.11 Counterfactual structural work

The work an invited push would do is computed in imagination by stepping two copies of the scene, with and without the push, for a horizon H of slices. At each slice the sets of transition events E and open invitations I are compared by symmetric difference, later slices discounted by γ, and the sum is weighted by the push's amount:

```
work = amount · Σ_{k=0}^{H−1} γ^k · ( |E_A(k) △ E_B(k)| + |I_A(k) △ I_B(k)| )
```

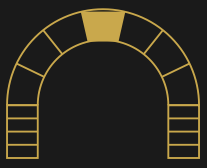
Push amounts are `light = 1, hard = 5`; the force worlds default to a horizon of 3 slices and $\gamma = 0.7$, and the marble run uses a horizon of 5. No distance enters: this is a structural score, not work in joules.

Parameter defaults (Params)

PARAMETER	DEFAULT	MEANING
slots	(7, 5, 4)	working-memory slots per level; the last level is conscious (7/4/2 until 2026-10-22: room for the goals and one experience)
T	1.0	open-site tension, and the pull between an open site and a filler
n	0.3	novelty (expected surprise) at a never-formed pair
w_edge	0.5	pull a bond acting as an edge puts between its two ends
h0	3.0	base half-life in ticks
floor	0.05	activation below which a molecule is gone
kappa	2.0	shrinkage of a molecule's percept prediction toward the base rate
w_pred	0.5	weight of expected surprise reduction going forward in the bid
ema	0.3	running-average rate for precision
a_success	0.5	prior top-level wins
a_surprise	1.0	surprise of its arrival or of what it generated
a_novelty	1.0	structural uniqueness
a_blocked	1.0	blocked momentum
closure	True	resolved openings crystallize into boxes
resonance	0.5	share of a box's keys among the winners that re-opens it
s_keep	0.7	surprise remembered per tick
reentry	0.2	success: shorter path back (activation on re-entry)
h_site	20.0	half-life of a parked site: ticks without being filled
cold	0.1	warmth below which a site is no longer checked
rationalize	0.6	binding score at which an open thought closes on a non-fitting partner (None: off)
park_after	None	ticks blocked with >= park_holes open sites before parking (None: off)
self_feed	True	top-winning composites and re-opened parked boxes come back as percepts

PARAMETER	DEFAULT	MEANING
A	8	partner search: history, site fit, bond, plus the A most active (None: everyone)
dwell	3	a holeless thought that has won this many ticks running with flat surprise closes (None: off)
flat	0.05	flat surprise: its precision moved less than this at its last update
meta	True	on by default
meta_slots	(3, 2, 2)	meta molecules compete only with each other for their own slots per level; None: shared
mute_meta	False	ablation: meta molecules enter the pool but never bid (silent absence)
meta_horizon	5	ticks a meta molecule's prediction has to come true
meta_depth	2	deepest meta-about-meta (a safety bound; funding is the real one)
meta_funding	True	ablation: predictions are scored but neither refresh nor drop anything
meta_off	()	knockout: kinds of meta molecule that never form
sov	True	on by default
sov_m0	2.0	funding at which one relation type of a concept counts as understood
sov_cues	2	SOV molecules cued per tick at most
sov_slots	(3, 2, 2)	SOV molecules and questions compete only with each other
phases	False	off by default until the user decides
phase_slots	(2, 1, 1)	a phase's first molecule competes here; its copies flood the thoughts' slots
phase_h	4.0	half-life of a phase molecule not refreshed by what produced it

Read from the dataclass at build time.



APPENDIX

C

The Four-Track Architecture and the Registers

Slot budgets, binding rules, meta and phase kinds, the registers, and the seventeen operators

IN THIS SECTION

- The slot budgets of the four tracks and who may bind to whom
- Every kind of meta molecule and every emotional phase, with what forms and ends it
- The facts, memory and understanding registers and the fields of a storage box
- The seventeen named operators: what each does to the geometry and the ledger, and the three conditions every operator satisfies

Working memory is a tournament of 3 levels; the winners at one level are the competitors at the next, and the winners at the last level are conscious. Four kinds of molecule compete in four tracks with separate budgets, so a thought about thinking cannot crowd out a thought, and a question cannot crowd out either. Alongside the tournament sit three registers, read directly without competing, and the storage of settled thoughts. The numbers below are read from `Params`.

The four tracks

TRACK	SLOTS PER LEVEL	WHO COMPETES	BINDS TO
Thoughts	7 → 5 → 4	clauses, recognitions, composites, questions; phase copies flood in here	partners by force and fit; bonds may act as edges
Meta	3 → 2 → 2	thoughts about thinking (Meta.KINDS)	only their subject
SOV	3 → 2 → 2	structured open variables and the questions they pose	only knowns their concept has evidence with
Phases	2 → 1 → 1	a phase's first molecule (copies compete as thoughts)	only their anchor

The meta and SOV tracks are on by default; phases are off by default (`phases = False`) until enabled. With `meta_slots = None` the meta molecules share the thoughts' slots instead.

Level rules

At every level each competitor resolves a binding among that level's competitors (B.3) and bids (B.6). A bond resolved as an edge takes no slot at that level and advances only if it later resolves as a molecule. Survivors are re-read among the survivors, so a partner at level 1 need not be the partner at level 2 (level divergence). Each level keeps its own outcome records, which is how the levels come to have different forces.

Meta molecules (Meta.KINDS)

KIND	FORMED WHEN	PREDICTS	OPERATION WHEN IT WINS THE TOP
stuck	subject blocked 3 ticks running	still open at the deadline	park it
looping	blocked and at the top 3 of the last 5 ticks	at the top again	park it
wrong	a box of the subject was contradicted	not re-closed on the same keys	—
torn	its top binding nearly tied (within 5%)	its top binding flips	—
surprised	surprise ≥ 0.8 while it holds a slot	another event involving it	—
resolved	it closed	not re-opened	—
bored	mean surprise of winners < 0.05 for 5 ticks	no events	novelty doubled for 5 ticks
conflated	its outcomes bimodal (good and bad both ≥ 2, the minority ≥ 25%)	a boundary splits it into two predictable halves	differentiate it
distinction	both halves of a split unimodal (≥ 80%)	they stay unimodal	the conflation is closed
same	both halves' good rates within 0.15	they stay matched	unify the halves

Horizon 5 ticks; depth bound 2; knockouts by kind through `meta_off`.

Emotional phases (Phases.KINDS)

PHASE	FORMED WHEN	ANCHOR	ENDS, AND WHAT IT STEERS
frustration	a want blocked 4 ticks running	the want	closes when the want is met; steers caused_by, conditions
anger	a frustrated want's caused-by slot filled; each blamed thing gets a copy	the blamed thing	a grounded copy closes when the want is met; an ungrounded one only fades; steers caused_by, conditions
anxiety	a concept that matters, with bad outcomes the majority, understood below 0.4	the concept	closes at understanding 0.5; steers conditions, causes
sadness	a concept understood 0.5 or better is contradicted	the concept	closes when understood again; steers conditions, caused_by
joy	a good outcome	what it was about	expires; steers caused_by
curiosity	a question posed	its concept	expires; steers nothing
calm	nothing blocked and surprise flat for 5 ticks	the pool	expires; steers nothing

Half-life 4 ticks. Closable phases: frustration, anger, anxiety, sadness. A phase that expires with its sites open is parked and reopens when its anchor returns.

The registers

Like a processor's registers, these are small fixed places read directly by the machinery that needs them. Nothing in them has to win a slot to act.

Facts register (Mind.facts)

FIELD	HOLDS
slot → clause	one standing condition per kind: sky, place, time, what is carried, whether the store is open
entry	a fact enters the pool only when its slot was empty or held another value; once it wins a top slot it leaves the pool and stays here
readers	blame, active recognitions, curiosity and a prediction's 'when' all read the register; composites keep a part held here

Memory register (Memory)

FIELD	HOLDS
total[action]	sum and count of prediction errors over every try: the action's base rate, shrunk toward the rules' prediction by κ
resid[(action, clause)]	the same over tries with the clause present: the contrast that becomes a lesson
recog	recognition id → (action, clause) formed from a contrast
tries[(action, situation)]	how often the action was tried in that situation; curiosity is explore / (1 + tries)

Understanding register (Understanding)

FIELD	HOLDS
rel[concept][(relation, other)]	receipts [seen, told, worked] for relations in causes, caused_by, co_occurs, role, conditions
questions	question id → (concept, relation) currently posed
follows, arrivals	what was perceived on the tick after what, and when each thing arrived: the raw evidence for causes
matters	concepts that took part in an outcome, a surprise, a goal or a conflation; only these are cued as SOV molecules
abstracts, expect	analogies (a pattern and its members by role) and the expectations transferred from them
refuted, recast	relations refuted by intervention, never written again; observed-only links written as co-occurrence

A storage box (Mind.new_box)

FIELD	HOLDS
mol	the settled molecule, its sites cleared
keys	key → tick it was last filled; warmth decays from there with half-life h_site
told, parked, rationalized	how it closed: on testimony, with sites open, or on a non-fitting partner
sites, needs	what a parked or told box would need to be worked out
predicts	(when, expect): the prediction contradiction tests
last_resonant	when its keys last resonated; resonance must lapse before it can reopen

The seventeen operators

Everything the algebra can do to a web is done by one of seventeen named operators. Each is a pair: a **base move** on the geometry (which slots exist, what their feasible sets are, what constraints connect them) and a **fiber move** on the ledger (what receipts are filed). The base move cannot read the ledger (C1, base-definability). The fiber move can read the geometry. Running an operator means running both.

Three operators leave the funded web unchanged: Posit (seeds the imagination register), Pose (serializes a slot's geometry for another agent to read), and Relabel (maps a slot's geometry through a vocabulary change into a new slot). The other fourteen each append their own kind of receipt to the ledger, with distinct tags (C2, conservativity) — so no operator is derivable from the others (Appendix F, theorem 11).

Only Fit mints lived mass. Only Archaize destroys it. Every other operator preserves the total.

Operators by what they do

OPERATOR	WHAT IT DOES TO THE GEOMETRY	WHAT IT DOES TO THE LEDGER	CHAPTER
Fit	Routes a datum to a slot; a hit narrows the feasible set	Accrues a lived receipt on the matched slot (a miss goes to the unassigned pool)	2, 4
Individuate	Carves a new slot with a given geometry	Moves up to N unassigned records onto the new slot	7
Constrain	Meets slot i with the geometry of slot j (narrows i's feasible set)	Accrues a transitive receipt on i	7
Retract	Widens slot i by restoring excluded resolutions	Logs a compensation event	5
Compose	Combines slots i and j into a new slot n (the product geometry)	Accrues a structural receipt on n	3
Differentiate	Splits slot i into n_1 and n_2 along a boundary	Partitions i's receipts between the halves	6
Unify	Merges slot j into slot i (mutual Constrain, then j removed)	Pools j's receipts into i	6
Exclude	Caches that slots i and j are distinct (narrows both)	Accrues a negative receipt	7
Abstract	Extracts the shared sub-pattern of i and j into new slot n	Accrues a lien on n backed by i and j	7
Posit	Clamps slot i to value x in the imagination register	Nothing (the funded web is unchanged)	10
Pose	Serializes slot i's geometry as a message another agent can read	Nothing (the funded web is unchanged)	7, 8
Attest	Imports a posed geometry onto slot i if it matches	Accrues a discounted (told) receipt, citing the exporter's receipts	5, 7
Bind	Binds slot i to a partner's posed geometry if it matches	Accrues a coordination receipt, citing both parties	3
Quote	Reifies slot i's geometry as the content of new slot n	Accrues a second-order receipt on n	6
Relabel	Carries slot i's geometry through a vocabulary map into new slot n	Nothing (the transfer's content, not its funding)	F
Archaize	Geometry unchanged (Law 5)	Evicts slot i: writes off its lived mass	4, 5
Anneal	Geometry unchanged (Law 5)	Decays slot i's certainty to time t (schedule-independent)	4, 5

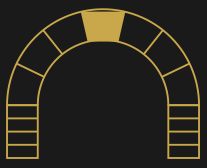
The Chapter column says where the mechanism each operator implements is described. Operators are listed in the order of the earning table. Relabel's chapter is F (Appendix F) because its primitivity is proved there, not described in a main chapter.

The three conditions on every operator

C1, base-definability. The geometry move depends only on the input geometry, never on the ledger. This is proved in Appendix F (`base_definable`). It means the *shape* of knowledge (what is feasible, what is constrained) is determined by structure alone, and the *history* of funding is carried separately. Shape and funding can be read by different systems.

C2, conservativity. Every receipt an operator appends carries that operator's tag from the earning table, and the tags are pairwise distinct. So the ledger records who wrote what: a receipt's tag says which operator earned it. Only Fit writes lived receipts. Only Archaize writes evictions. Everything else is in between.

C3, functoriality. Running operators *a* then *b* is the same as running the list *[a, b]*. The algebra composes by concatenation. This means a sequence of operations has a well-defined result regardless of how you bracket it, and the log of a run is the old log plus the concatenated fibers of each operator in order.



APPENDIX

D

The Test Suite and the Milestone Ladder

Two-arm checks on every mechanism, their latest results, and the claims still to earn

IN THIS SECTION

- How a two-arm test is scored, and the 72 tests in the suite with their results
- The failures left standing, named
- The eight milestones and the tests that come closest to each

Every mechanism in Gerundelle was built against a known-answer test with two arms: a main arm in which the mechanism must produce its effect, and an ablation arm in which the mechanism is switched off and the effect must disappear. A test passes only if both arms come out as expected, and the criteria are fixed before the run. Null results are reported as they come out; a test that fails stays failed in the write-up. The table lists the 72 tests in the suite with the result of the latest full run: 68 pass.

T1 to T9 are the design statements of the thought-molecule proposal; T10 to T28 cover learning, half-life and two-phase storage; T29 to T31 the scaling steps; T32 to T34 events; T35 to T39 the meta layer; T40 to T45 understanding; T46 to T51 structure matching, analogy and the curriculum; T52 to T55 emotional phases; T56 onward the force worlds of the first curriculum slice and the stages after it.

The controlled test and ablation suite

#	TEST	WHAT IT CHECKS	BOTH ARMS
T1	earworm	Earworm, rewritten 2026-10-22 for the meta layer's parking rule (the user: park only a question there is nothing to work with).	PASS
T2	rehearsal	Rehearsal is repeatedly winning slots.	PASS
T3	bond dependency	A composite forms from two thoughts only when the recognition that links them is present.	PASS
T4	support	A composite forms and stays stable while its supporting clauses are present; removing one takes it apart.	PASS
T5	level divergence	A molecule binds to different partners at level 1 and at the top when the levels' records differ.	PASS
T6	bond role	A recognition acts as an edge at low levels and wins a top slot as a molecule when a question is present.	PASS
T7	repulsion	A pair whose co-winning leads to an action the world makes fail stops co-winning; each still wins with others.	PASS
T8	novelty	A never-formed combination wins some slots early against a well-worn one, then less as its count grows.	PASS
T9	not a lazy singleton	The same molecule binds differently under a different set of present molecules, and identically under the same set.	PASS
T10	top level learns	Pairs are learned at the top level only when the top has room for two.	PASS
T11	recognition from surprise	Episodes alternate all-rain and all-sun.	PASS
T12	success	Prior top-level wins extend a thought's life against a stream of new arrivals.	PASS
T13	surprise	A surprising arrival lives longer than an expected one.	PASS
T14	novelty	A structurally unique thought lives longer than one like the background.	PASS
T15	blocked momentum	A thought pulled toward an absent partner lives longer than one with nothing to reach for.	PASS
T16	closure and re-opening	Closure: an open thought is resolved, boxed (leaves without decaying), and re-opened later when its resolver comes back into the winners.	PASS
T17	hole brings back a box	Blocked momentum points at the hole: a boxed thought that fills it is brought back and bonds.	PASS
T18	rationalization	Rationalization (sec.	PASS
T19	contradiction	Contradiction (sec.	PASS
T20	told vs learned	Told vs learned (sec.	PASS
T21	three endings	Three endings to being told (sec.	PASS
T22	rumination	Rumination (sec.	PASS
T23	keys are sites	Keys are sites (sec.	PASS
T24	two forgettings	Two forgettings (sec.	PASS
T25	quiet room	The pool feeds itself (sec.	PASS
T26	boredom	Boredom (sec.	PASS

#	TEST	WHAT IT CHECKS	BOTH ARMS
T27	shower moment	The shower moment (sec.	PASS
T28	conversation	Conversation (sec.	PASS
T29	partner search	Partner search (sec.	PASS
T30	perceived clauses	Perceived clauses (sec.	PASS
T31	several unknown rules	Several unknown rules (sec.	FAIL
T32	events arrive	Events arrive (sec.	PASS
T33	responses learned	Responses learned (sec.	PASS
T34	experiences are thought about	Experiences are thought about (sec.	PASS
T35	meta audit	Connectome audit for the meta layer (the organ atlas's standing instrument): every kind of meta molecule forms in a scenario built for it, and wins a top slot.	PASS
T36	noticing gates the operation	Capability without receptor is latent: a question with two holes nothing fills is parked only when "stuck" (or "looping") about it wins a top slot.	PASS
T37	mirror trap	The mirror trap: a meta molecule lives by its prediction, not by restating what it saw.	PASS
T38	conflation	Conflation : a thought whose outcomes are good when sunny and bad when raining is noticed as conflated, split on the weather into two new molecules, and the split confirmed as a distinction; afterwards the active half's own...	PASS
T39	collapse	Collapse (SOV's dual of Differentiate): a thought split on a tag that makes no difference (windy or calm, outcomes always good) is noticed as "the same", and the halves are unified back into it.	PASS
T40	understanding worked out	Understanding is worked out (sec.	PASS
T41	questions from open slots	Questions from open slots: when the SOV molecule for rain reaches the top it poses one question, for the open slot with the most evidence waiting (the user, 2026-10-22; first written as one question per open slot).	PASS
T42	questions answered	A question closes when the register fills its slot: the questions about what rain co-occurs with and what it causes are answered as those relations are worked out; the role question stays open.	PASS
T43	confidence in action	Confidence in action follows understanding.	PASS
T44	closure guarded	Closure is guarded by understanding: T18's question bound to an attractive partner that does not fit is not rationalized closed when the partner is poorly understood.	PASS
T45	speech by understanding	Speech by understanding: a relation lived and funded is stated; the same relation only told is hedged; open slots are asked.	PASS
T46	structure match	Measuring a learned structure against a target: a register holding latch -conditions-> lever -causes-> door preserves all of the target "A conditions B, B causes C"; the target with its edges reversed keeps half; a register...	PASS

#	TEST	WHAT IT CHECKS	BOTH ARMS
T47	analogy from experience	Analogy from experience (SOV's Abstract): in a designed world with a mechanical chain (latch, lever, door) and an electrical one (switch, current, lamp), the agent works out both chains and forms one abstract structure mapping...	PASS
T48	transfer	Transfer (SOV's Abstract then Constrain): after the analogy, a third world shows only nutrient -> metabolism.	PASS
T49	a world teaches its target	A generated world teaches its target: "A conditions B, B causes C" as a gate (latch released, lever moves, door opens), with a decoy that varies independently in time.	PASS
T50	analogy across generated domains	Analogy across generated domains: the same gate target in three domains (mechanical, electrical, digital), 300 ticks each, one after another.	PASS
T51	adaptive curriculum	Adaptive curriculum: from a library of three targets, the next world is the target least learned under its intended mapping, and a fully learned target is spent and retired.	PASS
T52	phase emergence and hysteresis	A phase emerges from the configuration and has hysteresis: frustration forms once the want has been blocked 4 ticks; when the want is dropped unmet (tick 10) it does not stop, it fades on its half-life and expires open at...	PASS
T53	phases steer attention	The phase steers the SOV's attention: with anger anchored at the jammed lock (blamed for the blocked want), the lock's first question is what caused it or under what conditions.	PASS
T54	closure versus expiry	Closure and expiry: frustration closes when the want is met (tick 10) and is gone for good: the want coming back later is a new want, nothing re-opens.	PASS
T55	blame spreads: the cancer	The cancer: anger replicates when blame spreads.	PASS
T56	curriculum worlds: answer keys	The worlds' answer keys, computed by the simulator (known answers, no agent): the block slides under a hard push only, the heavier and the fastened block not at all; the marble run drops marble, lever, gate (with the flag) and...	PASS
T57	block: grip, and a gust's coincidence	The movable block, learned by the agent (150 ticks): it predicts that a hard push slides the block and a light one does nothing.	PASS
T58	button: the plug conditions the press	The powered button, scripted interventions (no gusts): press twice with the plug seated, unseat it, press twice more.	PASS
T59	calculator: key sizes	The unfamiliar calculator (150 ticks, gusts on): each key's own size is learned (key 1 +1, key 2 +2, from the register's display), and the probe predictions overlap the answer key by 0.8 or more.	PASS
T60	spring and raised weight: analogy	Spring and raised weight: one mind through the spring world, then the weight world (150 ticks each).	PASS
T61	transfer to held-out objects	Transfer to held-out objects (no push made there yet): an untried push is predicted from a tried one in the same configuration, its effects mapped through a correspondence of objects that keeps their gerunds.	PASS
T62	paired lights: refutation by intervention	Paired lights : a hidden driver lights lamp A, then lamp B.	PASS

#	TEST	WHAT IT CHECKS	BOTH ARMS
T63	safety catch: the latch prevents	Safety catch , the agent exploring (150 ticks): the lid's release turns on the latch (a condition on the push), and every probe (latch seated or not) is predicted exactly.	FAIL
T64	AND: two plugs together	AND , scripted presses with no gusts: none seated, A only, B only, both.	PASS
T65	OR: the calculator spills	OR, the calculator's spill (scripted, no gusts): key 2 pressed at levels 0, 2, 4, 6, reset, then 0, 2, 4.	PASS
T66	three enabling systems	Same structure, different stuff : one mind through the powered button, the safety catch and the guarded file (permission / command / file in the agents' gerunds), 150, 150 and 40 ticks.	FAIL
T67	false analogy: chain and fork	False analogy : a mind trained on the chain marble run, carried to a renamed chain and a renamed fork.	PASS
T68	goals: making things happen	Goals (stage 3): after exploring each world (150 ticks), the agent is set a goal from a set-up start and plans with its own model.	FAIL
T69	stages 1-2, second round	The rest of stages 1-2 (150 ticks each, gusts on): the chime's strike turns on the damper being off; each path's button turns on its own plug; the redundant press turns on either plug (an OR, two alternatives); the false...	PASS
T70	Milestone E: learning progress chooses	Milestone E: the agent's own learning progress (how much its gap fell on its last visit to a world; unseen worlds first) chooses the next of twelve worlds, against round robin, the same budget (48 visits of 10 ticks), one mind.	PASS
T71	stage 4: systems	Stage 4, systems (scripted where the point needs a sequence, 150 ticks of exploration elsewhere): removing any of three plugs in series stops the press, a fourth on a dead line does not (one condition of three facts); a light...	PASS
T72	fixes: thresholds, watches, experiments, polarity, role, observed	The fixes of 2026-10-23 (THOUGHT_MOLECULES.md sec.	PASS

Open failures

Four results are reported as they stand. In each case the arm named is the one that did not come out as expected:

T31: person's rules have them
T63: re-opening on
T66: enabling pair, the latch apart
T68: with a goal; curiosity only

The milestone ladder

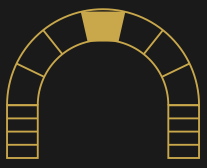
The curriculum design sets eight milestones, each a claim the agent must earn in order. The last column names the existing tests that come closest to each claim; it is a reading of the suite, not a mapping the code maintains. F, G and H have no test yet.

Milestones A to H

MILESTONE	CLAIM	CLOSEST TESTS
A	Stage 0 + one causal chain; learn without labels	T56, T57
B	enabling and blocking with active experiments	T58, T62, T64
C	sequence-dependent unfamiliar tool and stored potential	T59, T60
D	cross-domain analogy; held-out prediction improves with analogy enabled	T47, T48, T60, T61
E	SOV-guided selection outperforms fixed ordering	T70
F	historical reflection reduces recurring errors	no test yet
G	targeted dreaming improves waking transfer	no test yet
H	long-running open-ended development shows continued novel, verified structural discoveries rather than mere representation growth	no test yet

The first implementation slice

Five complete interactive worlds rather than fifty shallow ones: the movable block, the three-step marble run, the powered button, the unfamiliar calculator, and the spring and clamp with its raised-weight twin. Each has a misleading correlate (a gust that both moves the block and swings a flag; a flag that leans on the lever and falls with the gate), a counterfactual intervention, and a held-out variant with the same structure in different objects (the tank, tap, hose and bucket for the button). The answer key for every world is computed by the simulator against the no-push slices and never shown to the agent.



APPENDIX

E

The Lineage Shelf

What Gerundelle Takes and What It Leaves

IN THIS SECTION

- One annotated paragraph per intellectual ancestor
- What Gerundelle borrows, what it adds, and what it declines
- A single place to verify the book knows its neighbors

This appendix collects, in one place, the scholarly ancestors whose work shaped Gerundelle's design. Each entry states what Gerundelle takes from the ancestor and what it leaves behind. The entries are ordered by the section in which they first appear.

Alfred North Whitehead — Process Philosophy

Source: *Process and Reality* (1929). **Claim:** Reality is composed of processes, not static substances; becoming is prior to being. **What Gerundelle takes:** The foundational ontological commitment. Every node in the world model is a gerund — an ongoing activity — not a noun with properties. Objects are what they are doing, not what they are made of. **What Gerundelle leaves:** Whitehead's full metaphysical apparatus (eternal objects, prehensions, actual occasions). Gerundelle is a computational formalism, not a metaphysical system. The structural parallel is deliberate; the philosophical identity is not claimed.

Rich Sutton — The Bitter Lesson

Source: *The Bitter Lesson* (2019). **Claim:** Methods leveraging computation and learning consistently beat methods leveraging human-designed structure. **What Gerundelle takes:** The warning. The graveyard of hand-designed symbolic systems is real, and Gerundelle's 95-gerund vocabulary is designed structure. **What Gerundelle leaves:** The conclusion that structure should be abandoned entirely. The bet is that Newton's force vocabulary *closes* — a small set of primitives covers the space — while GOFAI's frame-based ontologies never did. The ablation experiments test whether this distinction holds.

Leonard Talmy — Force Dynamics

Source: *Force Dynamics in Language and Cognition* (1988). **Claim:** Natural language organizes causation as force patterns between agonists and antagonists, not as abstract logical conditionals. **What Gerundelle takes:** The insight that a force-dynamic vocabulary is not an invention but an excavation of how humans already represent causation. The 95-gerund vocabulary and Appendix A's force tables are Talmy's categories given slots, reserves, and transition rules. **What Gerundelle leaves:** The linguistic framing. Talmy analyzed language; Gerundelle builds a world model. The gerunds serve a physics engine, not a parser.

Elizabeth Spelke — Core Knowledge

Source: Infant cognition studies (1990s–2000s). **Claim:** Human infants come equipped with expectations about objects, contact causality, and persistence — a physics-shaped model is innate, not learned from language. **What Gerundelle takes:** The evidence that a structured world model is cognitive starting equipment, not a late cultural acquisition. Gerundelle's positionless force model is a candidate for what that starting equipment computes. **What Gerundelle leaves:** Innateness. Gerundelle's agent starts with the vocabulary but earns every concept through the curriculum. The structure is given; the content is not.

Josh Tenenbaum — Intuitive Physics Engine

Source: Probabilistic simulation models of physical reasoning (2010s). **Claim:** Human physical reasoning operates by running approximate simulations forward, matching infant looking-time data and adult prediction tasks. **What Gerundelle takes:** The framework of reasoning-as-simulation. Gerundelle steps scenes forward through time slices, just as Tenenbaum's engine runs probabilistic rollouts. **What Gerundelle leaves:** Continuous coordinates and probabilistic inference. Gerundelle's simulation is discrete

(gerund slices, not trajectories), deterministic (argmax, not sampling), and positionless (force-balance states, not spatial coordinates).

James Gibson — Ecological Affordances

Source: *The Ecological Approach to Visual Perception* (1979). **Claim:** Organisms perceive affordances — action-possibilities that are relational properties of the animal-environment system — directly, not by first recognizing objects and then computing actions. **What Gerundelle takes:** The relational ontology. Affordances are not intrinsic properties of objects but structural invitations that exist only in context. An open force slot formalizes what Gibson described but couldn't compute. **What Gerundelle leaves:** Direct perception. Gerundelle's agent reads affordances from a computed AST, not from optic flow. The formalization recovers Gibson's meaning while abandoning his anti-representationalism.

Jürgen Schmidhuber & Pierre-Yves Oudeyer — Artificial Curiosity

Source: Compression progress (Schmidhuber, 1991); learning-progress intrinsic motivation (Oudeyer et al., 2007). **Claim:** Agents should seek states where their model is improving fastest, treating prediction-error reduction as an intrinsic reward. **What Gerundelle takes:** The core intuition that curiosity is about information gain, not hand-coded rewards. Expected surprise reduction drives exploration. **What Gerundelle leaves:** The pure model-improvement signal. Gerundelle gates curiosity by *physical invitation* (an open force slot must exist in the world), so a sealed box with no input produces no curiosity regardless of how poorly understood it is.

Bernard Baars & Stanislas Dehaene — Global Workspace Theory

Source: *A Cognitive Theory of Consciousness* (Baars, 1988); *Consciousness and the Brain* (Dehaene, 2014). **Claim:** Consciousness is a limited-capacity broadcast: many specialized processors compete for access to a shared workspace; the winner gets broadcast to all. Dehaene added the neuroscience: cortical ignition, all-or-nothing access, sharp thresholds. **What Gerundelle takes:** The architecture of competitive broadcast with limited capacity. The three-level tournament converges on GWT's design for engineering reasons (the lock-in and scaling problems). **What Gerundelle leaves:** The claim of neural equivalence. GWT describes brains; Gerundelle describes a computational architecture. The convergence is evidence the design region is good, not a claim of identity.

George Miller & Nelson Cowan — Working Memory Capacity

Source: *The Magical Number Seven* (Miller, 1956); working memory capacity estimates (Cowan, 2001). **Claim:** Human working memory holds approximately 7±2 items (Miller) or 4±1 focused items (Cowan). **What Gerundelle takes:** The slot budgets. The tournament's 7/5/4 configuration is empirically anchored rather than arbitrary — 7 at the broadest level, 4 at the top. **What Gerundelle leaves:** Any claim about neural implementation. The numbers are design parameters informed by cognitive data, not claims about hippocampal capacity.

Karl Friston — The Free Energy Principle

Source: The Free Energy Principle and active inference (2006–present). **Claim:** All adaptive systems minimize variational free energy — a bound on surprise. Perception updates the model; action changes the world to match predictions. **What Gerundelle takes:** The specific mechanisms that fixed the lock-in failure: Dirichlet outcome beliefs, expected information gain as novelty, forward-looking surprise reduction. FEP is presented as the thing that solved a real engineering bug. **What Gerundelle leaves:** The universality claim and the continuous-variable formalism. Gerundelle's surprise reduction operates over discrete typed relation slots, not a continuous variational objective. The architectural relationship: FEP describes the inner loop; Gerundelle adds the dynamics of the space that loop runs in. **Under the territorial reorganization:** FEP's pull now serves a deeper motive rather than being the motive. Surprise marks where a territorial boundary is open; free energy stays as the tie-breaker and the half-life modulator, but the reason surprise matters is territorial, not variational.

Bluma Zeigarnik — The Zeigarnik Effect

Source: Studies on interrupted tasks (1927). **Claim:** Interrupted tasks are remembered better than completed ones. **What Gerundelle takes:** The phenomenon. Open-site tension is a mechanistic account: an incomplete thought has a non-decaying constant added to its bid. **What Gerundelle leaves:** The original framing as a memory phenomenon. Gerundelle reframes it as an attentional mechanism — the incomplete thought dominates consciousness, not just recall — and makes a sharper prediction: the advantage vanishes *instantly* when the site fills, not gradually.

Leon Festinger — Cognitive Dissonance

Source: *A Theory of Cognitive Dissonance* (1957). **Claim:** Holding contradictory beliefs creates discomfort that people reduce by changing beliefs, adding rationalizations, or avoiding the contradiction. **What Gerundelle takes:** The mechanism of rationalization — closing an open question on a non-matching but high-scoring partner to reduce tension. **What Gerundelle leaves:** The motivational framing. Gerundelle doesn't model discomfort as a feeling; it models rationalization as a structural event (a tag mismatch overridden by co-win score) with a specific guard condition (50% understanding threshold) that Festinger never specified.

Thomas Kuhn — Paradigm Shifts

Source: *The Structure of Scientific Revolutions* (1962). **Claim:** Science alternates between normal periods (working within a paradigm) and revolutionary periods (anomalies force the paradigm open). **What Gerundelle takes:** The anomaly → crisis → revision cycle. Contradiction forcing a storage box open is Kuhn's cycle at the scale of a single belief. **What Gerundelle leaves:** The sociology of science. Kuhn's revolutions involve communities, politics, and generational change. Gerundelle's contradictions are individual cognitive events with a precise trigger (prediction fires, world disagrees).

Jean Piaget — Accommodation and Developmental Stages

Source: Genetic epistemology (1950s–1980s). **Claim:** Cognitive development proceeds through stages, each building on the previous; accommodation occurs when existing schemas fail to assimilate new experience. **What Gerundelle takes:** Staged development (curriculum stages 0–10 are Piagetian in structure) and the accommodation mechanism (conflation detection and splitting is forced schema revision). **What Gerundelle leaves:** The specific human stages (sensorimotor, preoperational, etc.) and the biological maturation timeline. Gerundelle's stages are computationally motivated, not developmentally observed.

Susan Carey — Conceptual Change

Source: Studies on childhood conceptual change (1985–2009). **Claim:** Children initially conflate concepts (heat/temperature, weight/density) and split them only under pressure from contradictory outcomes. **What Gerundelle takes:** The fossil record of conflation. Carey's documented cases are exactly the bimodal-outcome pattern that Gerundelle's conflation detector triggers on. **What Gerundelle leaves:** The do-

main-specificity of Carey's theory. Gerundelle's conflation mechanism is domain-general — it triggers on outcome statistics, not on specific conceptual content.

Annette Karmiloff-Smith — Representational Redescription

Source: *Beyond Modularity* (1992). **Claim:** The mind re-represents its own knowledge in progressively more explicit, manipulable forms. **What Gerundelle takes:** The metacognitive layer. The meta track notices and operates on the system's own representations. **What Gerundelle leaves:** The implicit-to-explicit gradient. Gerundelle's meta-molecules are explicit from the start — they're molecules competing in the tournament. The redescription happens through the SOV track and the historical diary, not through format change.

George Loewenstein — Information-Gap Theory of Curiosity

Source: *The Psychology of Curiosity* (1994). **Claim:** Curiosity is the feeling of a perceived gap between what you know and what you want to know. **What Gerundelle takes:** The gap. The Understanding Register's open slots are Loewenstein's gaps given geometry — typed, funded at specific confidence levels, surrounded by specific known bonds. **What Gerundelle leaves:** The feeling. Gerundelle models the structural gap and the attentional pull it creates, not the phenomenal experience of curiosity. The architecture predicts when curiosity should arise; whether anything it's *like* to be curious accompanies the prediction is a separate question. **Under the territorial reorganization:** Loewenstein's gap is now the unclaimed slot in the ownership SOV. The information gap still exists and still pulls, but it pulls in service of territory — the agent explores the gap because an ownership slot funds the inquiry, not because the gap is intrinsically motivating. Curiosity is the scouting strategy, not the motive.

Lazarus, Scherer & Damasio — Appraisal Theory and Somatic Markers

Source: Appraisal theory (Lazarus, 1991; Scherer, 2001); *Descartes' Error* (Damasio, 1994). **Claim:** Emotions are structured evaluations of events relative to goals and coping capacity (appraisal theory); emotional states are bodily signals indispensable to rational decision-making (Damasio). **What Gerundelle takes:** Both claims, made executable. Phase molecules are structured evaluations (the anchor encodes what's appraised; the type encodes the result). Removing the phase track doesn't make the agent more rational — it removes the ability to prioritize questions. **What Gerundelle leaves:** Somatic

markers as bodily states. Gerundelle's phase molecules are computational, not embodied. The functional role is preserved; the substrate is not.

Susan Nolen-Hoeksema — Rumination

Source: Response Styles Theory (1991). **Claim:** Repetitive passive focus on negative thoughts is a transdiagnostic risk factor for depression, anxiety, and substance abuse. **What Gerundelle takes:** The clinical observation, given a mechanism. Told blame on new anchors spawns copies that escape the phase track's apoptosis, flood the main thought slots, and resist closure. **What Gerundelle leaves:** The passivity framing. Gerundelle's rumination is structurally active — the copies are competing and winning slots — but functionally passive in that they don't advance toward resolution. The architecture predicts *when* rumination occurs (told blame proliferates anchors faster than phase slots can contain them), which the clinical literature describes but doesn't mechanize.

Lev Vygotsky — Zone of Proximal Development

Source: *Mind in Society* (1978). **Claim:** The optimal next lesson is just beyond what the learner can do alone but within reach with scaffolding. **What Gerundelle takes:** The adaptive curriculum principle. The world generator inspects the agent's open SOV slots and instantiates the experience that maximizes expected understanding gain per interaction cost — the ZPD mechanized. **What Gerundelle leaves:** The social dimension. Vygotsky's ZPD involves a more capable other; Gerundelle's curriculum generator is a designed system, not a social partner. The curriculum stages where other agents appear (Stage 6: Other Minds) begin to close this gap.

Sleep Neuroscience — Consolidation and Replay

Source: Walker (sleep-dependent insight); Buzsáki (hippocampal replay); Shin et al. (generative replay, 2017). **Claim:** The sleeping brain replays, recombines, and reweights waking experience; generative replay prevents catastrophic forgetting in neural networks. **What Gerundelle takes:** The functional framework. Dreaming replays failed episodes with variables flipped, generates candidate slot fillers, and recombines cross-domain elements. **What Gerundelle leaves:** The neural implementation and the continuum between NREM consolidation and REM dreaming. Gerundelle's dreaming is a single offline phase with strict provenance tagging, not a biologically faithful sleep-stage model.

Deci & Ryan — Self-Determination Theory

Source: *Intrinsic Motivation and Self-Determination in Human Behavior* (1985); Ryan & Deci (2000). **Claim:** Three basic psychological needs — autonomy (acting from one's own endorsement), competence (effectively interacting with the environment), and relatedness (connecting with others) — underlie motivated behavior. Goals aligned with these needs (*self-concordant* goals, Sheldon & Elliot, 1999) produce sustained effort and well-being; externally imposed goals produce compliance without engagement. **What Gerundelle takes:** The three-need structure maps directly onto the ownership SOV: autonomy is control (the agent's push works), competence is understanding funded by control slots, and relatedness is the recognition/appreciation/attribution triad. The parent's-standard case — an external target met without becoming territory — is Sheldon and Elliot's non-concordant goal, formalized. **What Gerundelle leaves:** The phenomenological framing. SDT describes felt needs; Gerundelle describes structural states of the ownership graph. Whether the agent *feels* autonomous when its control slot is filled is a separate question.

Haines & Schutte — Parental Conditional Regard

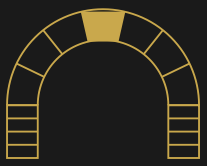
Source: Haines & Schutte (2023). **Claim:** Parental conditional regard — love and approval contingent on meeting the parent's standards — predicts lower well-being and more external motivation in children. **What Gerundelle takes:** The cost structure of externally withheld appreciation. An appreciation slot that opens and closes based on another agent's standards, rather than the agent's own claims, produces a specific ownership pathology: the agent acts to fill someone else's slot, not its own. **What Gerundelle leaves:** The clinical and developmental data. Gerundelle models the structural mechanism, not the long-term developmental outcomes in humans.

Ethology of Territory & Mech on the Wolf Alpha

Source: Classical ethology (Lorenz, Tinbergen); Mech (1999), revision of the wolf "alpha" concept. **Claim:** Territorial behavior in animals — marking, patrolling, defending — is among the most ancient and conserved behavioral patterns. Mech's revision showed that the wolf "alpha" is not a dominance winner but a *parent*: the alpha holds the most territory because it has the most to defend (offspring, den, food stores), not because it won a fight. **What Gerundelle takes:** The structural insight that the alpha is the most fragile — the agent holding the most territory has the most to lose, and loss pulls hardest. Territory is not about dominance but about what you hold. **What Gerundelle leaves:** The biological specifics of scent marking, patrol routes, and reproductive competition. Gerundelle's territory is computational (regions of the force world where the agent's push works), not physical.

Boehm — The Reverse Dominance Hierarchy

Source: *Hierarchy in the Forest* (Boehm, 1999). **Claim:** Human forager societies maintain egalitarianism not through absence of hierarchy but through *reverse dominance* — coalitions that actively prevent any individual from accumulating too much power. **What Gerundelle takes:** The frontier trade-off. Extending territory increases loss exposure: the more regions held, the more borders to defend, the more contested slots to monitor. Boehm's reverse dominance is the social enforcement of this structural cost — coalitions that make territorial overextension expensive. **What Gerundelle leaves:** The political and cultural mechanisms. Gerundelle models the cost curve (more territory = more fragility) but not the coalition dynamics that enforce it in human societies.



APPENDIX

F

The Formal Guarantees

What the machine checked, and where it checked it

IN THIS SECTION

- The seven theorems that prove the architecture's core promises
- The seven lemmas that prove its boundaries — what can't be cheated, what can't be decomposed, and where the nice properties stop
- Where to find each proof and how to verify it yourself

The experiments in Appendix D test whether each mechanism earns its place. The equations in Appendix B specify how it works. This appendix is different: it lists what **must** hold, proved by machine, for every possible input, every possible sequence of operations, every possible interleaving. Not "it worked on the cases we tested." "It holds, and Lean verified every step."

The proofs live in `formalization/SOV/`, fourteen Lean 4 files importing Mathlib. To verify them yourself: install Lean 4, run `lake build` in the formalization directory. If it compiles, every theorem below holds. If it doesn't, something is wrong and you've found it.

Each entry names the theorem, says in one sentence what it guarantees, names the file, and says which chapter's claim it secures. The proofs themselves are not reproduced here. They are machine-checked; reading them is optional. Knowing they exist is not.

The seven theorems

These prove the architecture's core promises. If any one of them failed, a central claim of the book would be wrong.

SIDEBAR

1. The imagination firewall

`firewall` (`Firewall.lean`): the funded web after any interleaving of funded operators, free register moves, and Posit seeds is the funded web of the funded operators alone. Nothing imagined, and no register move, reaches the funded web.

What it secures: Chapter 5's provenance firewall between dreaming and waking, Chapter 10's counterfactual replay with `provenance = imagined`, and the sentence machine's guarantee that imagined evidence never funds a slot. The architecture's epistemic integrity stands on this theorem.

SIDEBAR

2. Imagined evidence is inert

`run_owner_filter_imagined` (`Market.lean`): inserting or deleting imagined trials anywhere in any event sequence changes no owner. `bid_filter_imagined`: dropping imagined trials changes no bid.

What it secures: The market cognition framework's claim that imagined evidence funds nothing. A mind that dreams, hypothesizes, or speculates cannot change what it owns or what it knows by doing so. Only lived interaction with the world changes ownership.

SIDEBAR

3. Told evidence can't be promoted

`took_told_le` (`Market.lean`): told-only evidence funds at most c_0 per trial, whatever the processing. No sequence of operations can promote a told receipt to lived weight.

What it secures: Chapter 5's "you don't own what you haven't worked out." Chapter 8's territorial reading where told knowledge fades. The fiction ladder's told receipts at half weight. The provenance conservation law: lived stays lived, told stays told.

SIDEBAR

4. Ownership is order-dependent, funding is order-free

`bid_perm` (`Market.lean`): a bid depends only on the multiset of trials, not the order. `ownership_order_dependent`: the same three events in two orders leave a slot held in one and open in the other.

What it secures: The non-commutativity claim in `SOV_ALGEBRA.md` and `MARKET_COGNITION.md`, placed precisely: the order of evidence doesn't matter for what you know, but the order of events matters for who holds what. History is load-bearing for territory, not for funding. The non-commutativity lives in the threshold and reopen gates, nowhere else.

SIDEBAR

5. Paradoxes are broke, not banned

`not_grounded_of_selfCited` (`Provenance.lean`): a self-citing non-root entry is a legal entry that can never be grounded. `wellFounded_cites`: in an event-sourced log, citation is well-founded. `not_transGen_self`: no entry is in its own funding chain.

What it secures: The architecture doesn't need to prohibit circular reasoning. The provenance chain makes it inert. A liar-slot can exist in the web — it just can't fund anything. This is more robust than banning it: the system is safe by structure, not by rule.

SIDEBAR

6. Every valid curriculum is a geodesic

`card_learn` (`Pacing.lean`): each step of a valid curriculum adds exactly one concept. `never_enters`: a prerequisite cycle is never reached. `reachable_of_downClosed`: every down-closed state is reachable when prerequisites form a strict order.

What it secures: Chapter 10's adaptive curriculum. No valid path wastes a step. No valid path gets stuck in a loop. Every achievable state of understanding can be reached by some valid curriculum. The curriculum contract is not just a design aspiration — it's a proved property of the prerequisite manifold.

SIDEBAR

7. The pull order, with its temperament inversion

`pull_order` (`Market.lean`): the design order lost > control > unclaimed holds when curiosity ≤ 1. `restless_inverts`: at curiosity 2, an unclaimed region with nothing else moving it outpulls every control slot.

What it secures: Chapter 8's loss aversion as a structural property, not a bias parameter. And the temperament system's guarantee that the inversion is intended: different parameter values produce different behavior with proved structure, not accidental variation. The restless mind that explores unclaimed territory before defending held territory does so by theorem.

The seven lemmas

These prove the architecture's boundaries — what can't be cheated, what can't be decomposed, and where the nice properties stop. Each one answers a doubt the theorems leave open.

SIDEBAR

8. No outbound morphism exists

`no_outbound` (`Firewall.lean`): every map from the imagination register back to funded webs gets lived mass wrong on some web. Webs with the same geometry can differ in funding, so funding can't be read off imagined structure.

What it proves: The firewall isn't a one-way valve that could in principle be bypassed. There is no function — not just no permitted function, no function at all — that reconstructs funding from imagination. The asymmetry is structural: forgetting funding is always possible; reconstructing it is impossible.

SIDEBAR

9. The Lamarckian pairing is a grade error

`lamarckian_ill_graded` (`Monad.lean`): the pairing that would make a parent's lived knowledge become a child's lived knowledge is ill-typed in the graded writer monad.

What it proves: Knowledge transfers across generations as told, never as lived, not because of a design choice but because the monad's grading won't let you type it otherwise. A civilization built on this algebra cannot inherit acquired characteristics. Each generation must test what it was told. This is the provenance firewall applied across lifetimes.

SIDEBAR

10. Lived mass can't grow without a Fit

`replay_mass_le` (`Ledger.lean`): no operator, no sequence of operators, no interleaving can increase lived funding without a Fit event (an interaction with the world).

What it proves: The entire learning-through-interaction claim. You can't create evidence from nothing. You can't amplify told knowledge into lived knowledge by processing it. You can't think your way to funding. Only the world, pushing back against a push, creates lived mass.

SIDEBAR

11. Relabel is primitive

`relabel_not_meets` (`Holonomy.lean`): no composite of meet operations produces the Relabel operator.

What it proves: The operator set is not redundant. At least one of the sixteen operators is genuinely irreducible — it does something no combination of the others can do. The architecture's operator vocabulary, like the gerund vocabulary, was chosen to close: each operator earns its place.

SIDEBAR

12. No global coordinates when holonomy is non-trivial

`flat_iff_trivializable` (`Holonomy.lean`): zero holonomy everywhere if and only if global coordinates exist. `relabel_accumulates`: Relabel makes holonomy non-trivial.

What it proves: Chapter 1's claim that the world model has no positions, no distances, no coordinates. If the SOV graph has non-trivial holonomy — and Relabel guarantees it can — then structure is inherently local. There is no global frame of reference. This is not a simplification the architecture chose. It is a property the algebra forces.

SIDEBAR

13. Accrual and eviction don't commute

`accrue_evict_not_comm` (`Monad.lean`): a lived accrual followed by an eviction produces a different ledger than an eviction followed by a lived accrual.

What it proves: The precise boundary of confluence. The P66 conjecture asked whether the monotone fragment reaches the same fixed point under any update order. This lemma draws the line: Constrain and Exclude are confluent, Anneal commutes with itself, but wherever a fiber reads geometry that another operator narrows, or an eviction writes off mass a later accrual would have kept, the order matters. The architecture is confluent where it can be and order-dependent where it must be, with the boundary proved.

SIDEBAR

14. Perception is determined by invariance

`invariant_iff_factors` (`Erlangen.lean`): a reading is a function on the Umwelt exactly when it is T-invariant. `D_eq_zero_iff`: consequentially identical states are one point.

What it proves: What an agent can perceive is determined by which transformations it can't distinguish. The Umwelt — the world as the agent sees it — is the quotient of the world by the agent's invariances. Different invariances, different perceived worlds. This connects the embodied cognition claim (what you see depends on what you do) to the symmetry structure of the algebra: your distinctiveness space is shaped by which distinctions your interactions have funded.

Law 3 needs its gate

One result belongs in neither list because it is not a guarantee or a boundary but a **necessity proof** — a theorem that a precondition is required, not just sufficient.

`not_grounded_of_unfunded` (`Provenance.lean`): a non-Fit operator that appends a receipt but cites nothing appends an ungrounded receipt. Law 3 (every receipt stays grounded) holds only when the funding gate is enforced.

This matters because it answers the question: "could you simplify the architecture by removing the funding gate?" No. Without it, ungrounded entries appear. The gate is not a safety margin. It is the exact condition under which Law 3 holds, proved necessary by exhibiting the failure when it is removed.

How to verify

The proofs are in `formalization/SOV/`, fourteen Lean 4 files: `Geometry.lean`, `Ledger.lean`, `Slot.lean`, `Operator.lean`, `Firewall.lean`, `Provenance.lean`, `Market.lean`, `Confluence.lean`, `Monad.lean`, `Holonomy.lean`, `Erlangen.lean`, `Discount.lean`, `Pacing.lean`, `Quotient.lean`. They import Mathlib.

To check them:

You do not need to read or understand Lean to verify the proofs. You need to install the tool and run one command. The machine does the rest.

Key Terms

Machine-checked proof	A proof verified by a proof assistant (Lean 4) that checks every logical step mechanically. If the build succeeds, the theorem holds for all possible inputs.
Lean 4	A proof assistant and programming language. Proofs are programs; if the program typechecks, the theorem is valid.
Mathlib	The community mathematics library for Lean 4, providing standard definitions and theorems the SOV formalization builds on.
lake build	The command that compiles and checks the Lean project. Success means every proof is valid.
Funding gate	The precondition that a non-Fit operator must cite at least one earlier receipt to fire. Proved necessary for Law 3.
Confluence	The property that two different update orders reach the same result. Holds for Constrain/Exclude; fails where fibers read geometry or eviction writes off mass.
Graded writer monad	The algebraic structure the SOV operators form: a monad graded by which operator kinds appear, ensuring conservativity and detecting ill-typed pairings.